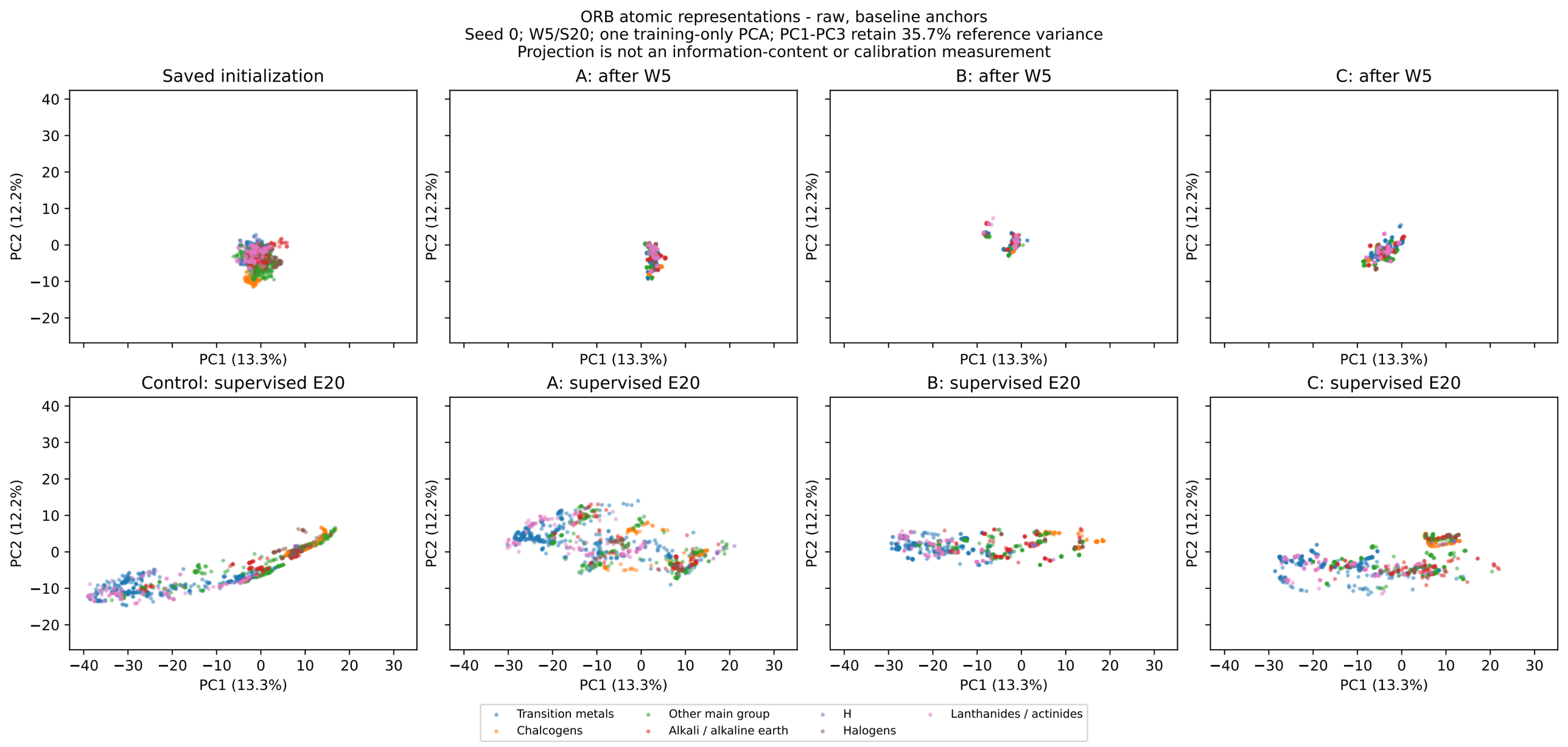
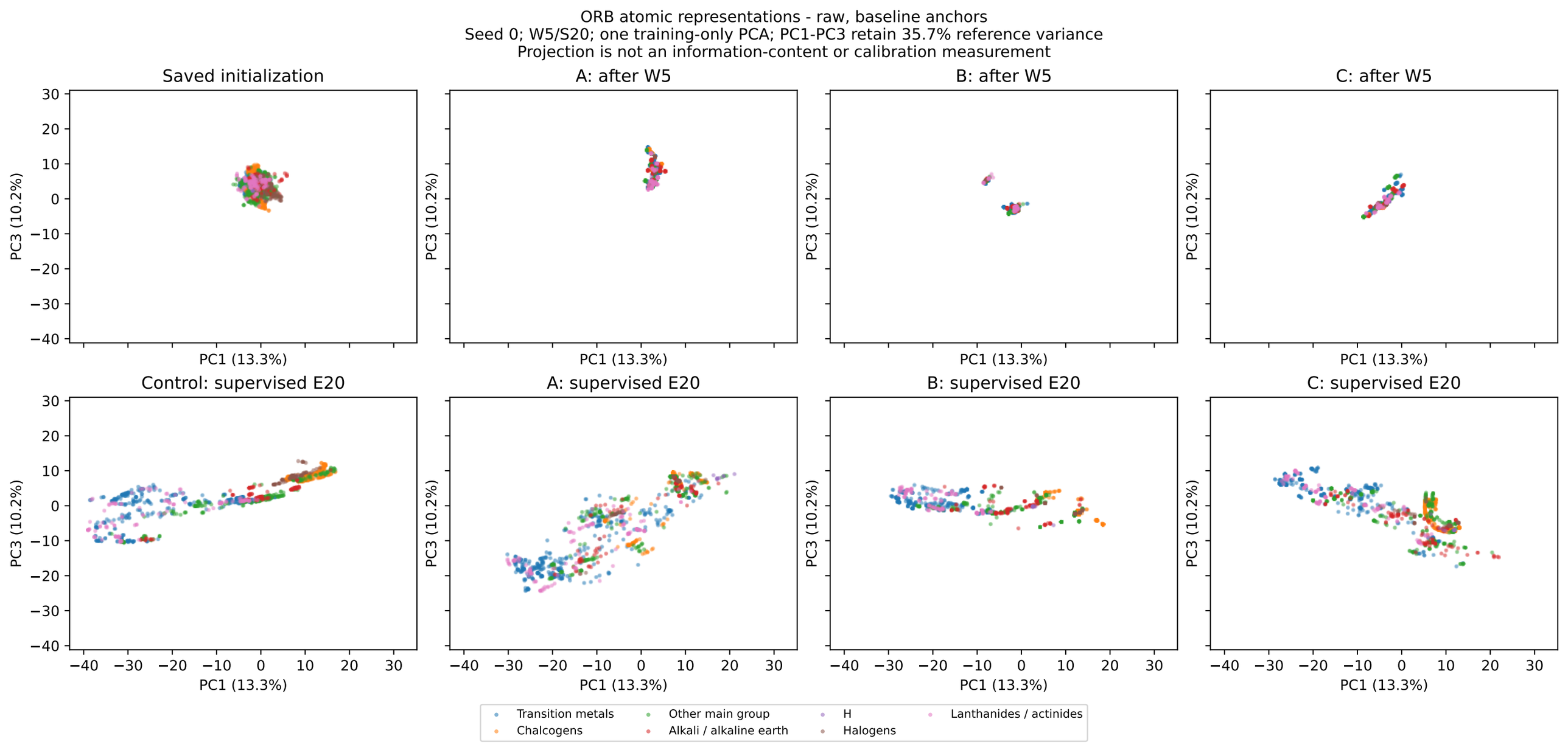
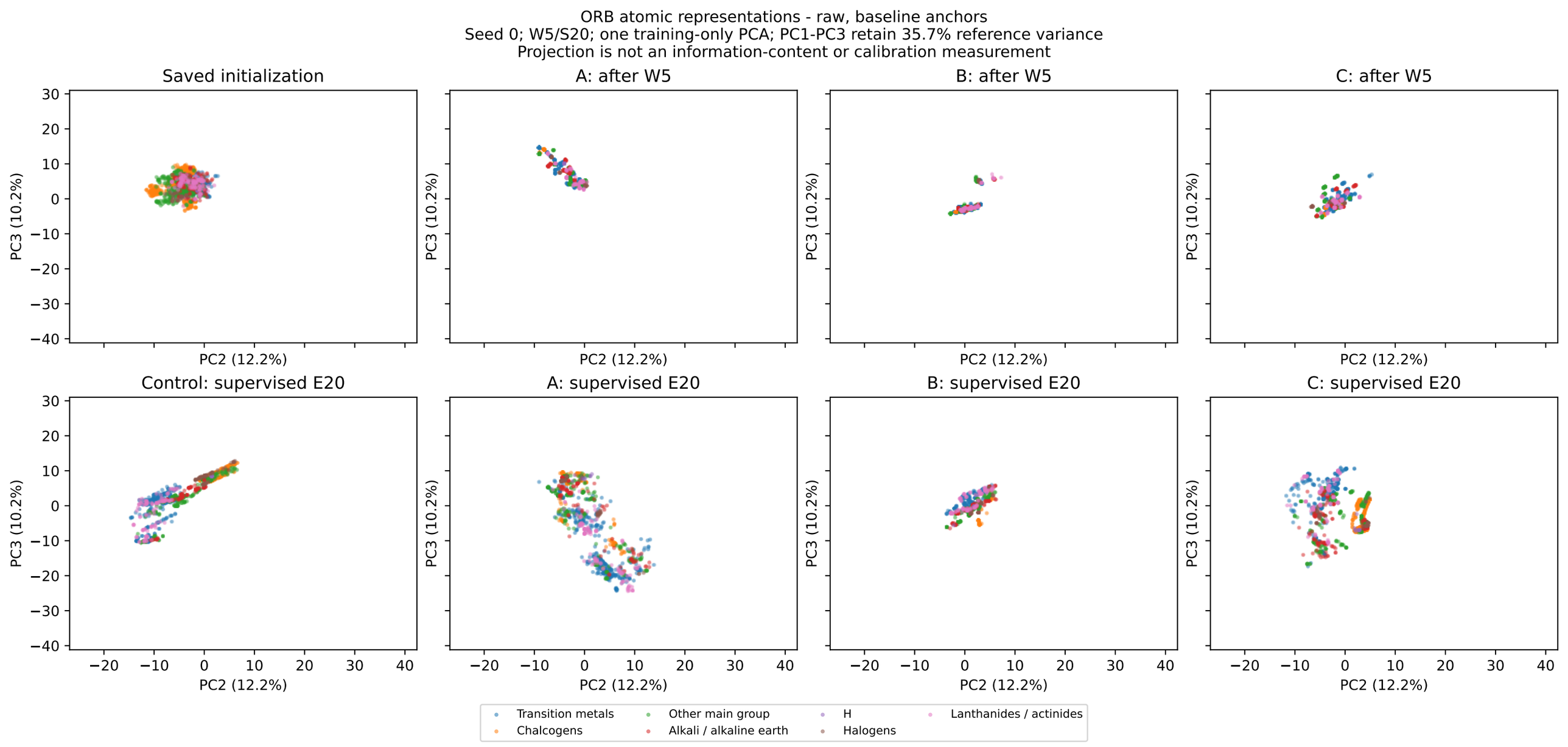
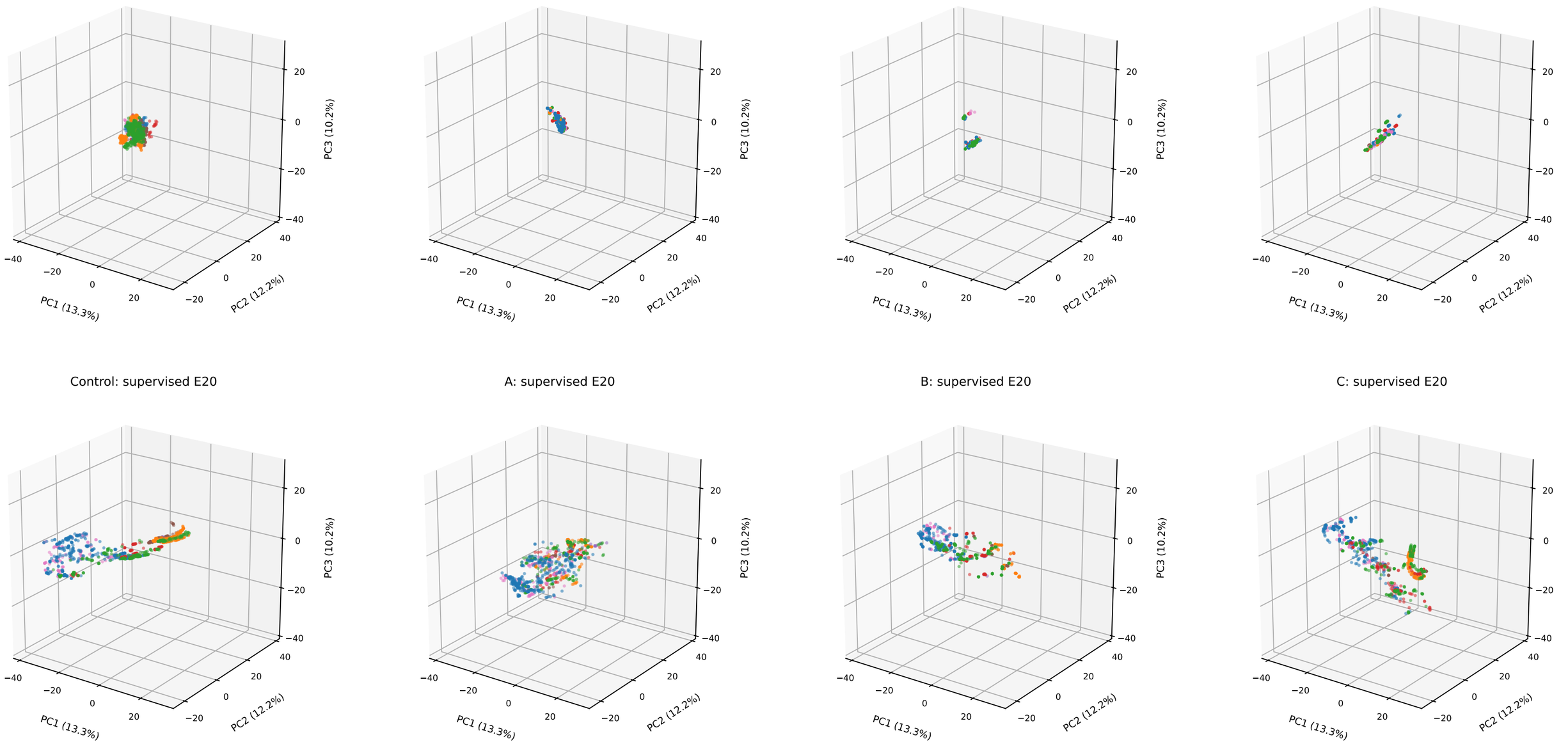

ORB atomic representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 35.7% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (13.3%)
PC2 (12.2%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

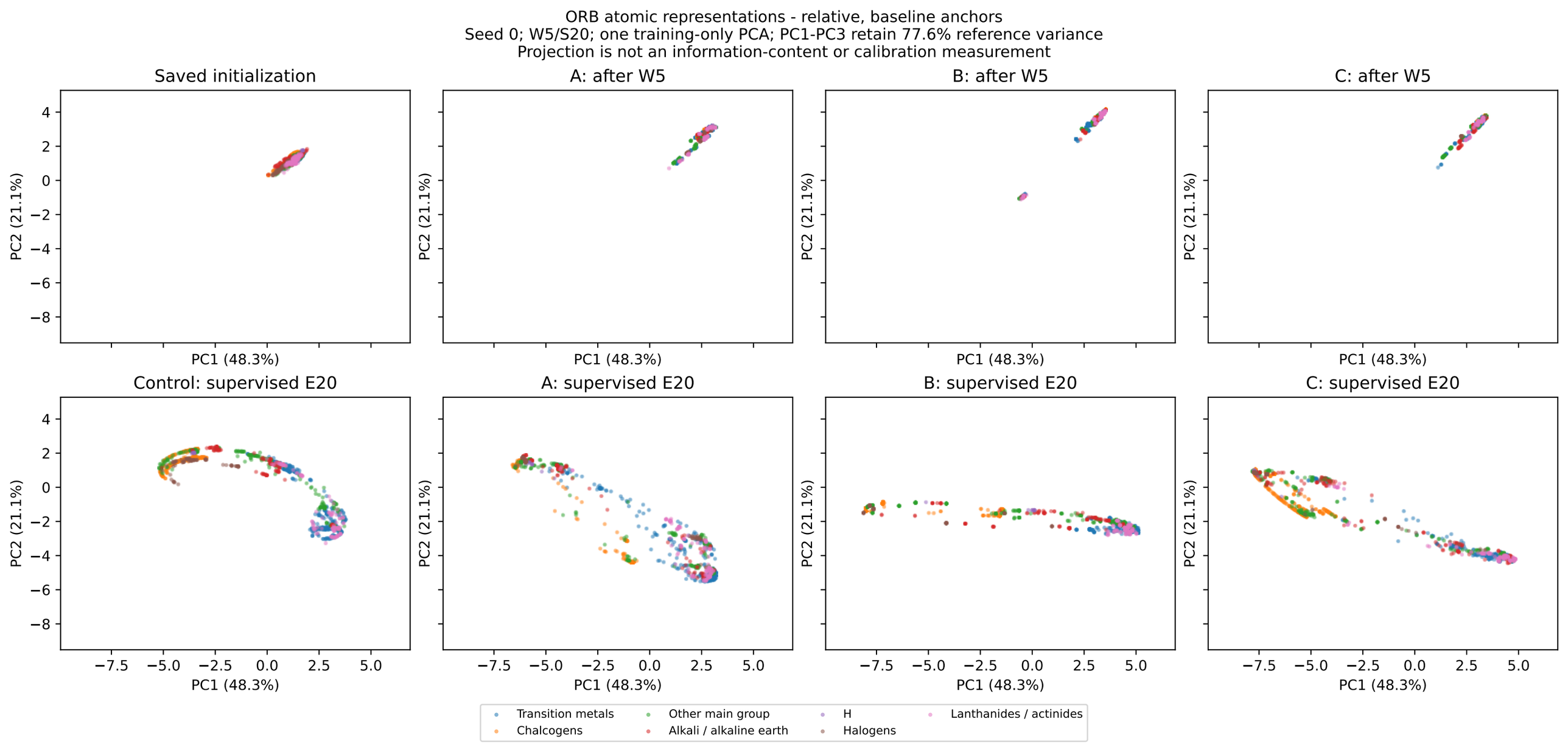

ORB atomic representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 35.7% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (13.3%)
PC3 (10.2%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

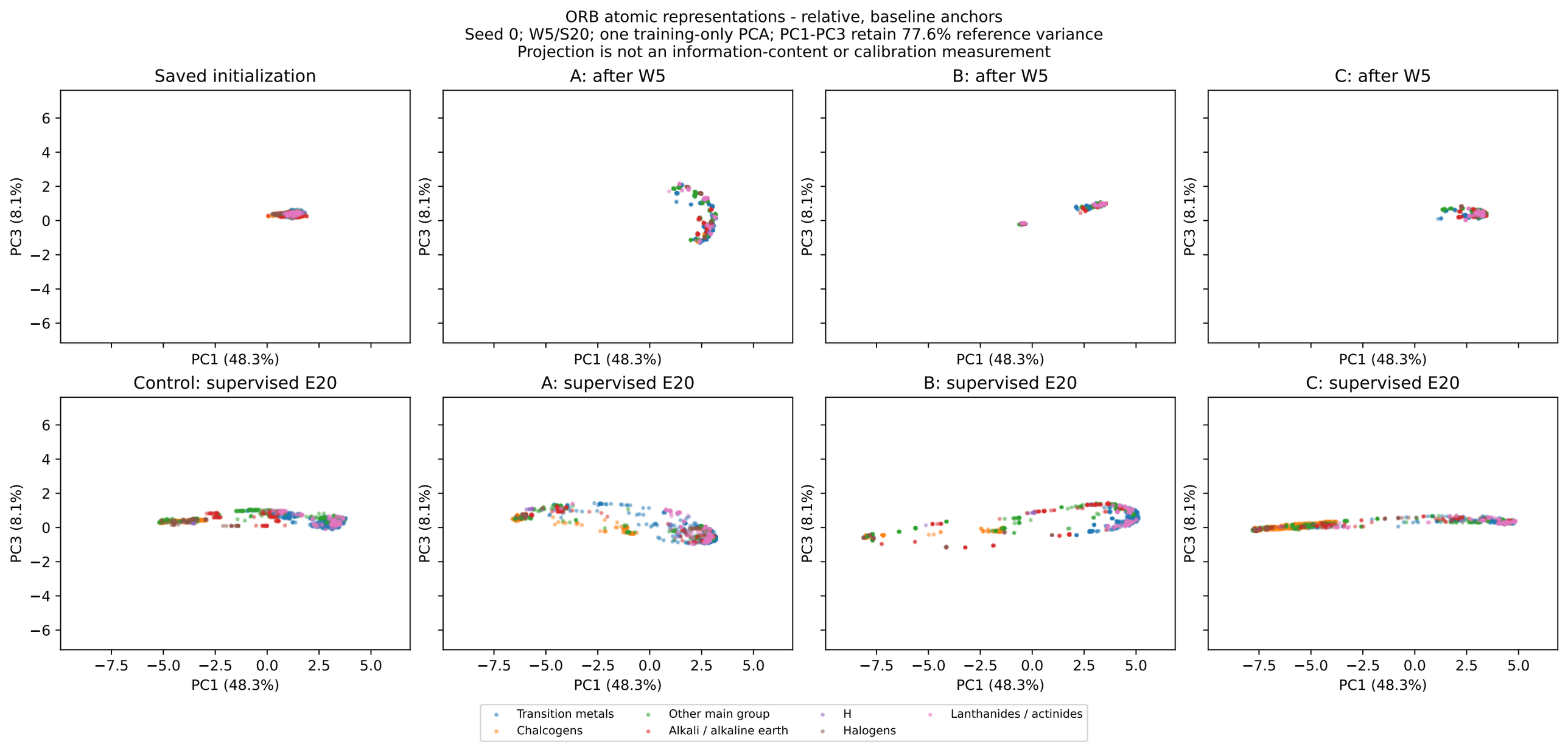

ORB atomic representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 35.7% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC2 (12.2%)
PC3 (10.2%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

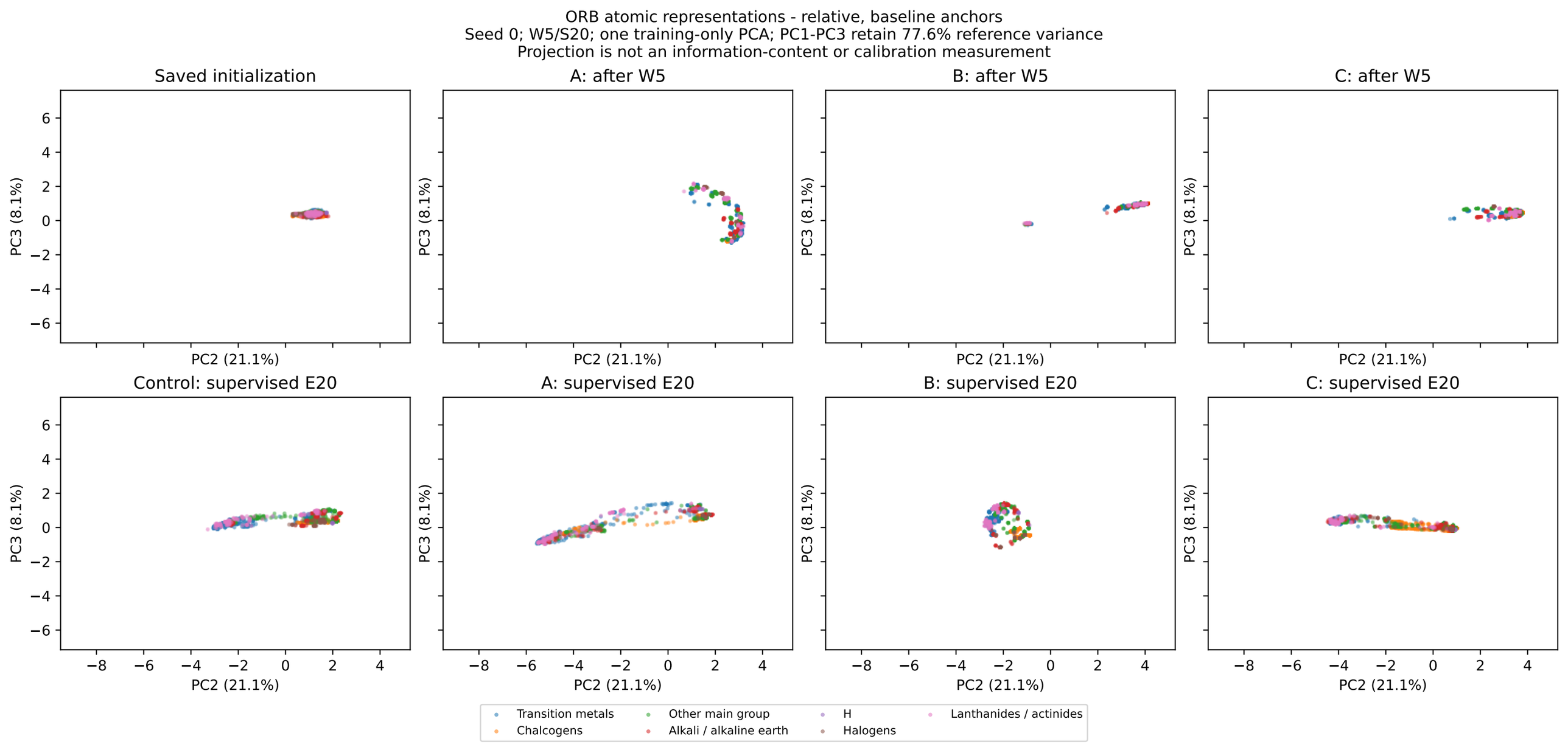
ORB atomic representations - raw: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 35.7% reference variance



Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

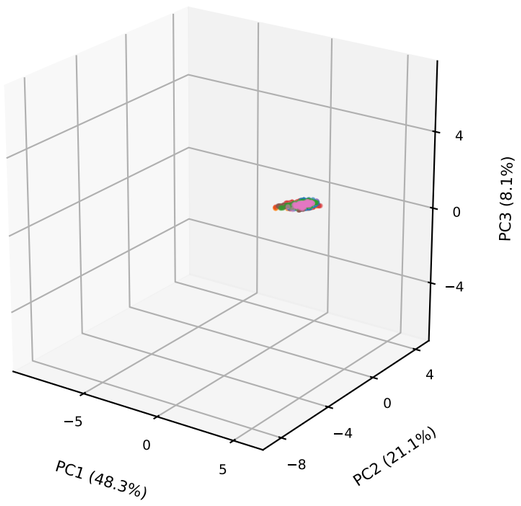

ORB atomic representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 77.6% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (48.3%)
PC2 (21.1%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides


ORB atomic representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 77.6% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (48.3%)
PC3 (8.1%)
Transition metals
Other main group
H
Lanthanides / actinides
Chalcogens
Alkali / alkaline earth
Halogens

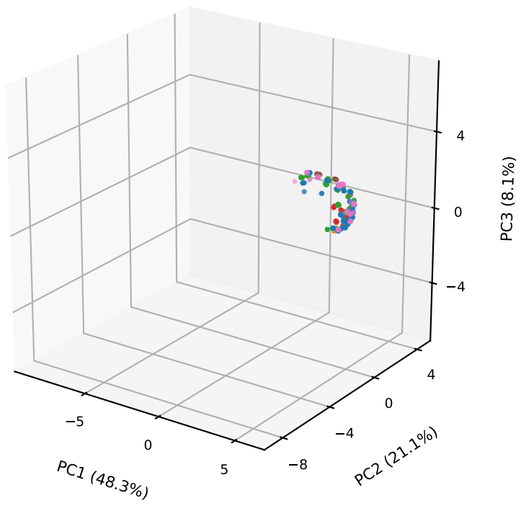

ORB atomic representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 77.6% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC2 (21.1%)
PC3 (8.1%)
Transition metals
Other main group
H
Lanthanides / actinides
Chalcogens
Alkali / alkaline earth
Halogens

ORB atomic representations - relative: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 77.6% reference variance

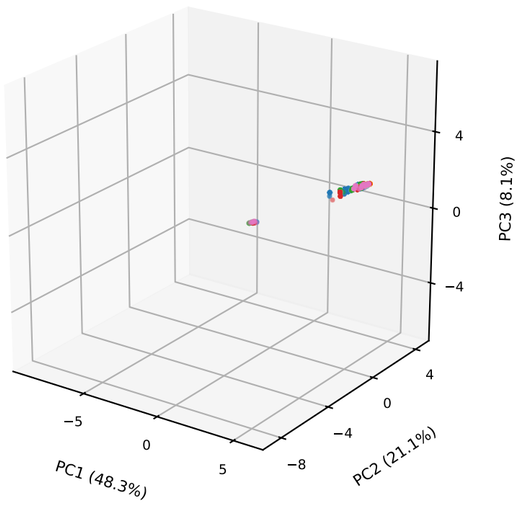
Saved initialization



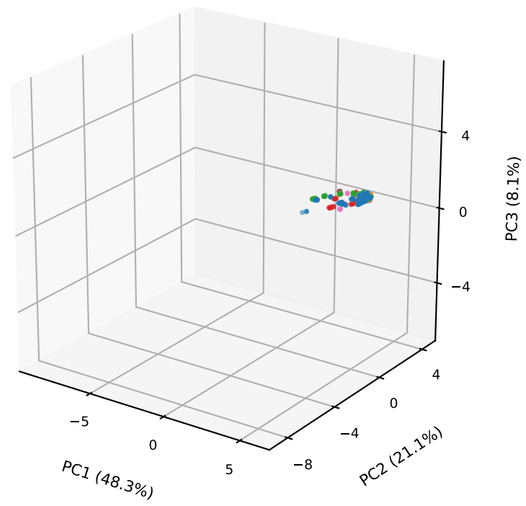
A: after W5



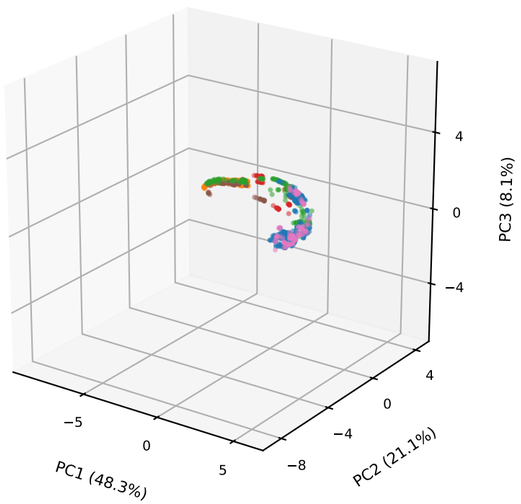
B: after W5



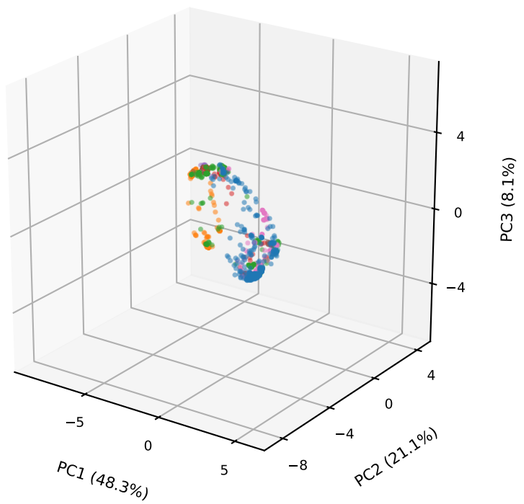
C: after W5



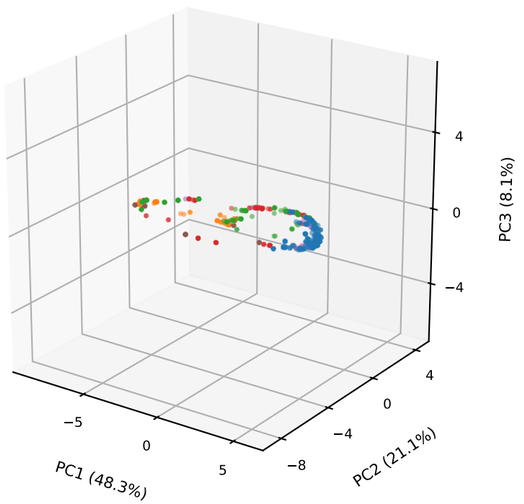
Control: supervised E20



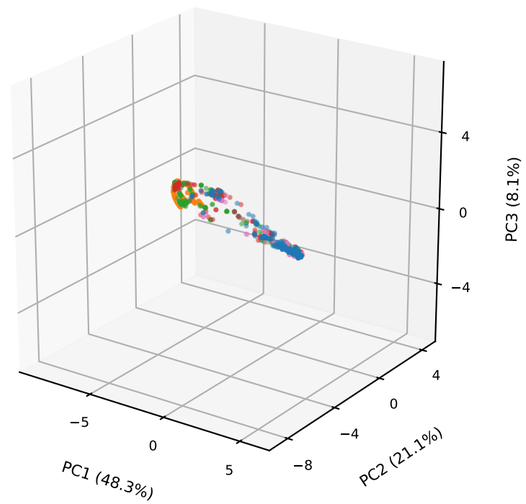
A: supervised E20



B: supervised E20

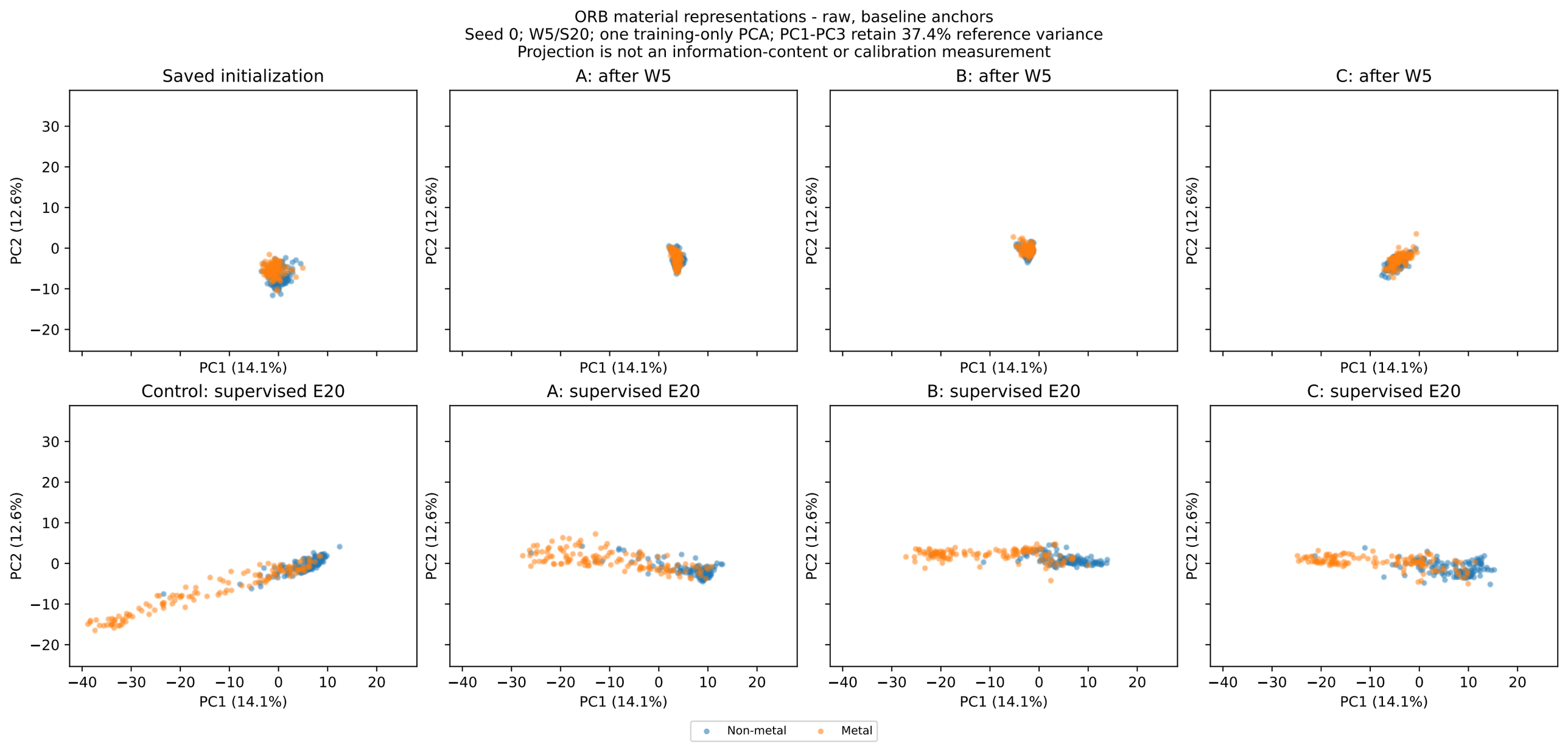


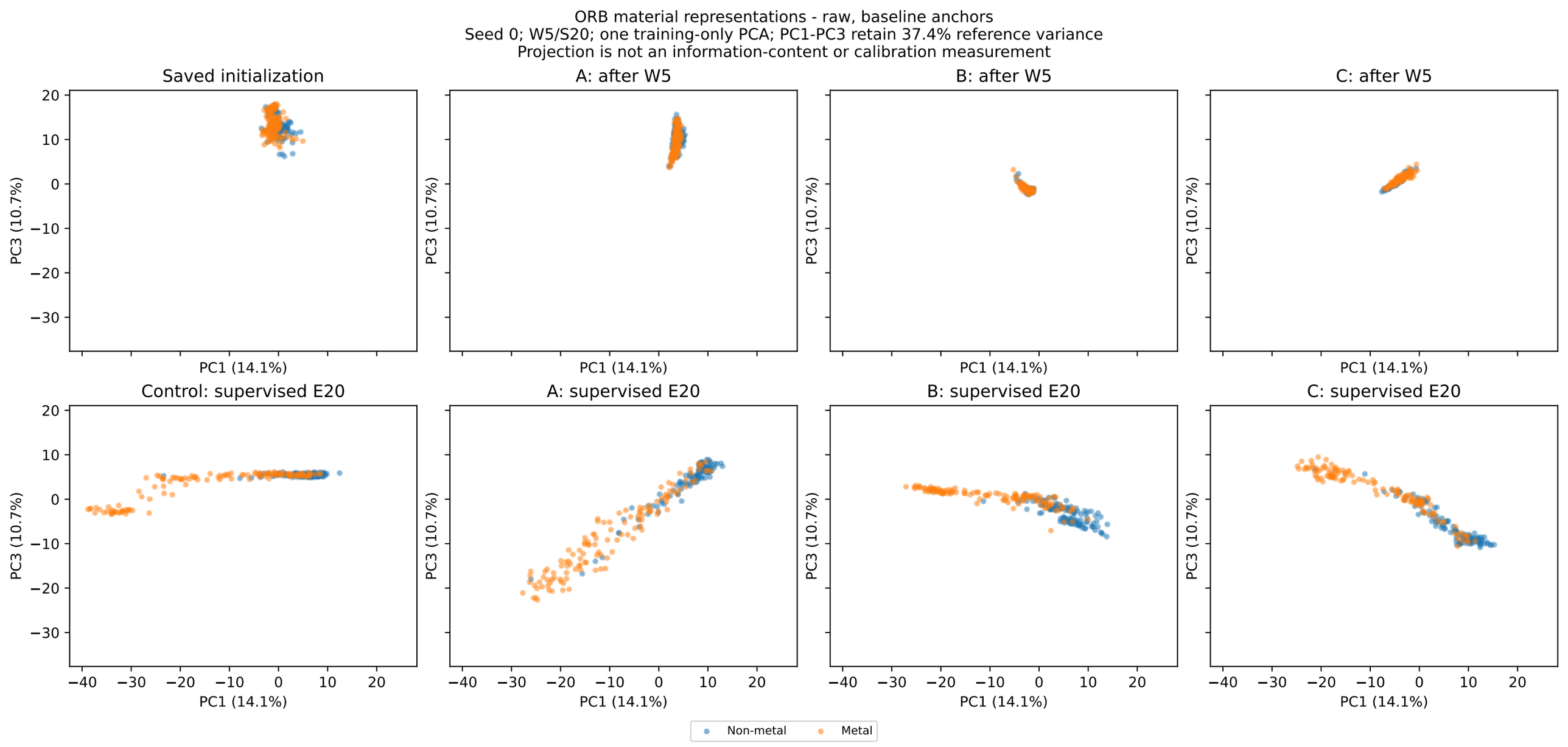
C: supervised E20

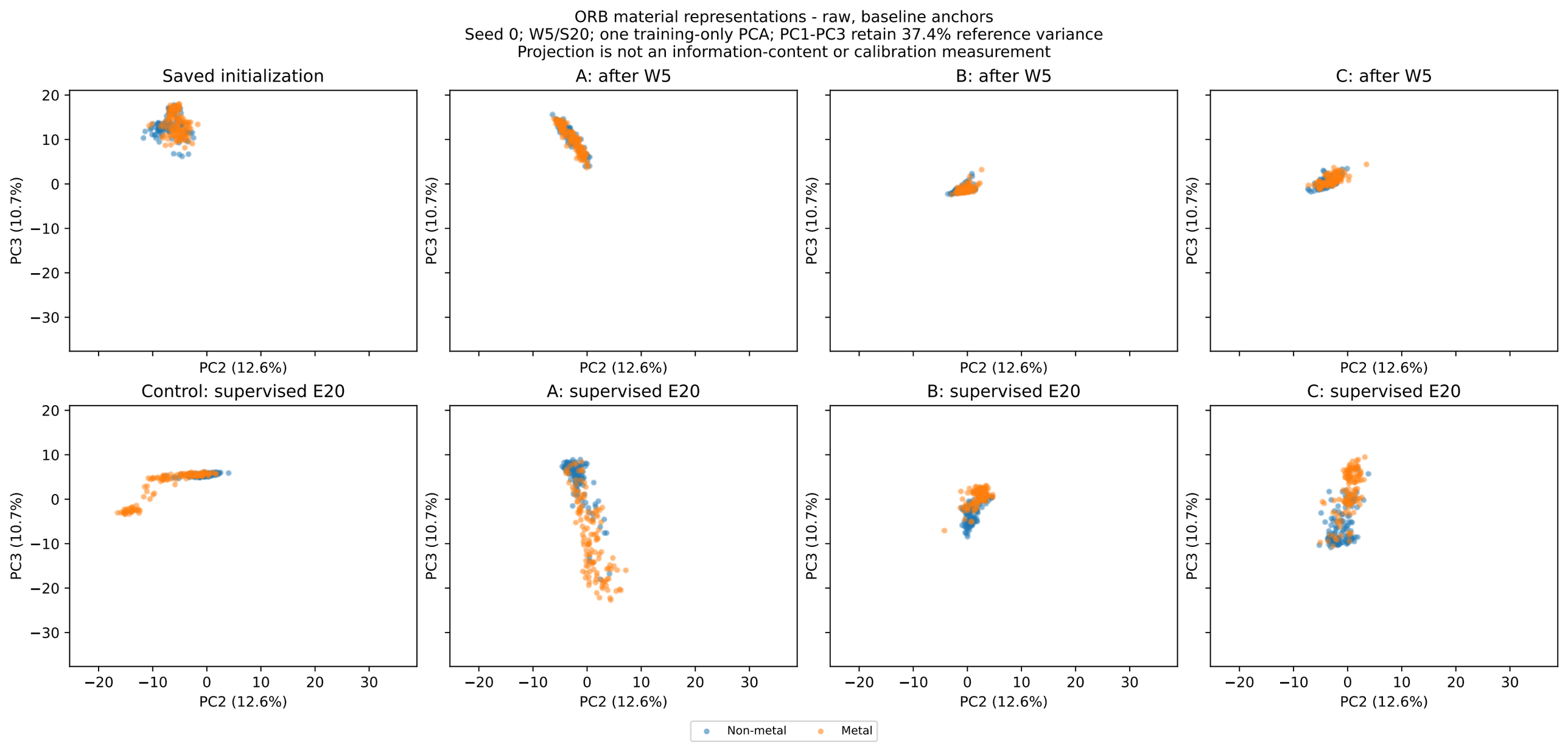


Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

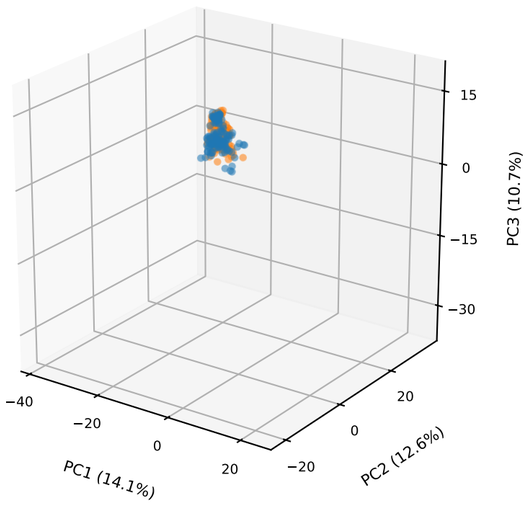

ORB material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 37.4% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (14.1%)
PC2 (12.6%)
Non-metal
Metal


ORB material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 37.4% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (14.1%)
PC3 (10.7%)
Non-metal
Metal

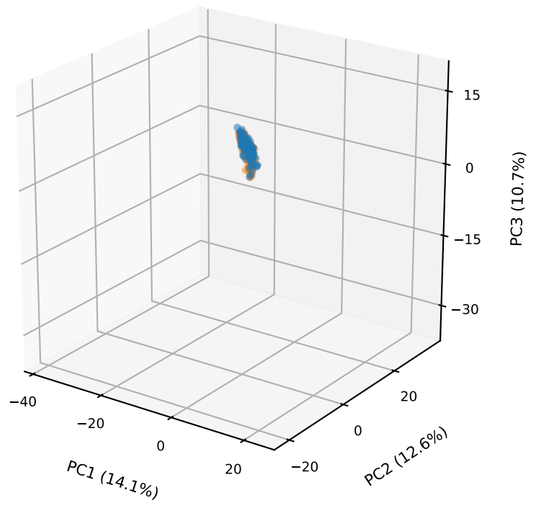

ORB material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 37.4% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC3 (10.7%)
PC2 (12.6%)
Non-metal
Metal

ORB material representations - raw: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 37.4% reference variance

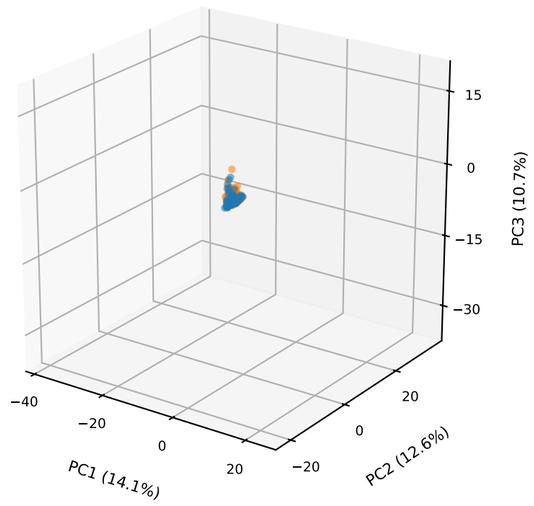
Saved initialization



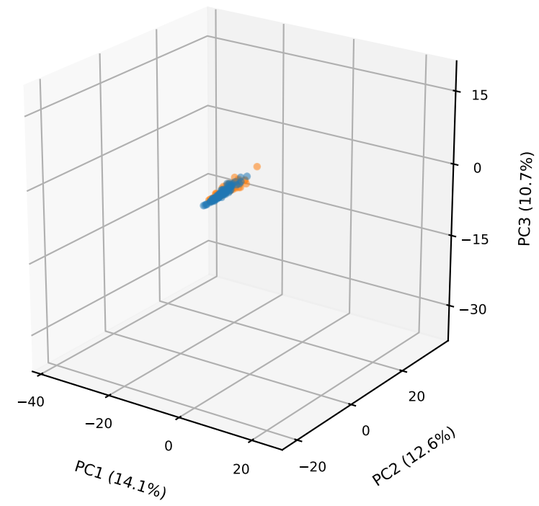
A: after W5



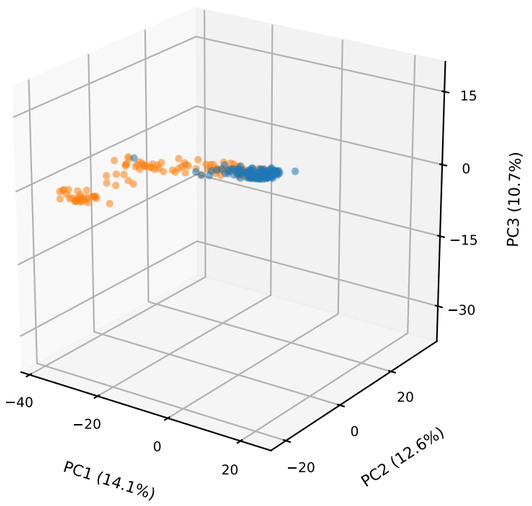
B: after W5



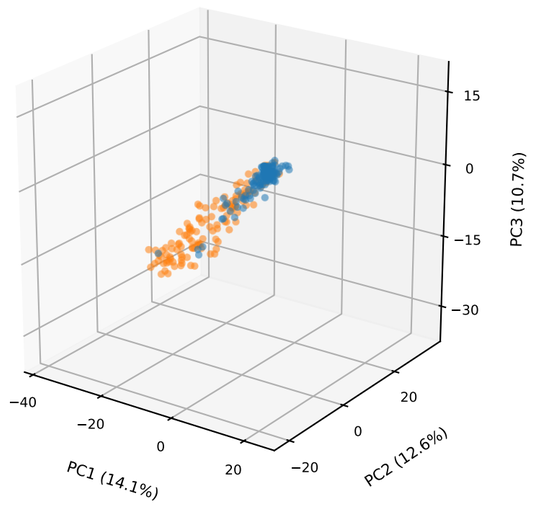
C: after W5



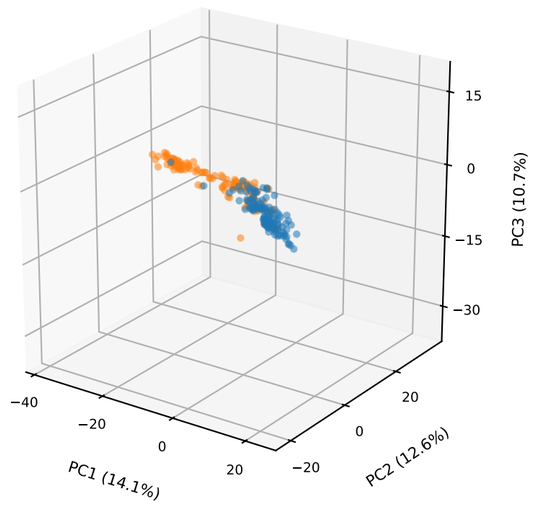
Control: supervised E20



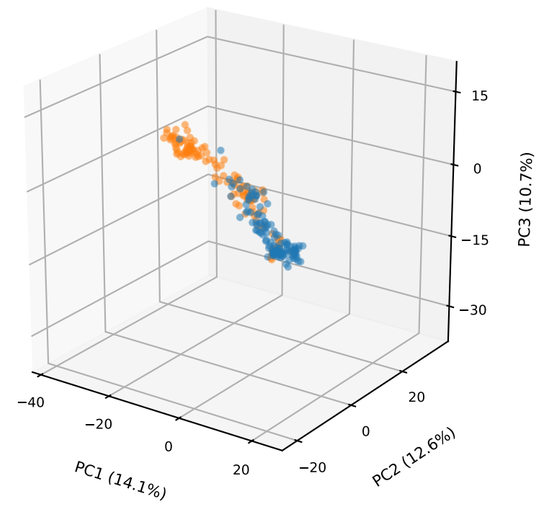
A: supervised E20



B: supervised E20

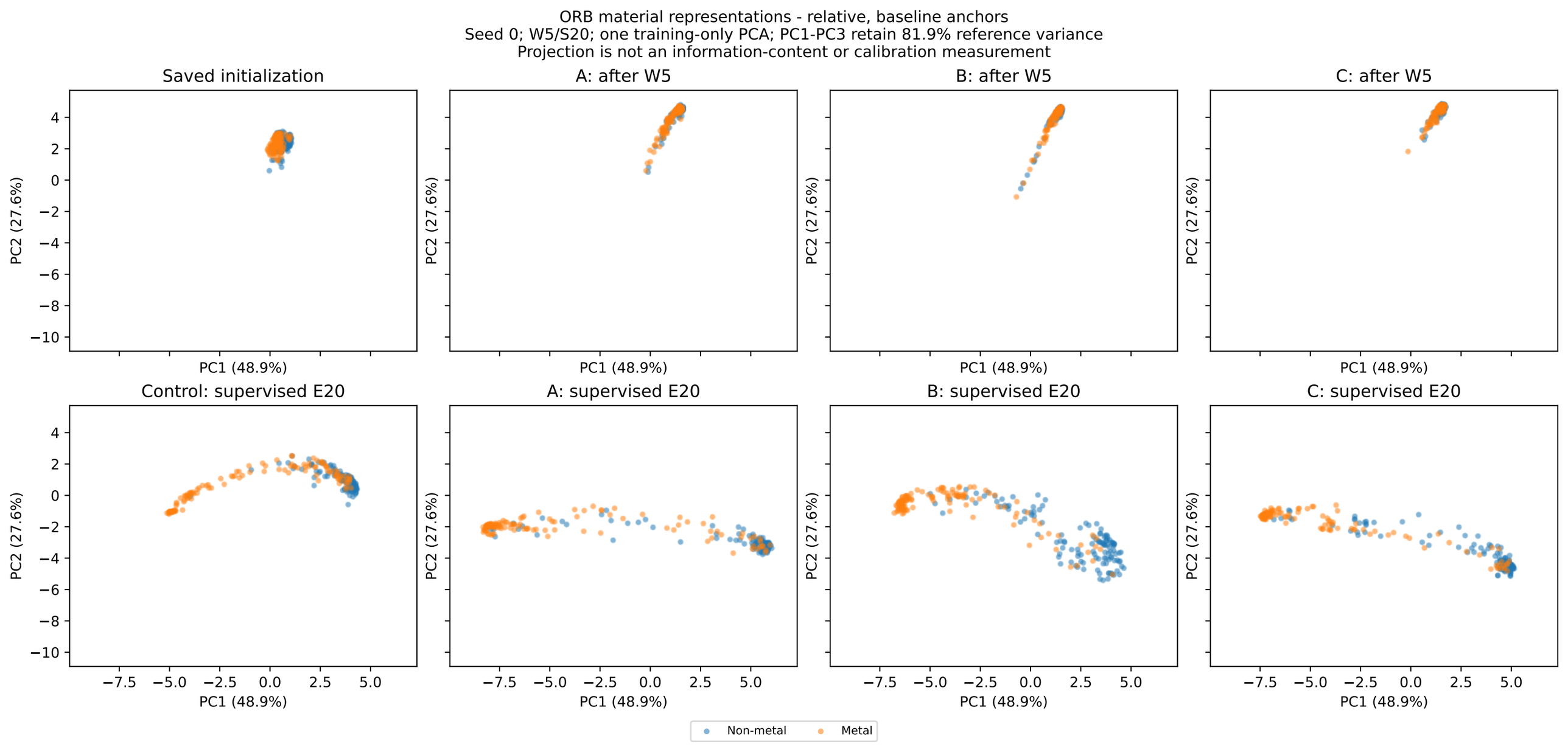


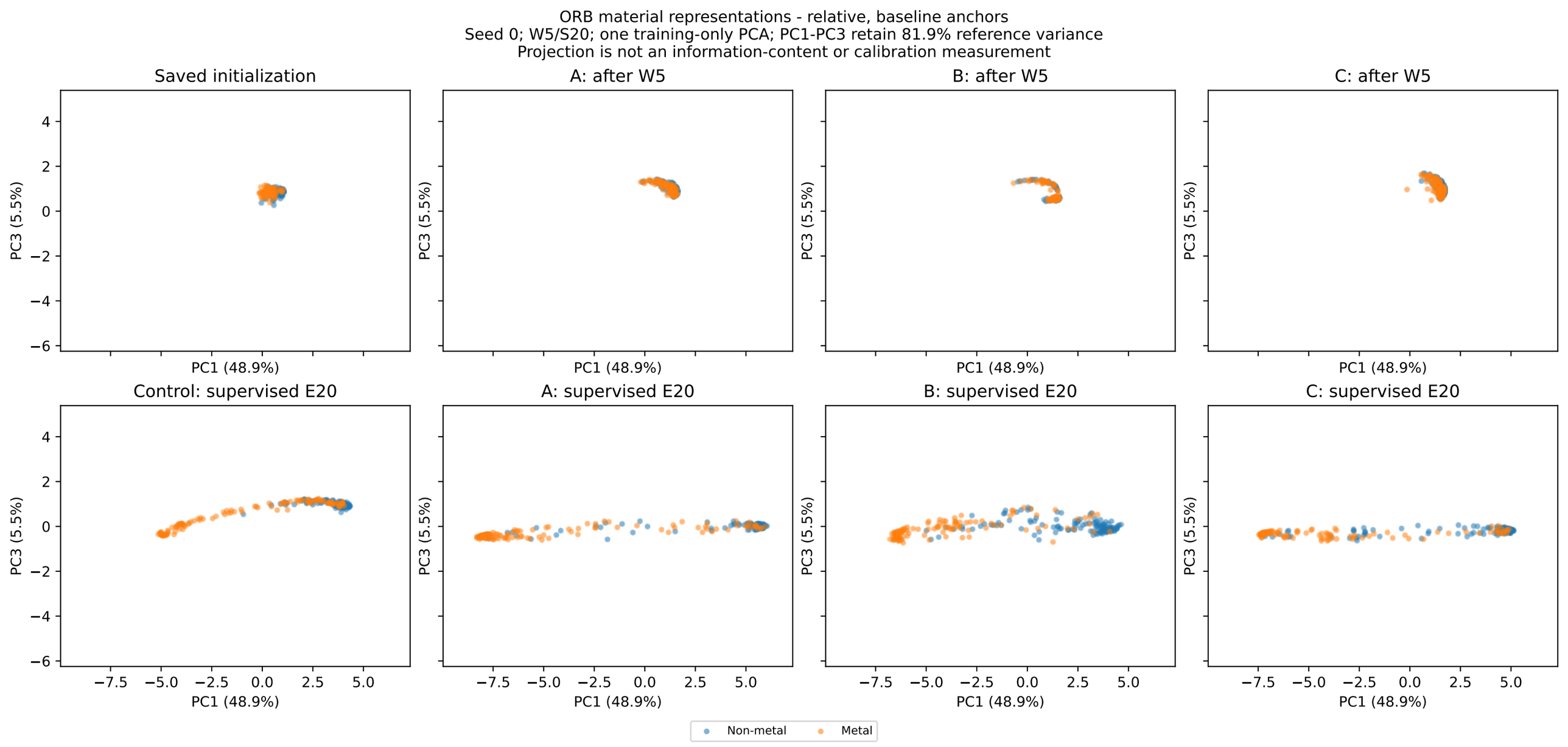
C: supervised E20

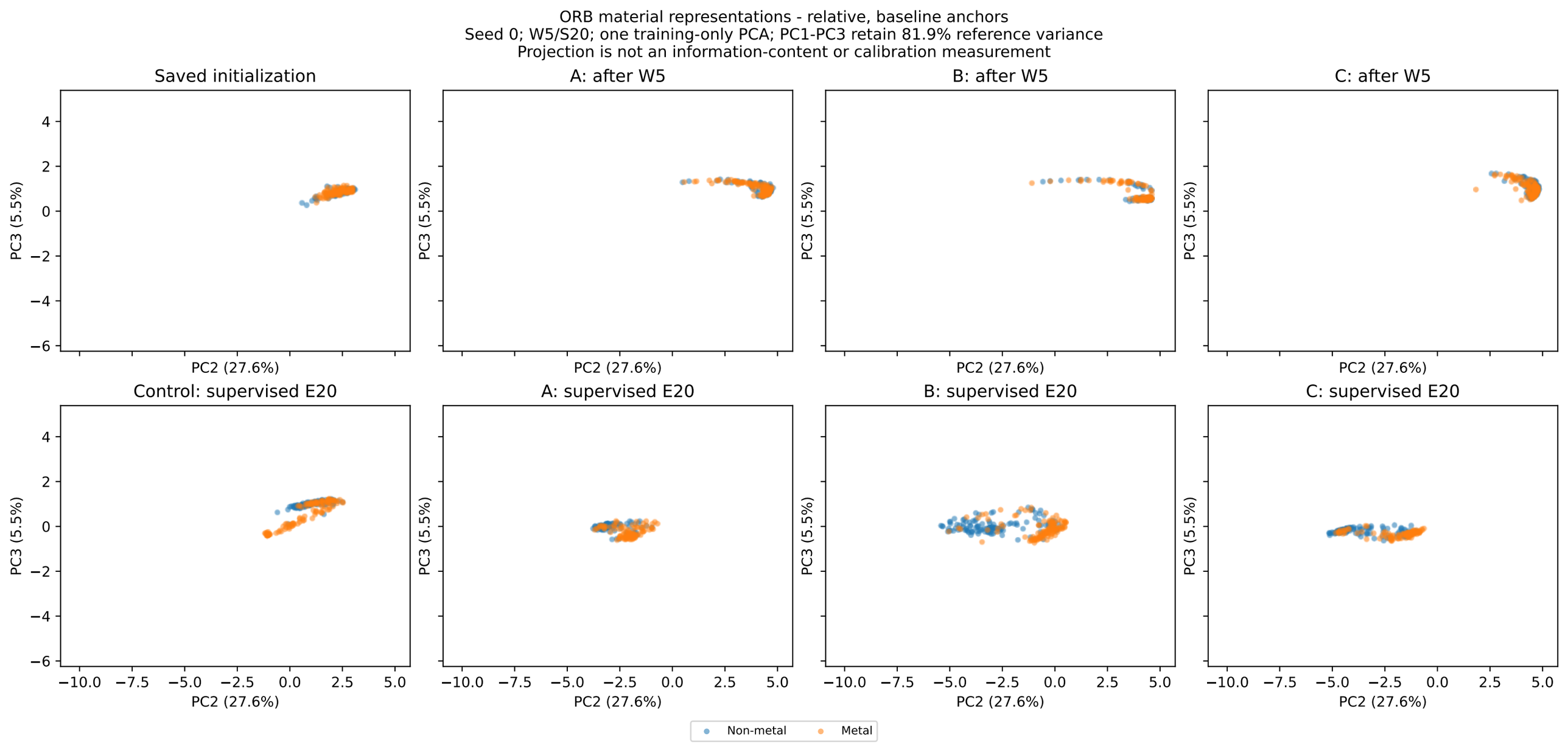


Non-metal Metal

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

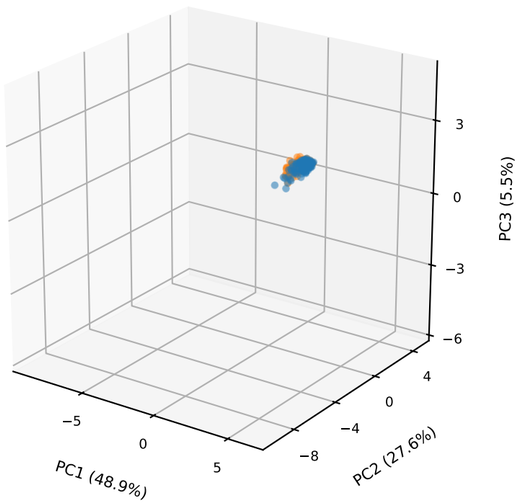

ORB material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 81.9% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (48.9%)
PC2 (27.6%)
Non-metal
Metal


ORB material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 81.9% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC1 (48.9%)
PC3 (5.5%)
Non-metal
Metal

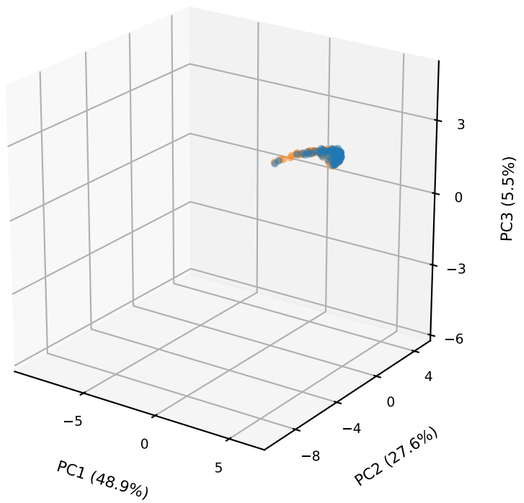

ORB material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 81.9% reference variance
Projection is not an information-content or calibration measurement
Saved initialization
A: after W5
B: after W5
C: after W5
Control: supervised E20
A: supervised E20
B: supervised E20
C: supervised E20
PC3 (5.5%)
PC2 (27.6%)
Non-metal
Metal

ORB material representations - relative: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 81.9% reference variance

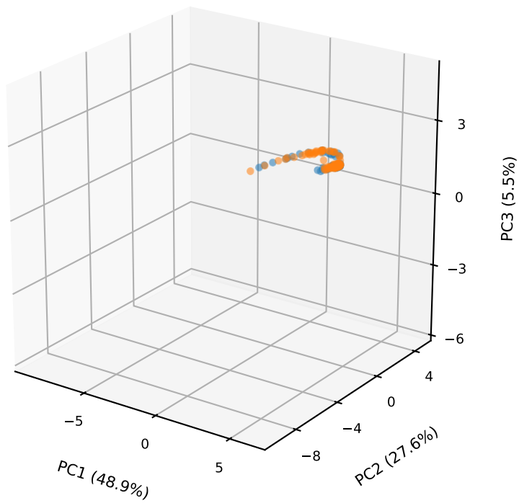
Saved initialization



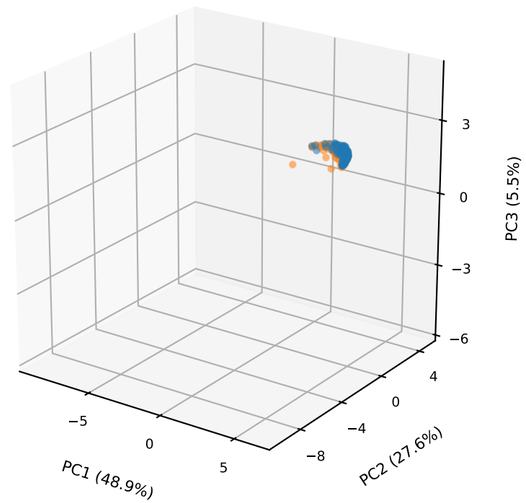
A: after W5



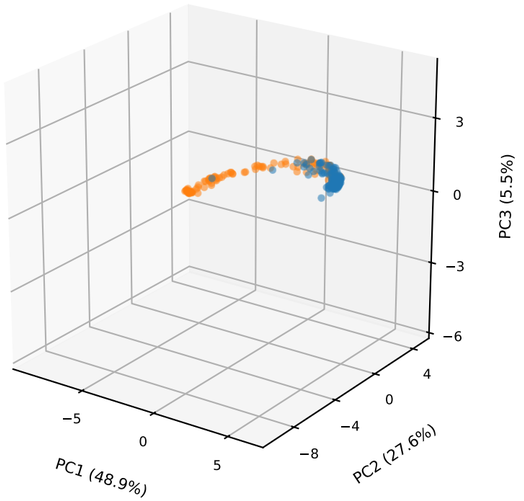
B: after W5



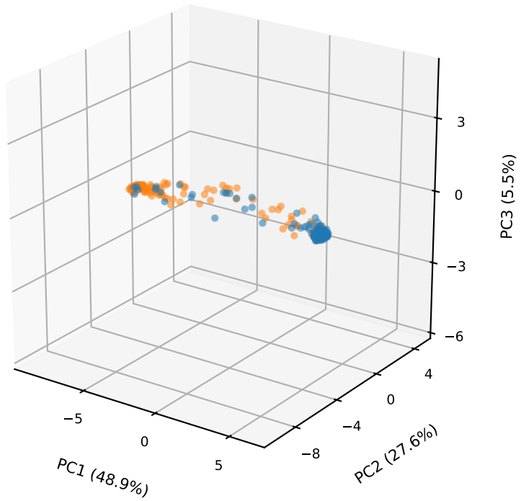
C: after W5



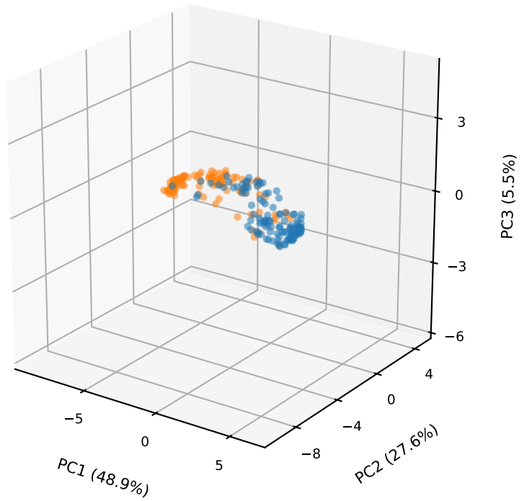
Control: supervised E20



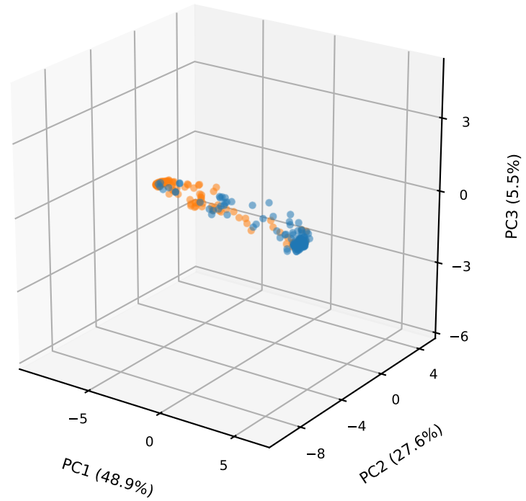
A: supervised E20



B: supervised E20

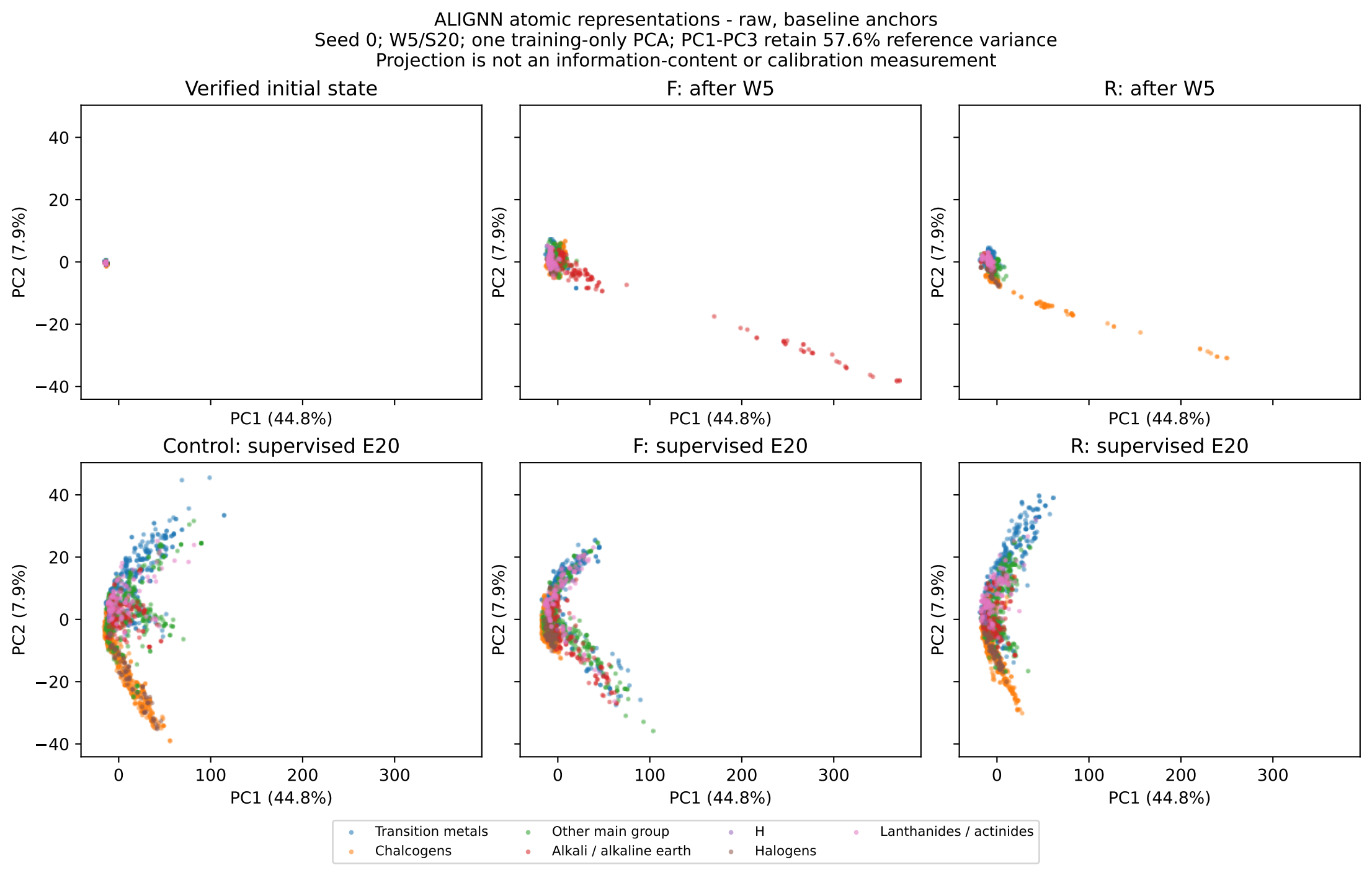


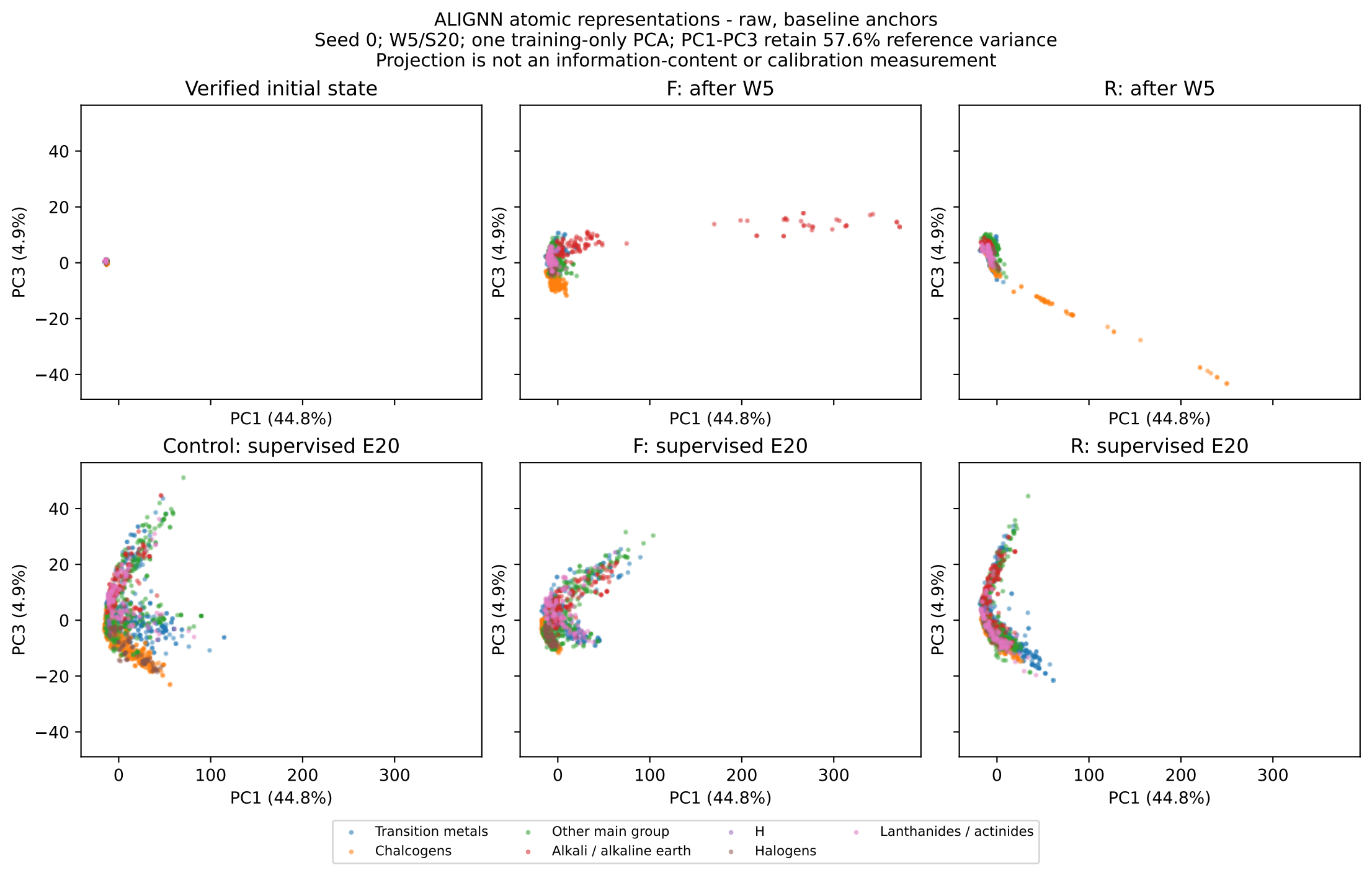
C: supervised E20

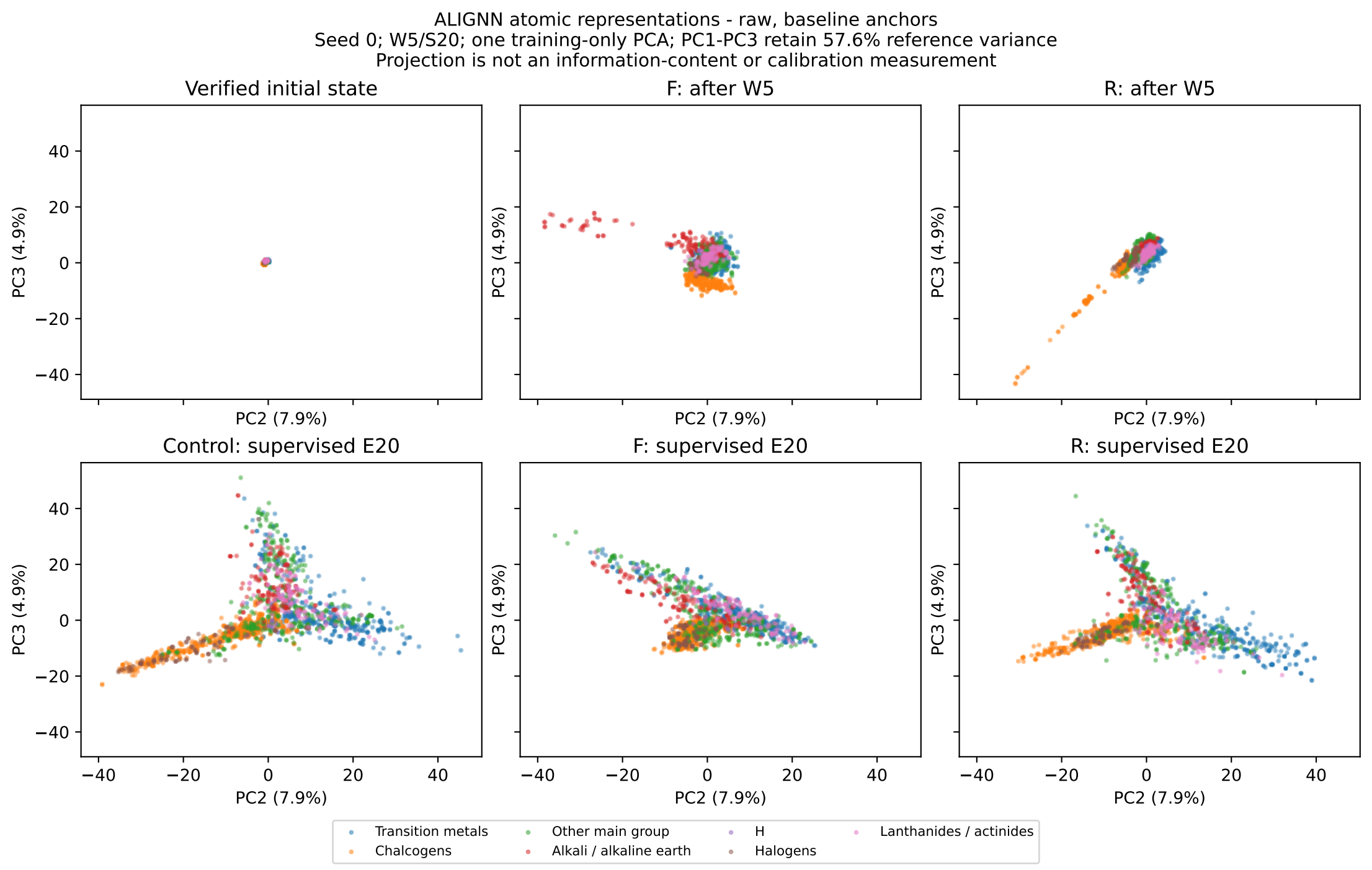


Non-metal Metal

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

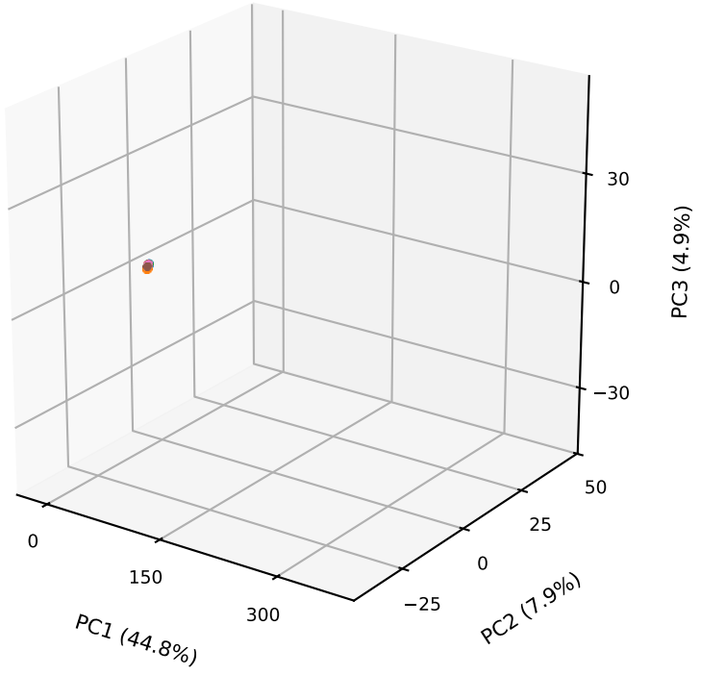

ALIGNN atomic representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 57.6% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (44.8%)
PC2 (7.9%)
Transition metals
Other main group
H
Lanthanides / actinides
Chalcogens
Alkali / alkaline earth
Halogens


ALIGNN atomic representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 57.6% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (44.8%)
PC3 (4.9%)
Transition metals
Other main group
H
Lanthanides / actinides
Chalcogens
Alkali / alkaline earth
Halogens

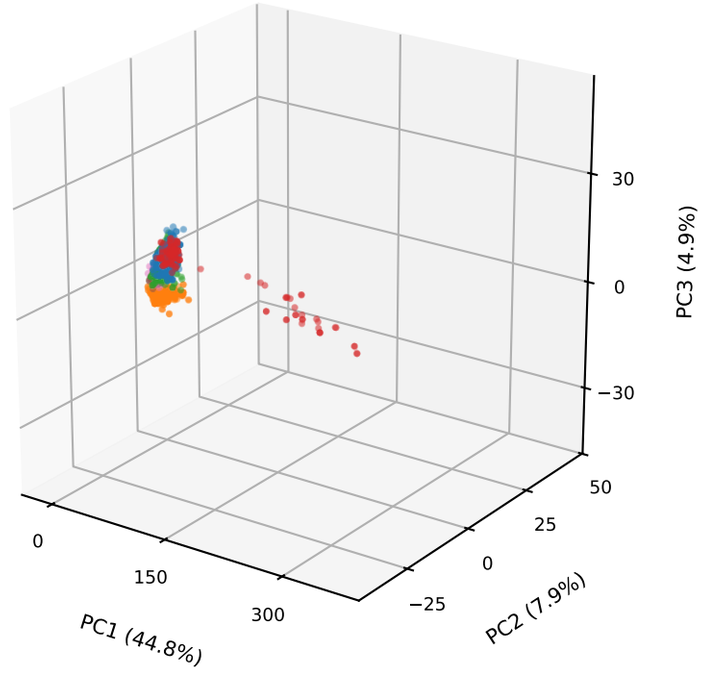

ALIGNN atomic representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 57.6% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC2 (7.9%)
PC3 (4.9%)
Transition metals
Other main group
H
Lanthanides / actinides
Chalcogens
Alkali / alkaline earth
Halogens

ALIGNN atomic representations - raw: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 57.6% reference variance

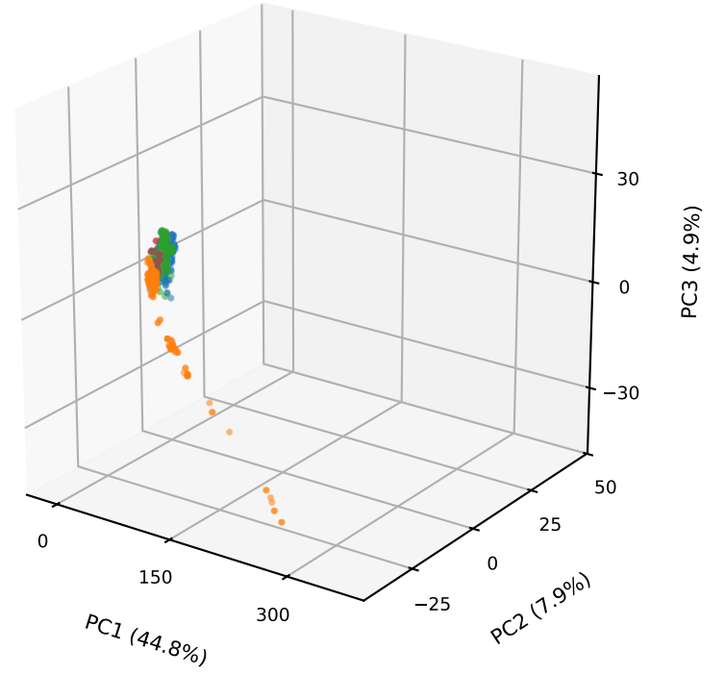
Verified initial state



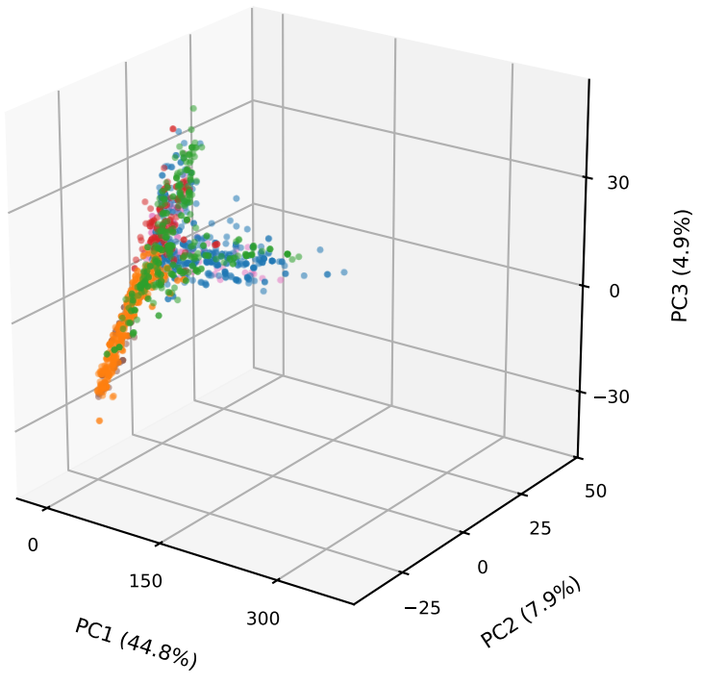
F: after W5



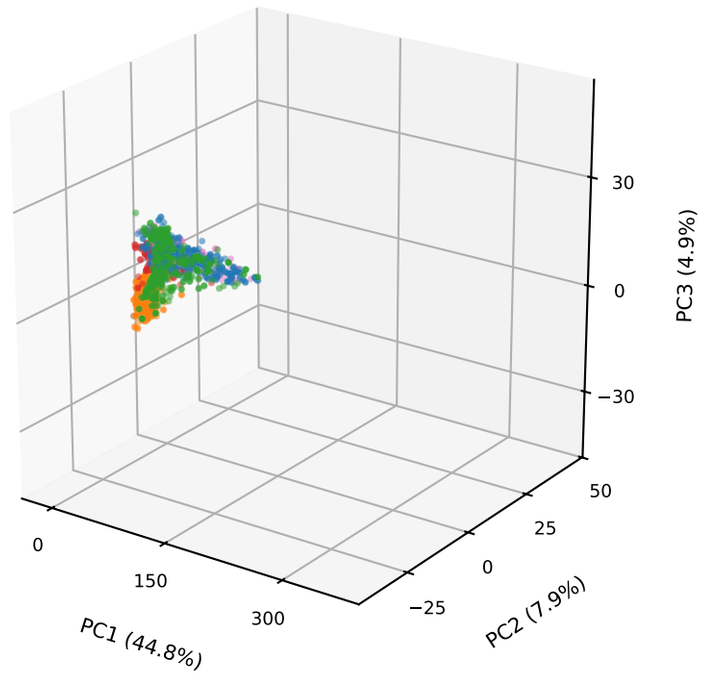
R: after W5



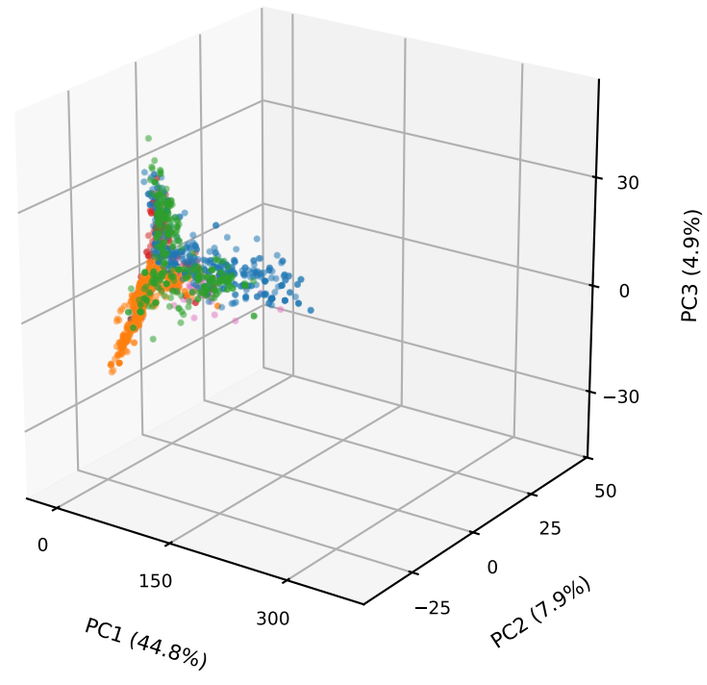
Control: supervised E20



F: supervised E20

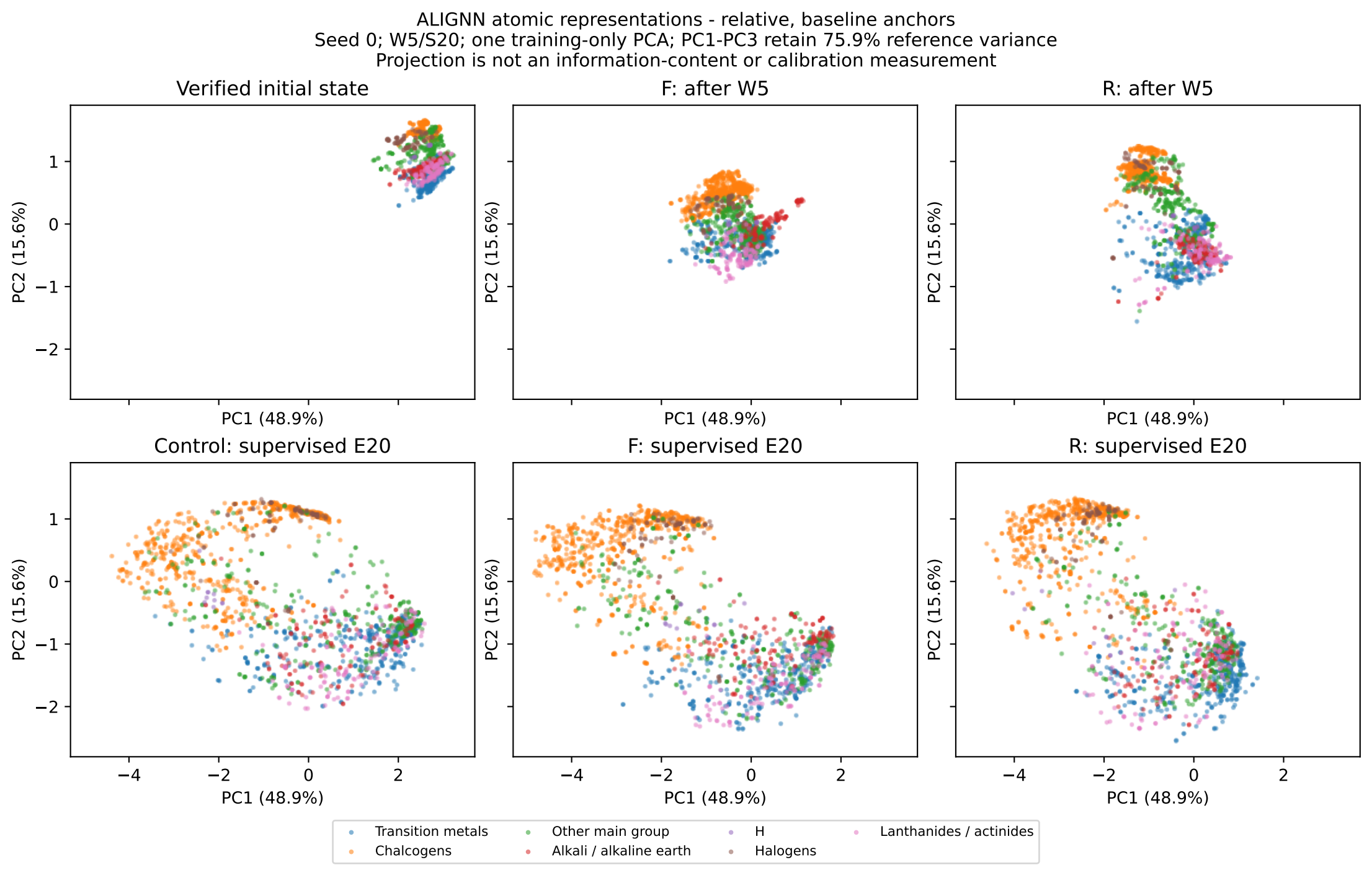


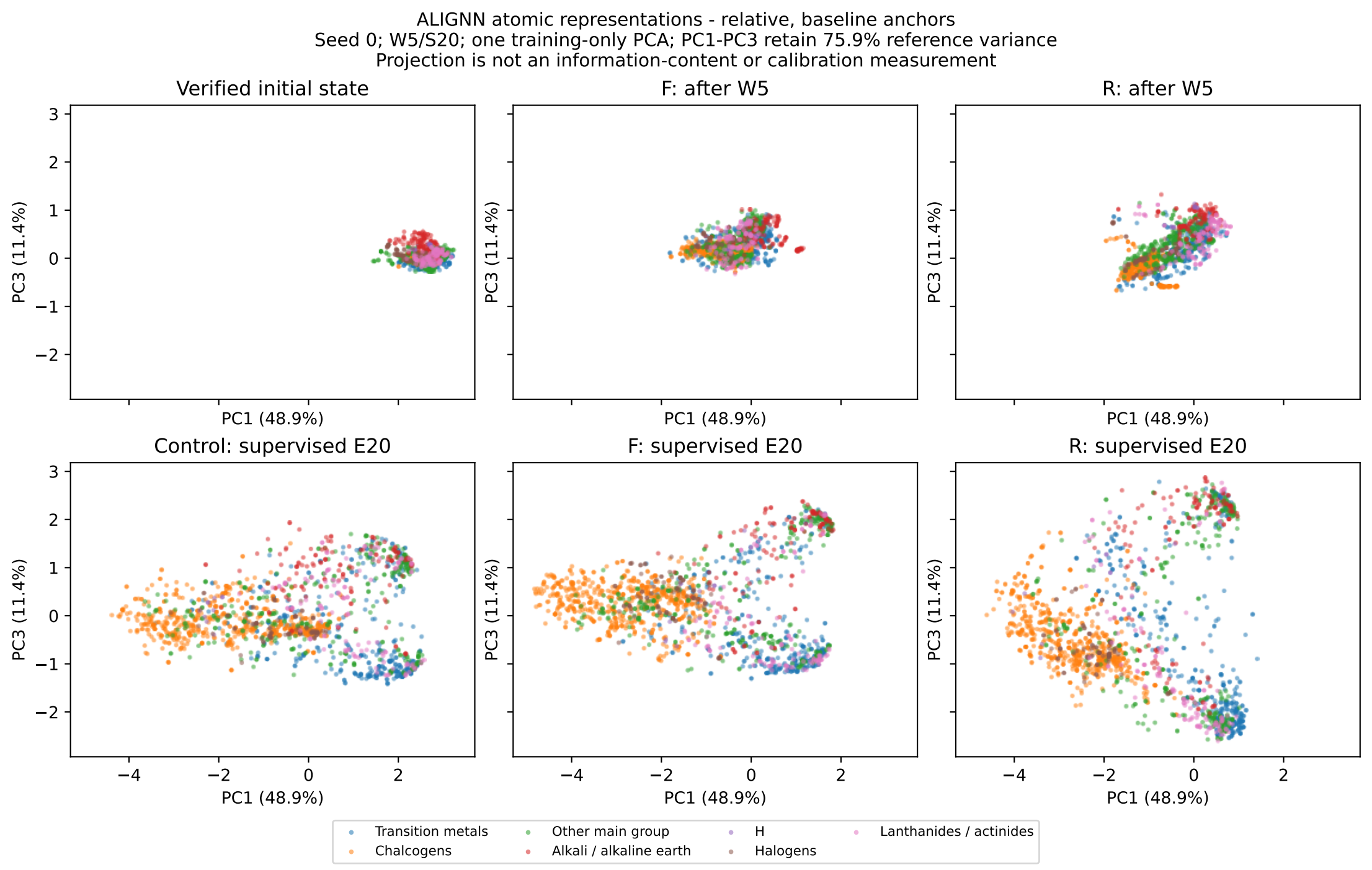
R: supervised E20

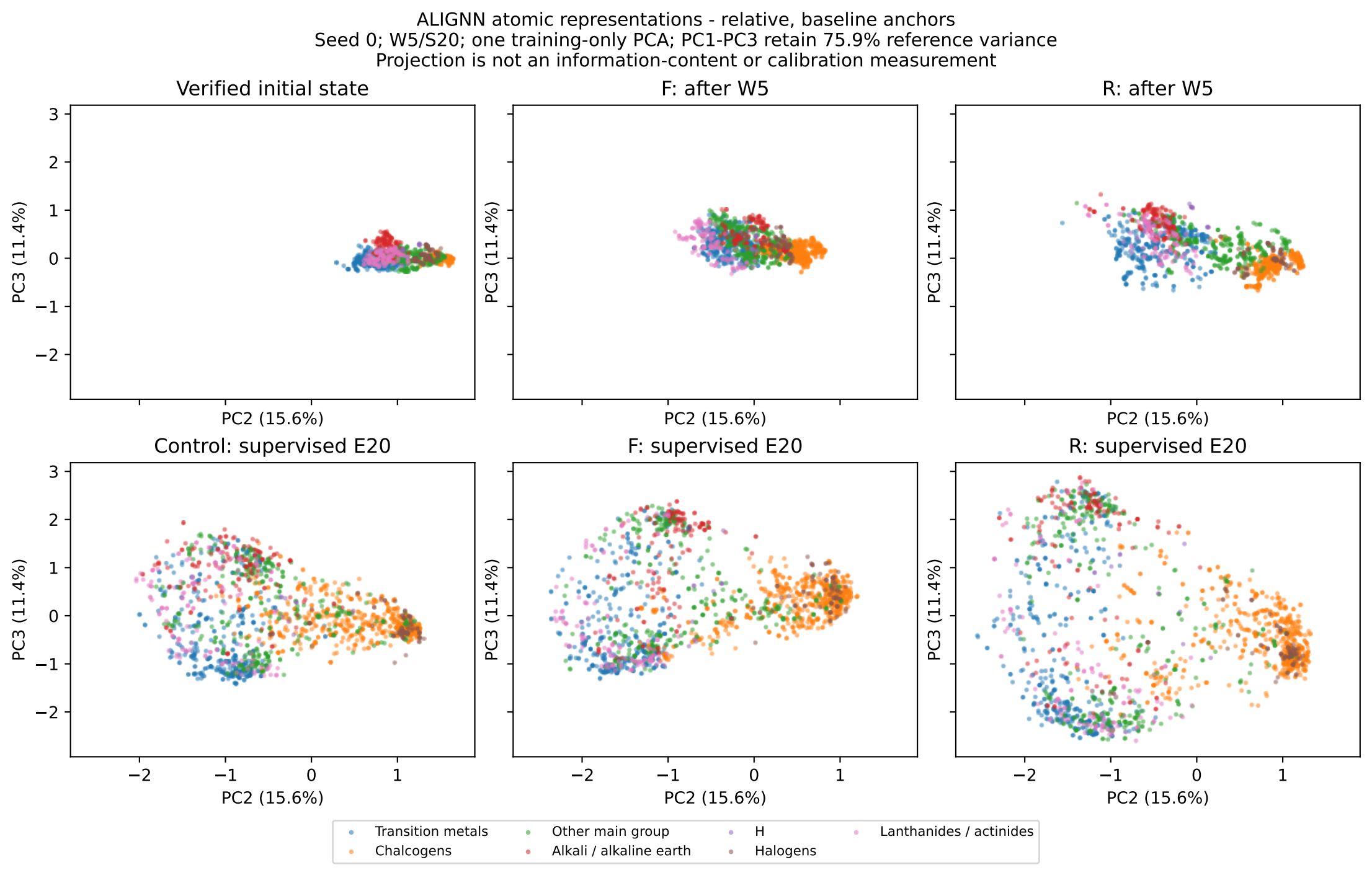


Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

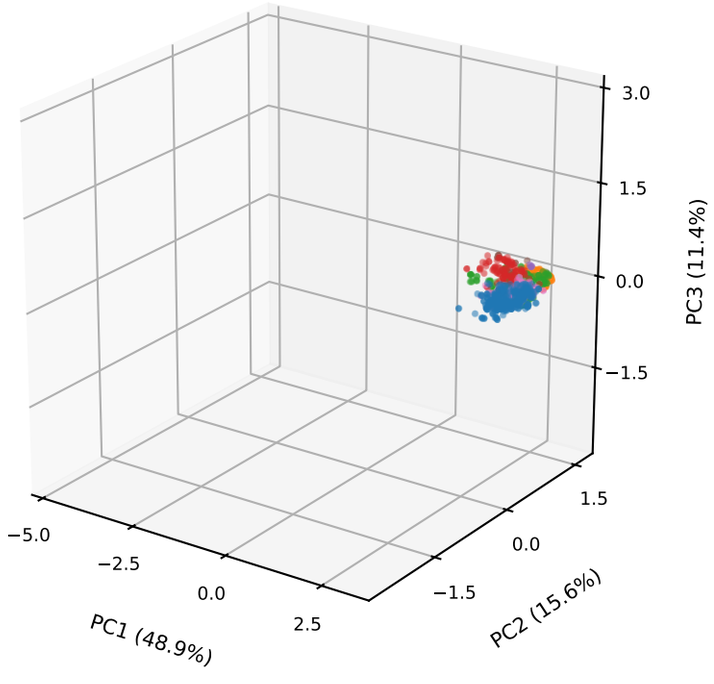

ALIGNN atomic representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 75.9% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (48.9%)
PC2 (15.6%)
Transition metals
Other main group
H
Lanthanides / actinides
Chalcogens
Alkali / alkaline earth
Halogens


ALIGNN atomic representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 75.9% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (48.9%)
PC3 (11.4%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

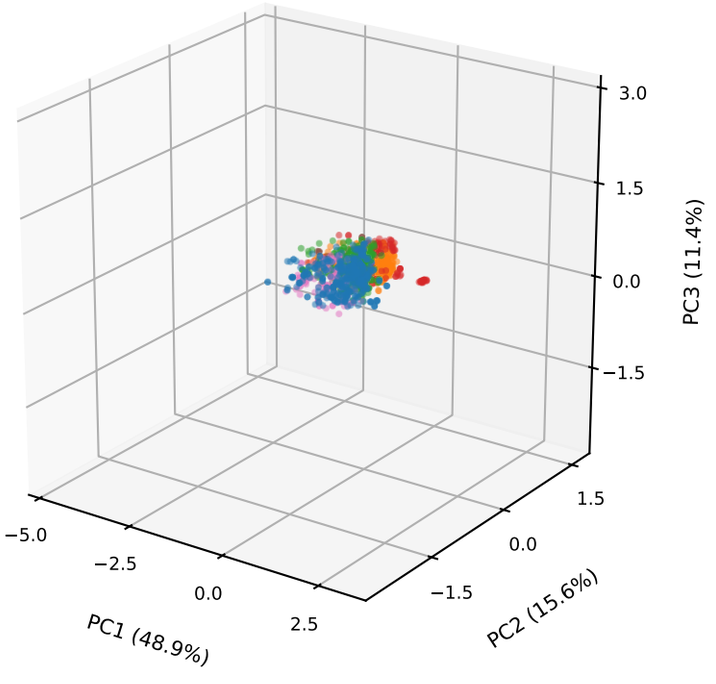

ALIGNN atomic representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 75.9% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC2 (15.6%)
PC3 (11.4%)
Transition metals
Other main group
H
Lanthanides / actinides
Chalcogens
Alkali / alkaline earth
Halogens

ALIGNN atomic representations - relative: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 75.9% reference variance

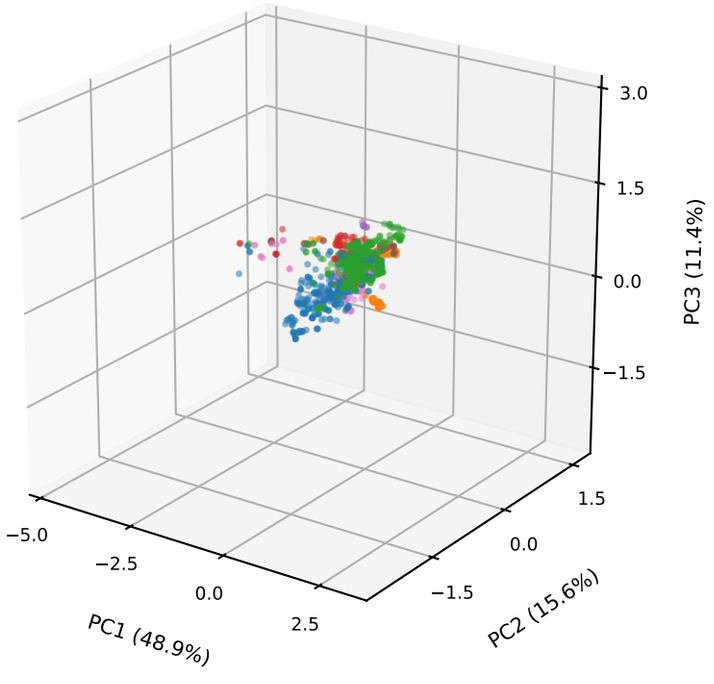
Verified initial state



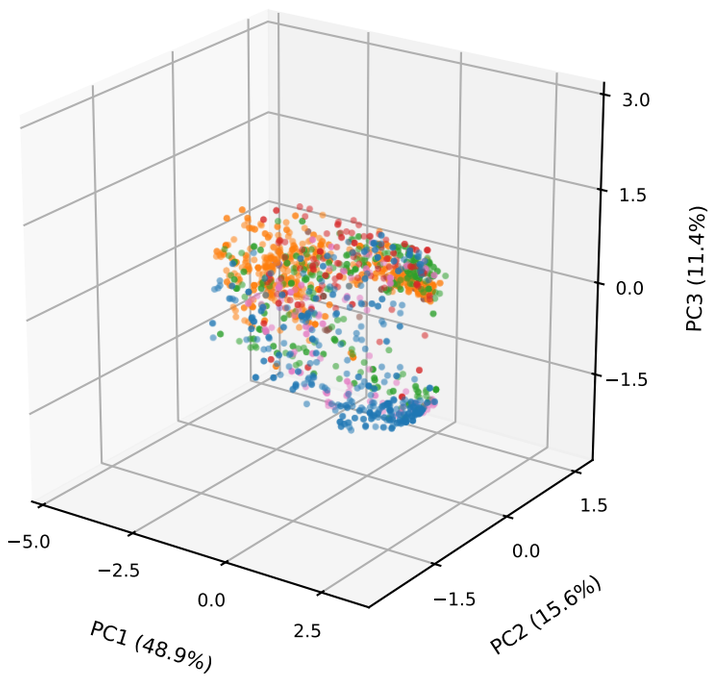
F: after W5



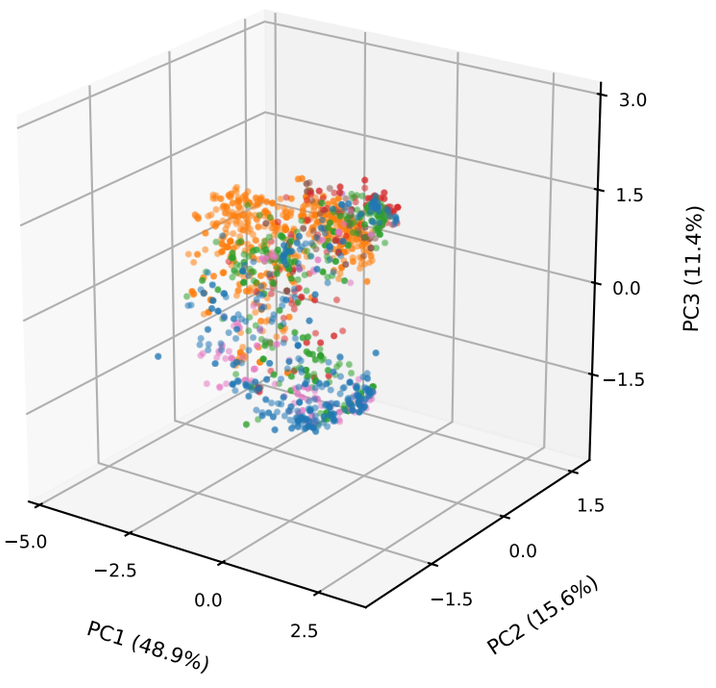
R: after W5



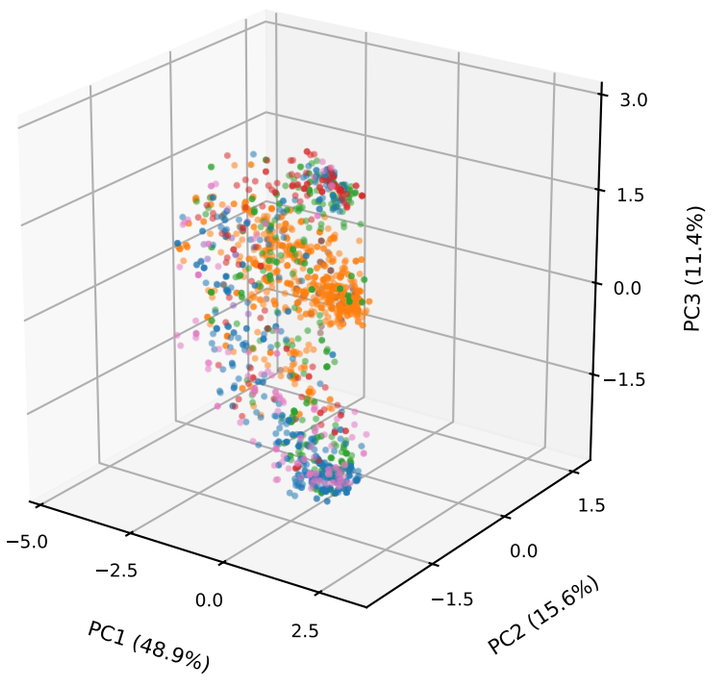
Control: supervised E20



F: supervised E20

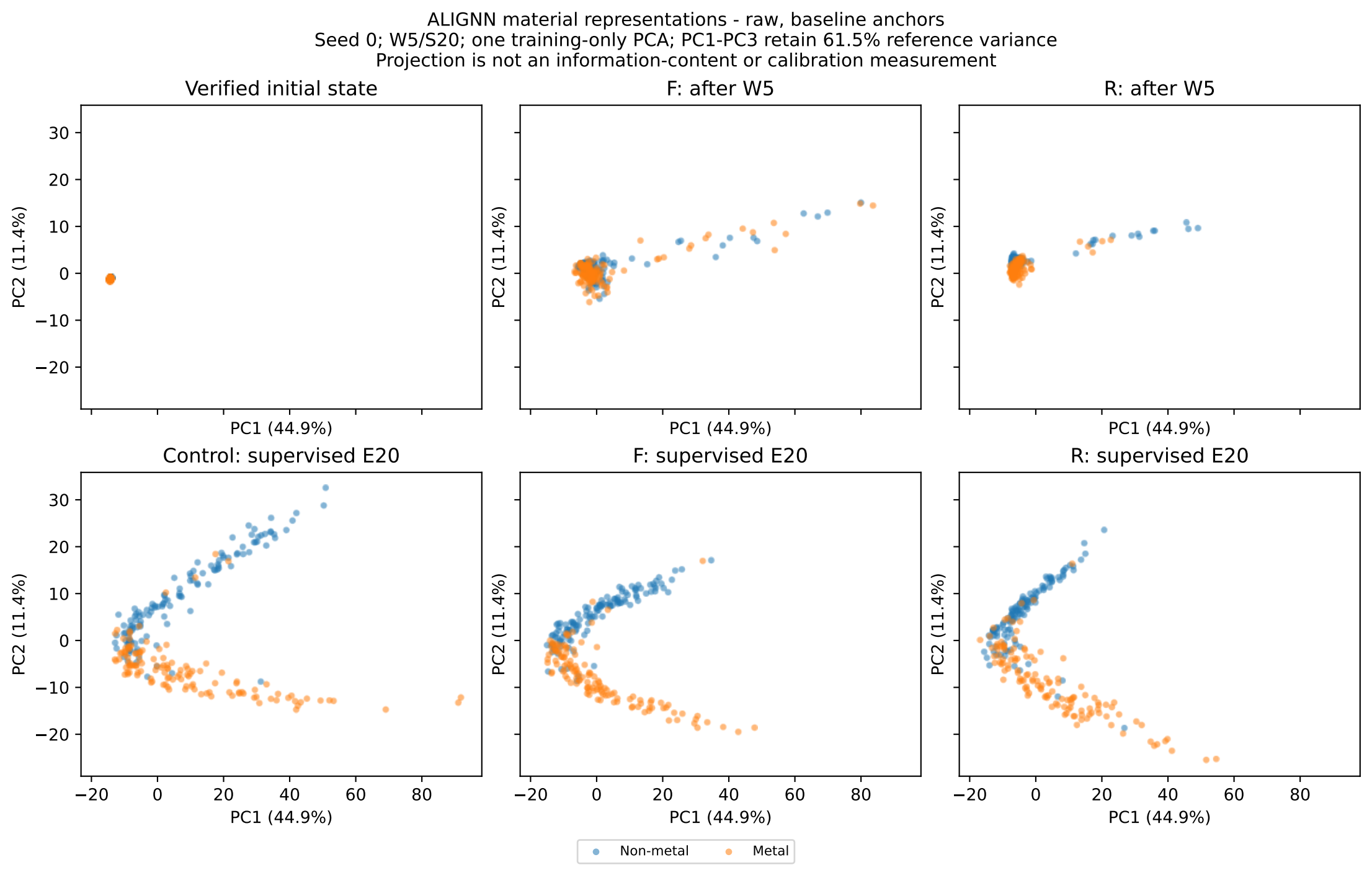


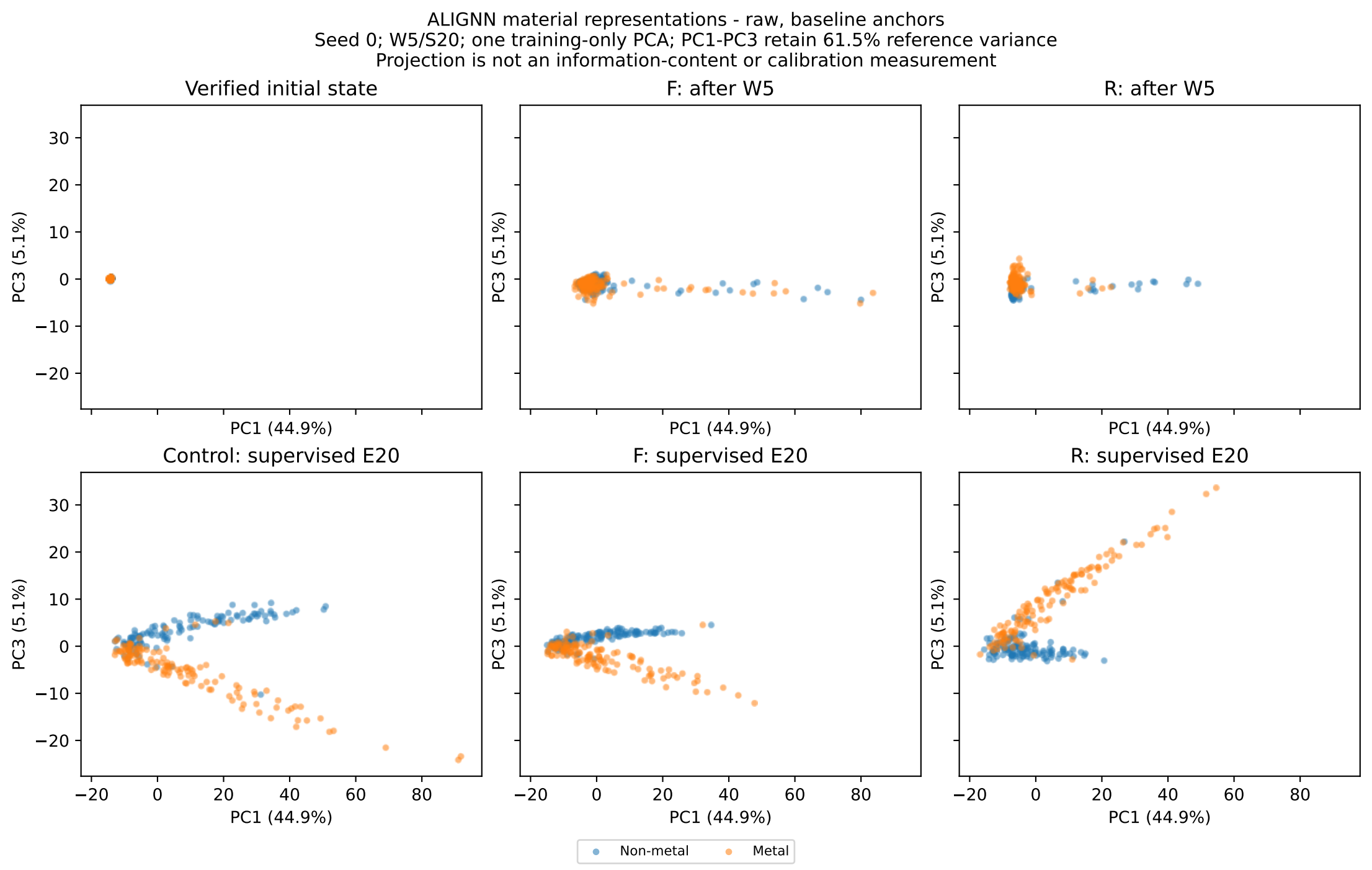
R: supervised E20

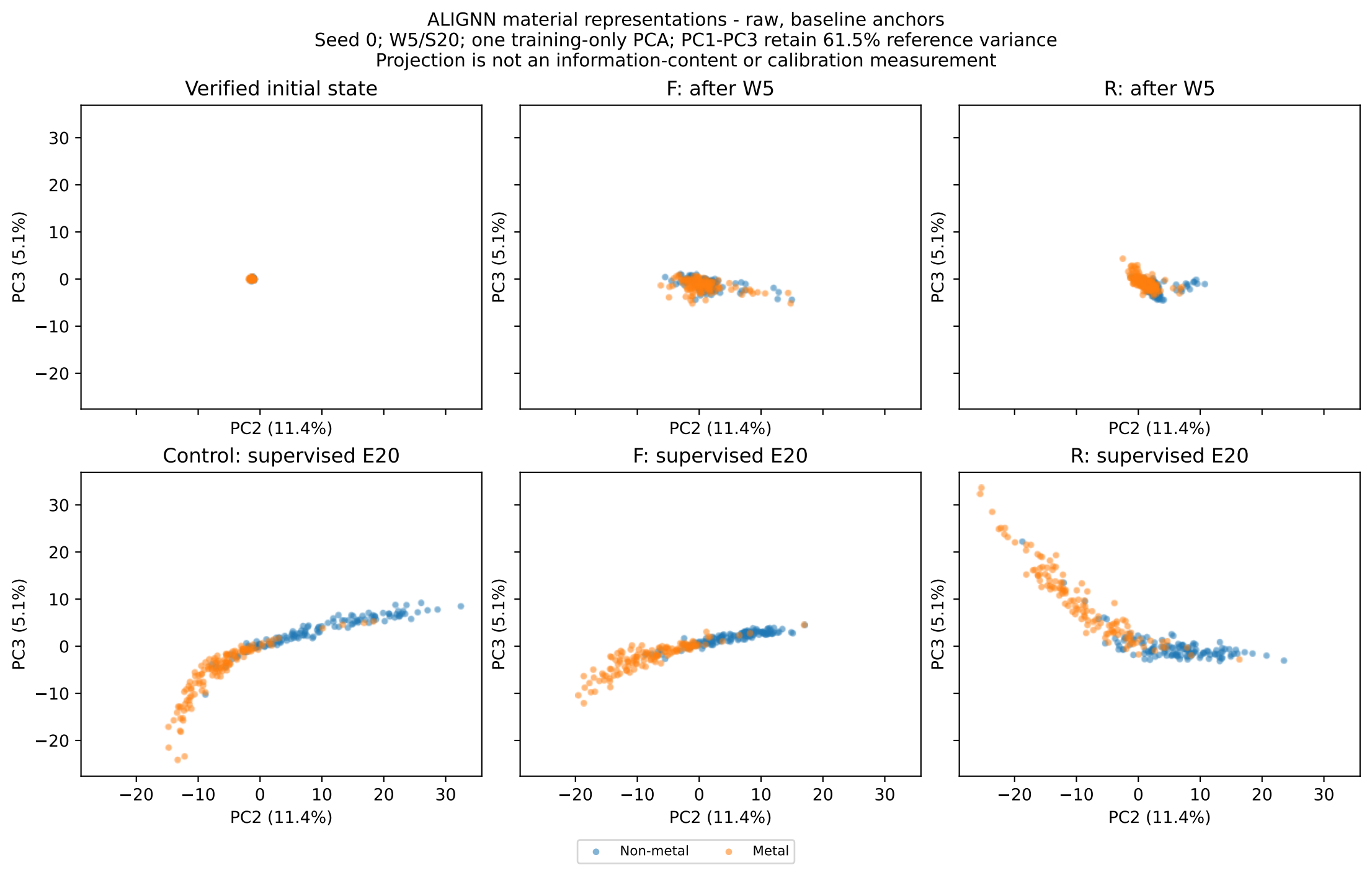


Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

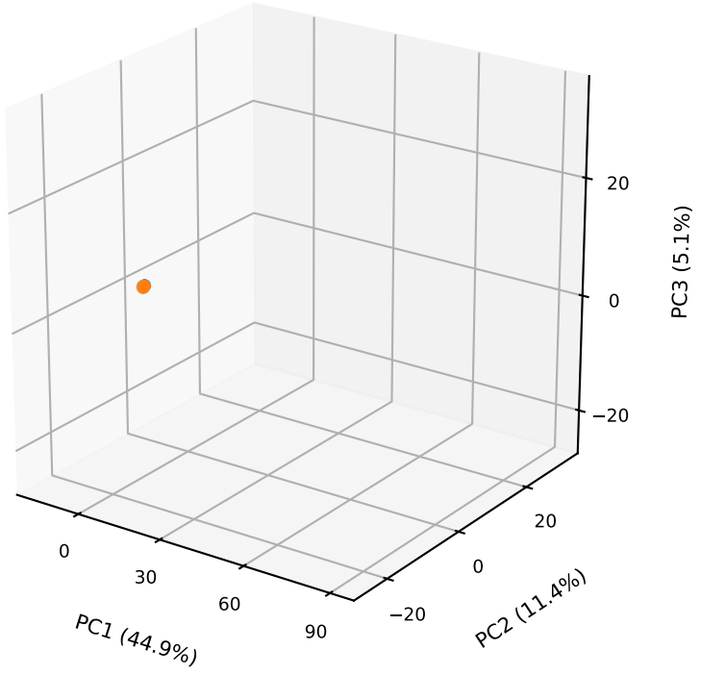

ALIGNN material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 61.5% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (44.9%)
PC2 (11.4%)
Non-metal
Metal


ALIGNN material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 61.5% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (44.9%)
PC3 (5.1%)
Non-metal
Metal

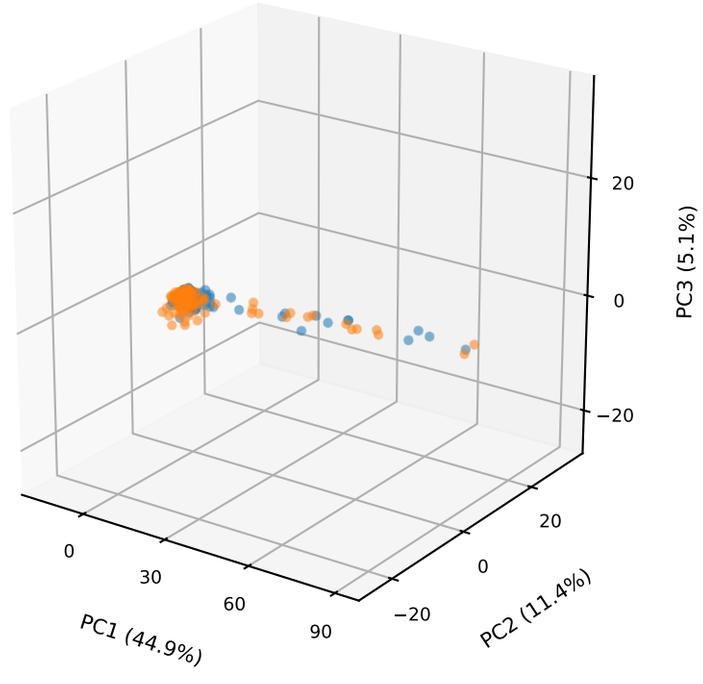

ALIGNN material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 61.5% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC2 (11.4%)
PC3 (5.1%)
Non-metal
Metal

ALIGNN material representations - raw: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 61.5% reference variance

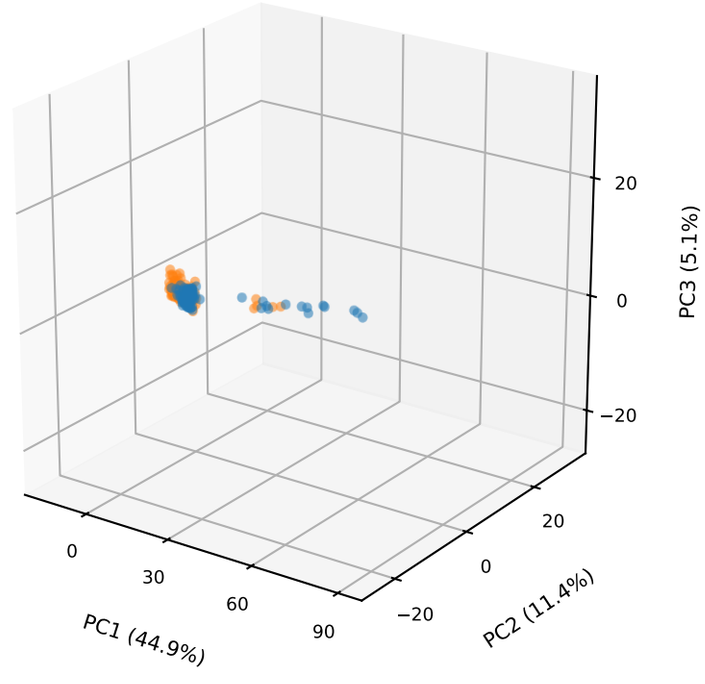
Verified initial state



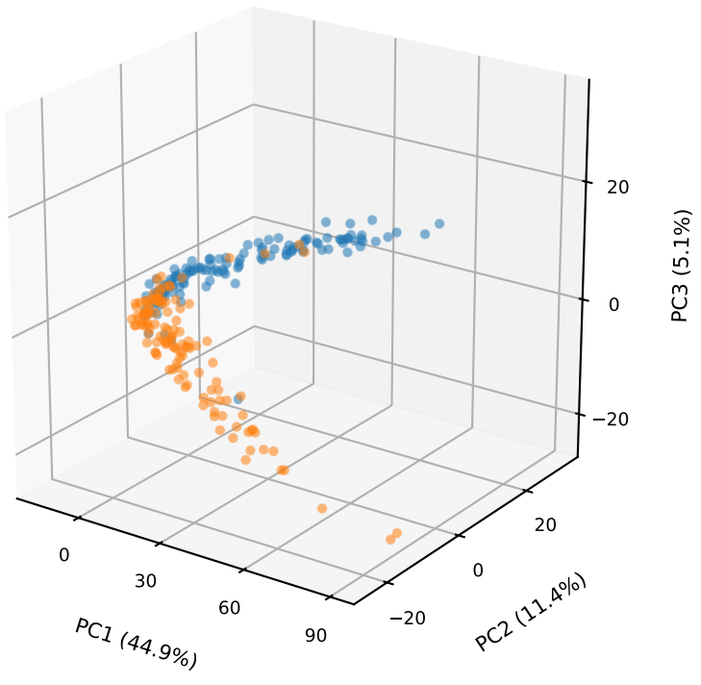
F: after W5



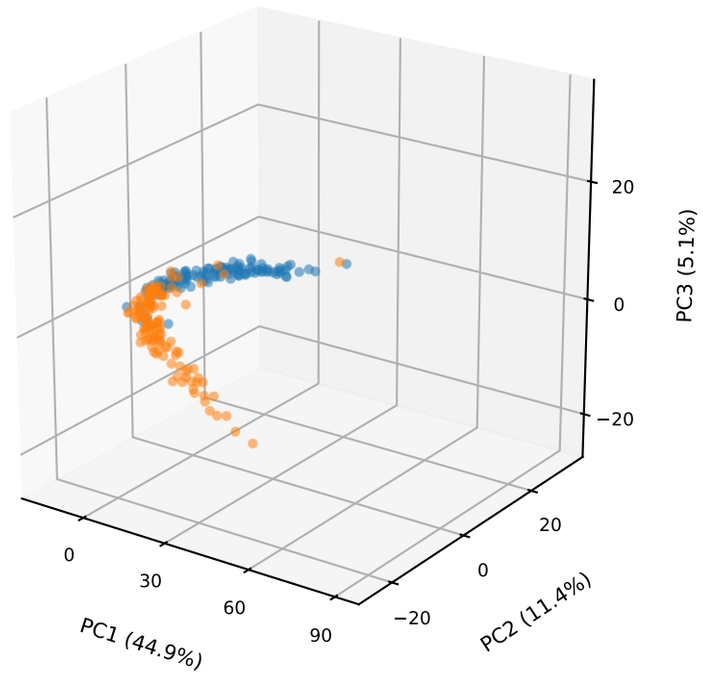
R: after W5



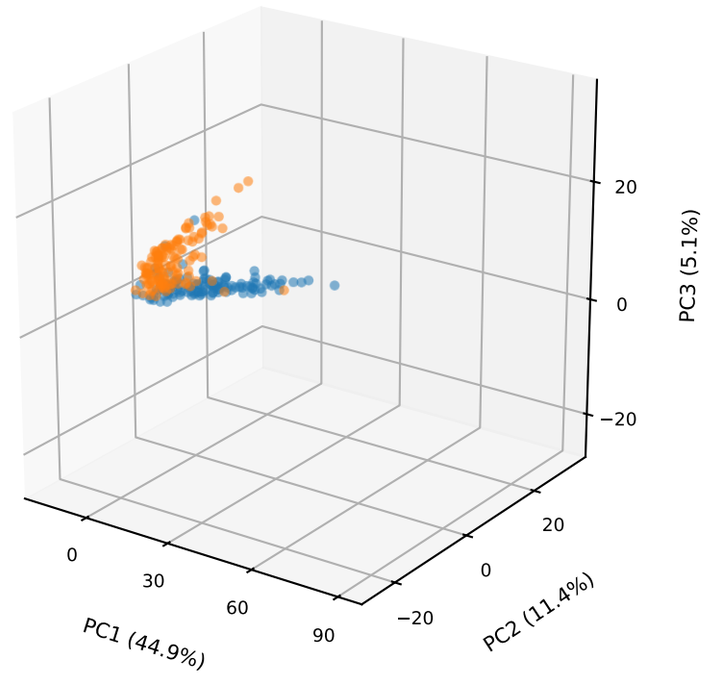
Control: supervised E20



F: supervised E20

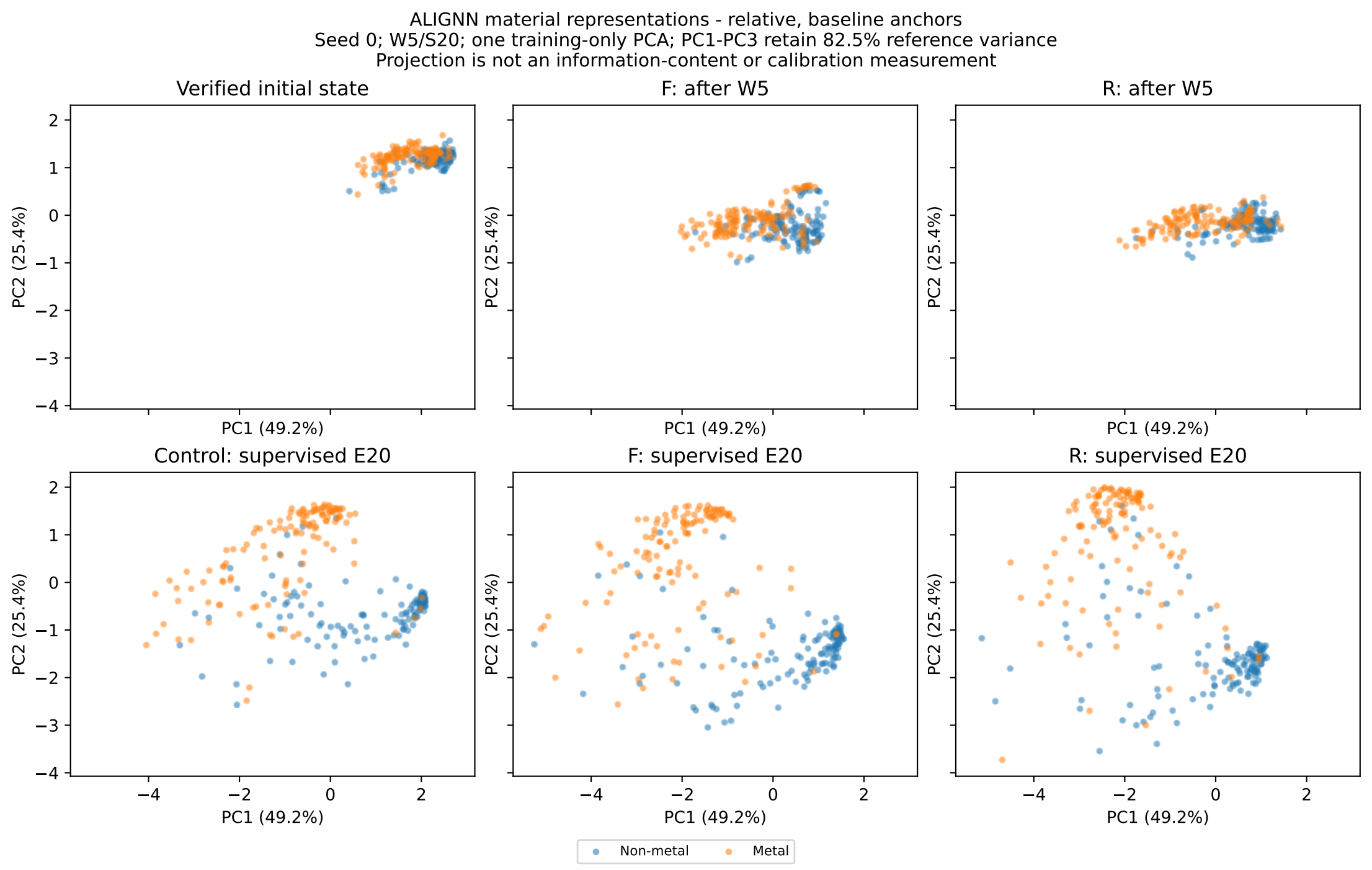


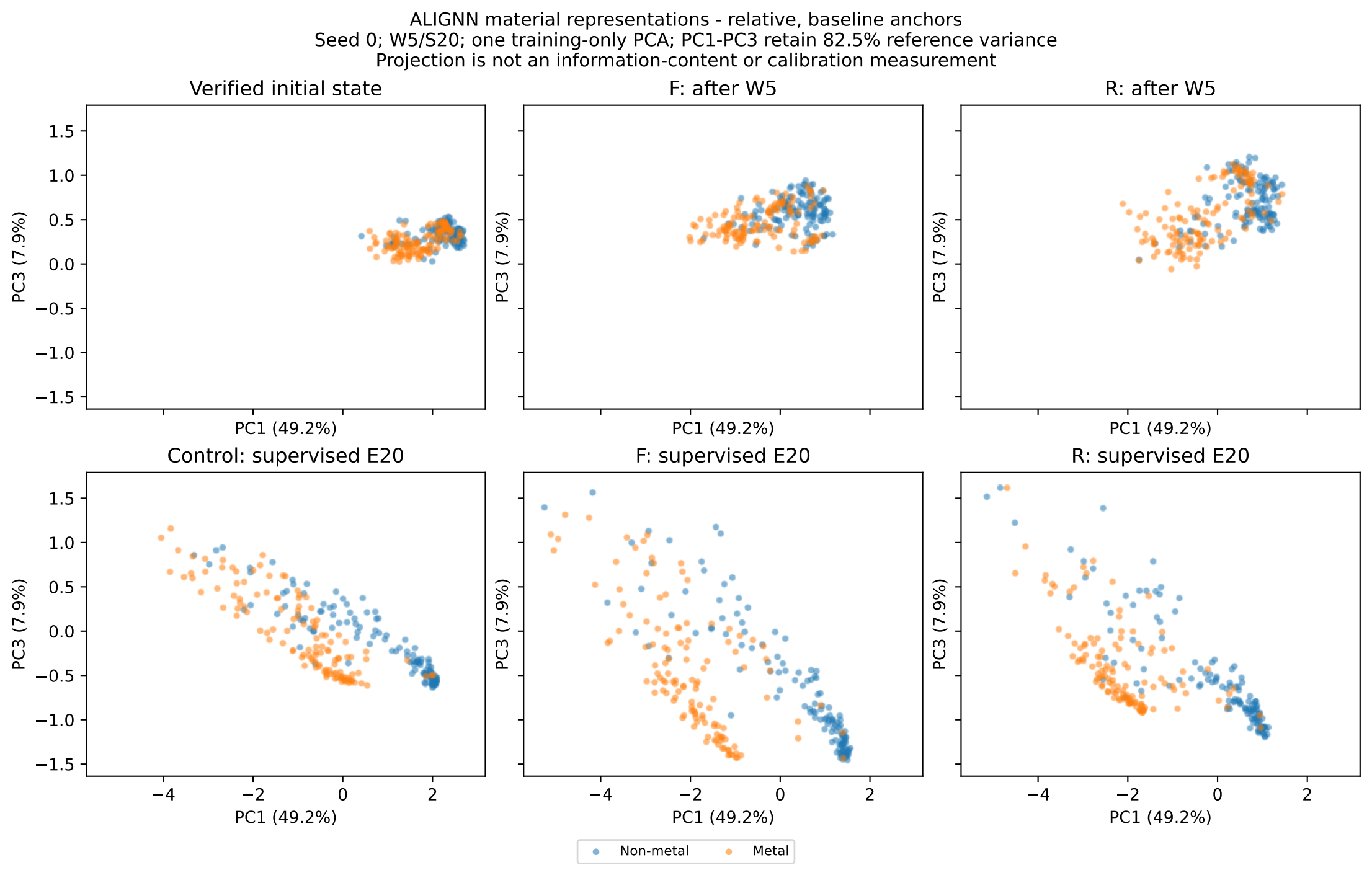
R: supervised E20

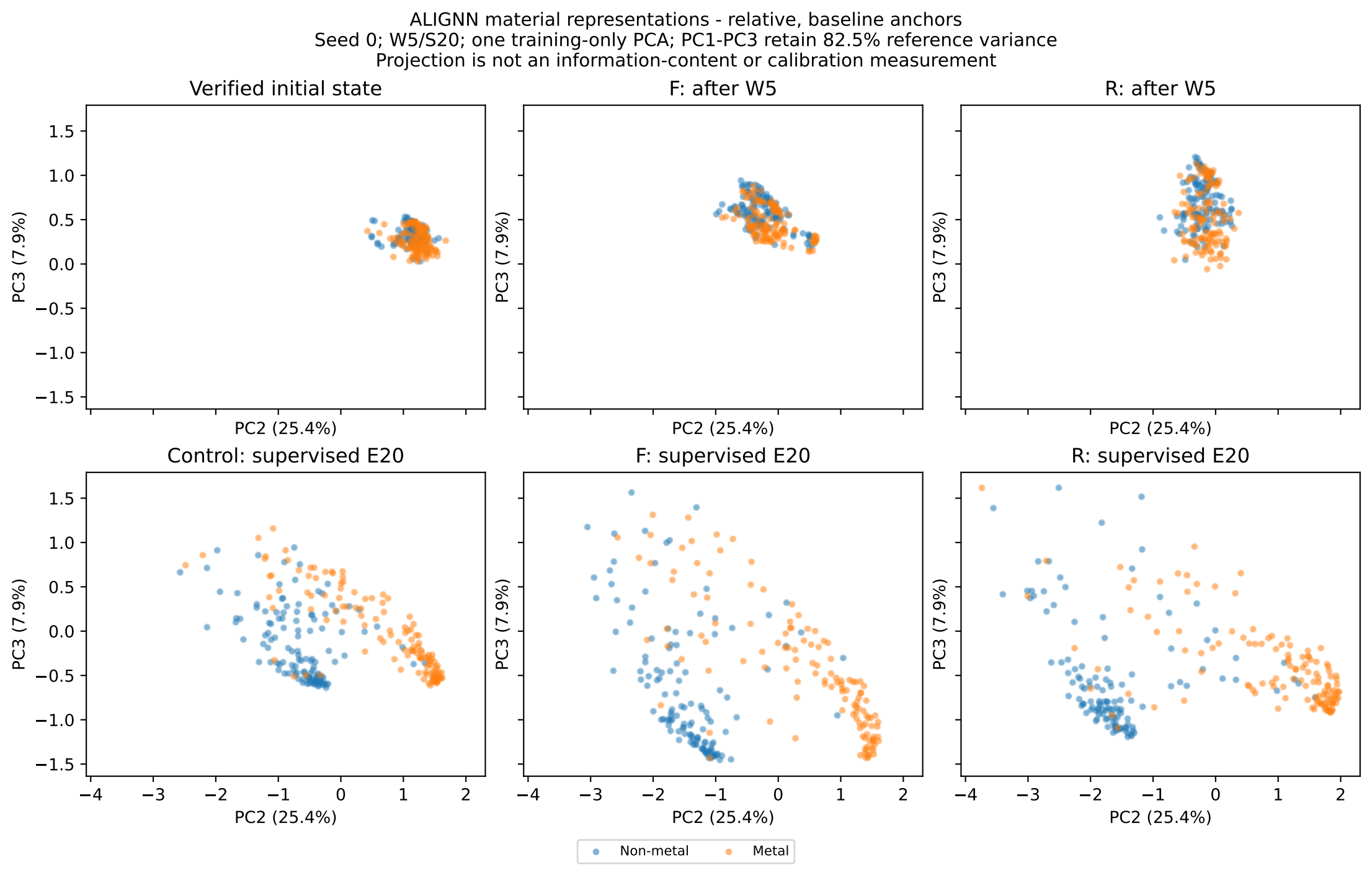


Non-metal Metal

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

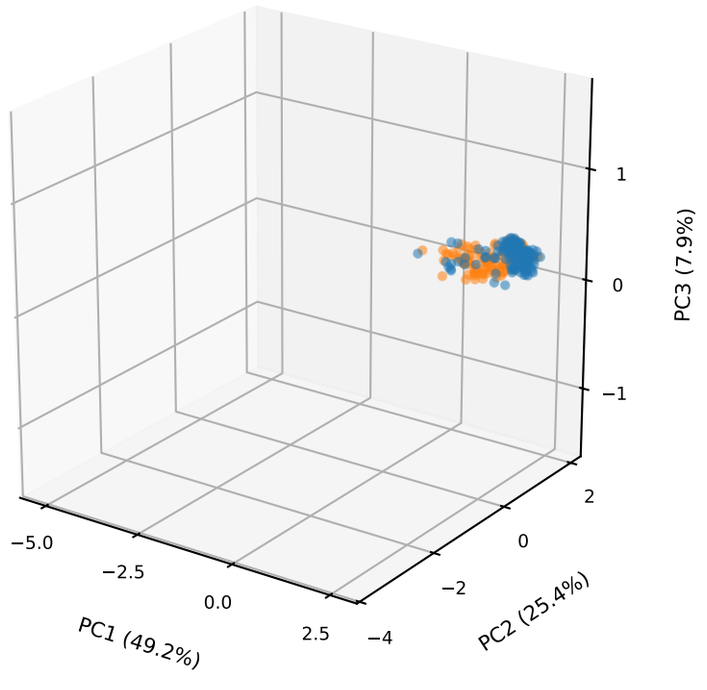

ALIGNN material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 82.5% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (49.2%)
PC2 (25.4%)
Non-metal
Metal


ALIGNN material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 82.5% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC1 (49.2%)
PC3 (7.9%)
Non-metal
Metal

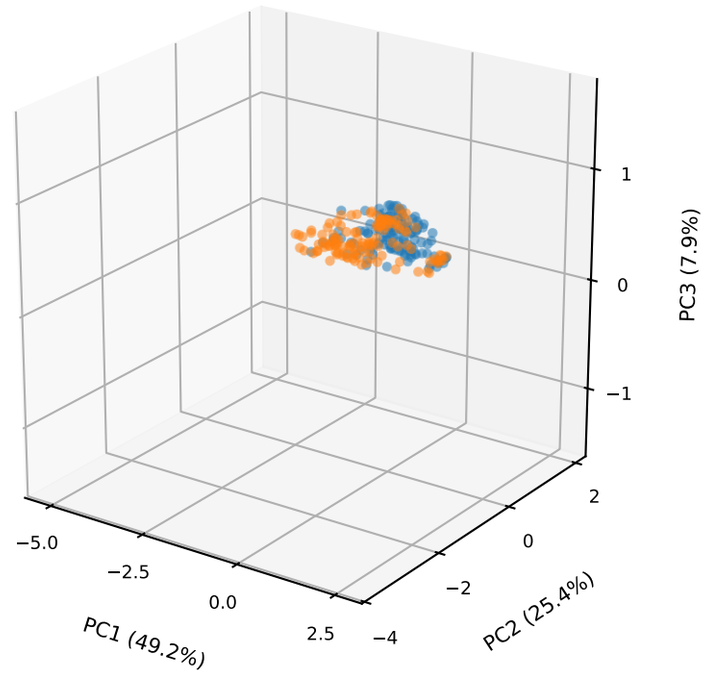

ALIGNN material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 82.5% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
F: after W5
R: after W5
Control: supervised E20
F: supervised E20
R: supervised E20
PC2 (25.4%)
PC3 (7.9%)
Non-metal
Metal

ALIGNN material representations - relative: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 82.5% reference variance

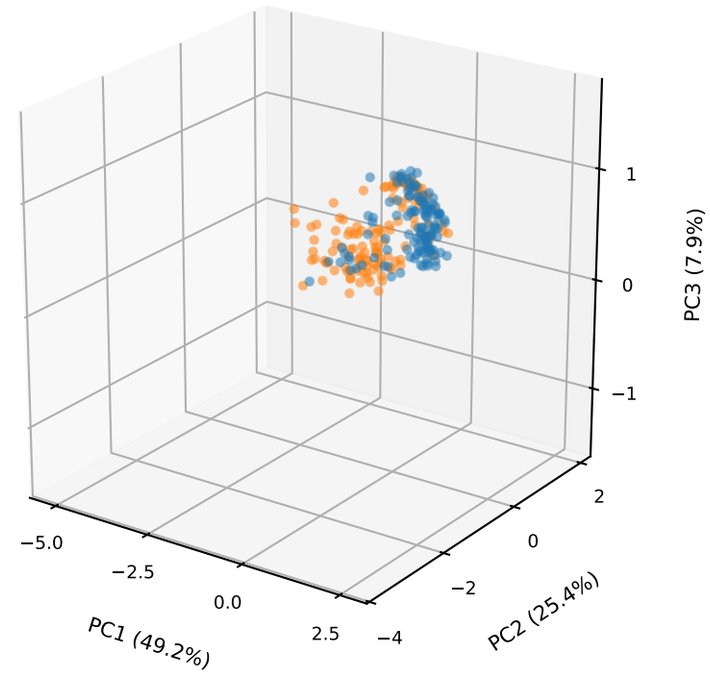
Verified initial state



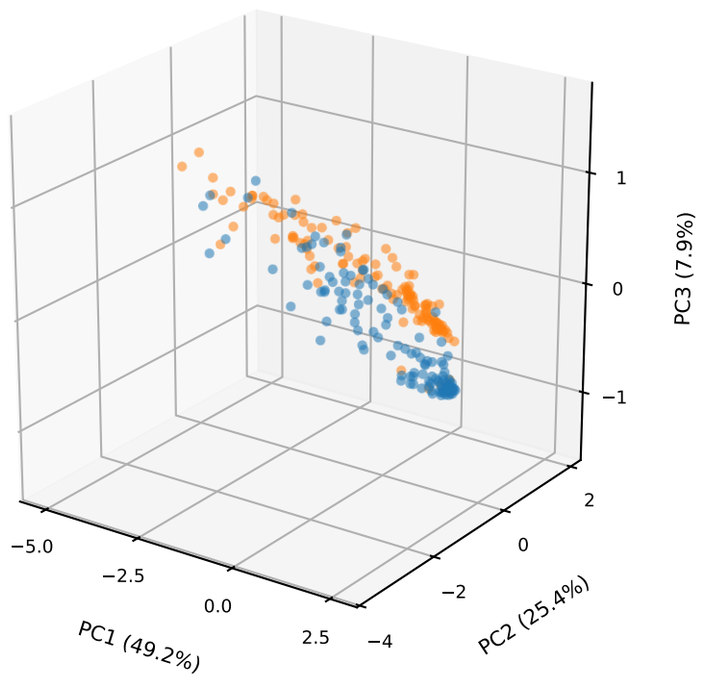
F: after W5



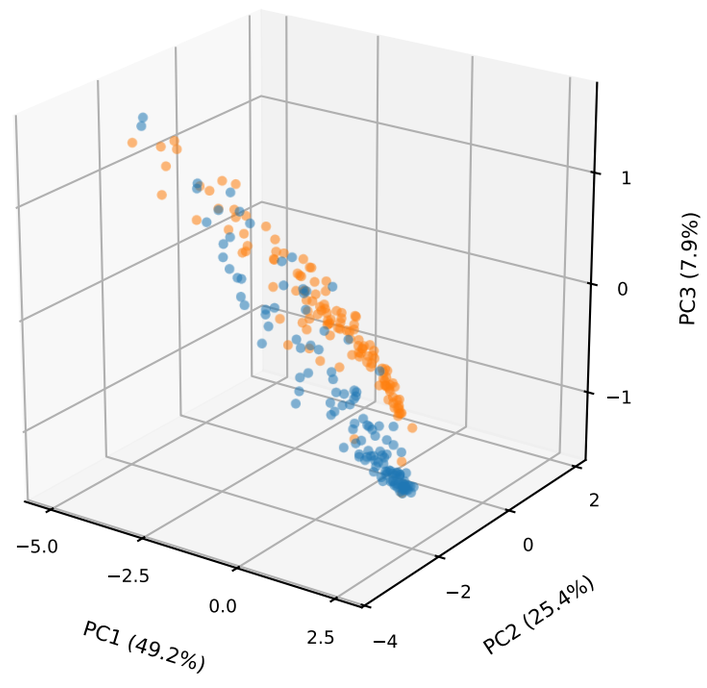
R: after W5



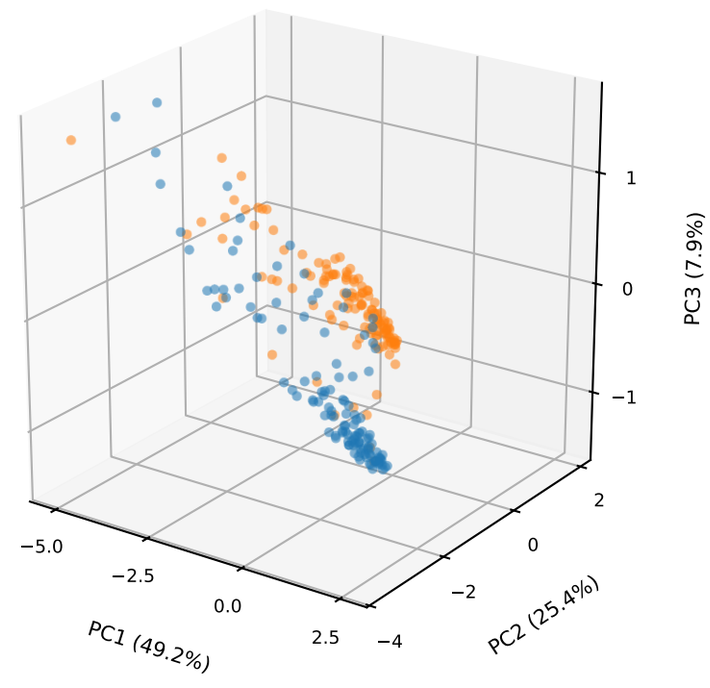
Control: supervised E20



F: supervised E20

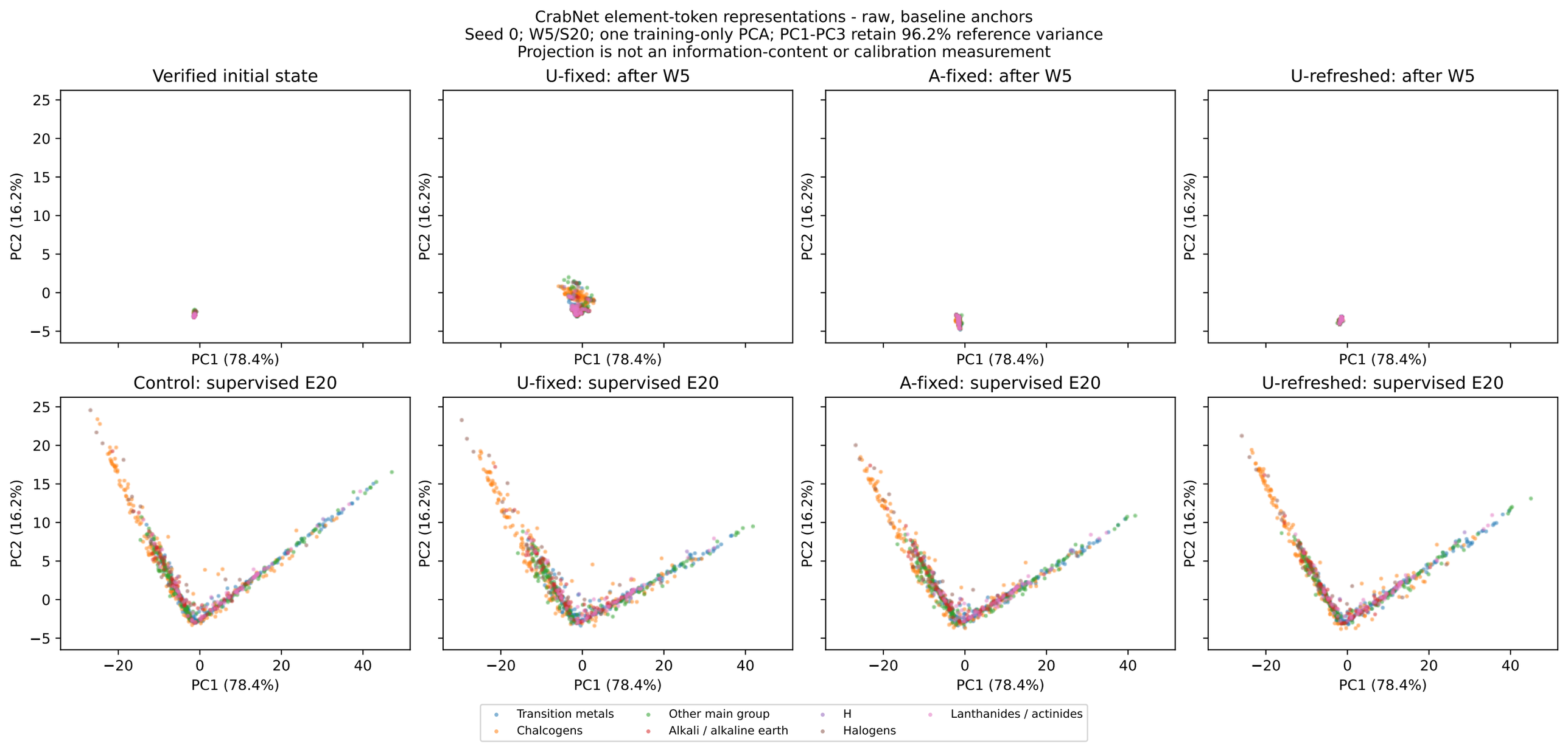


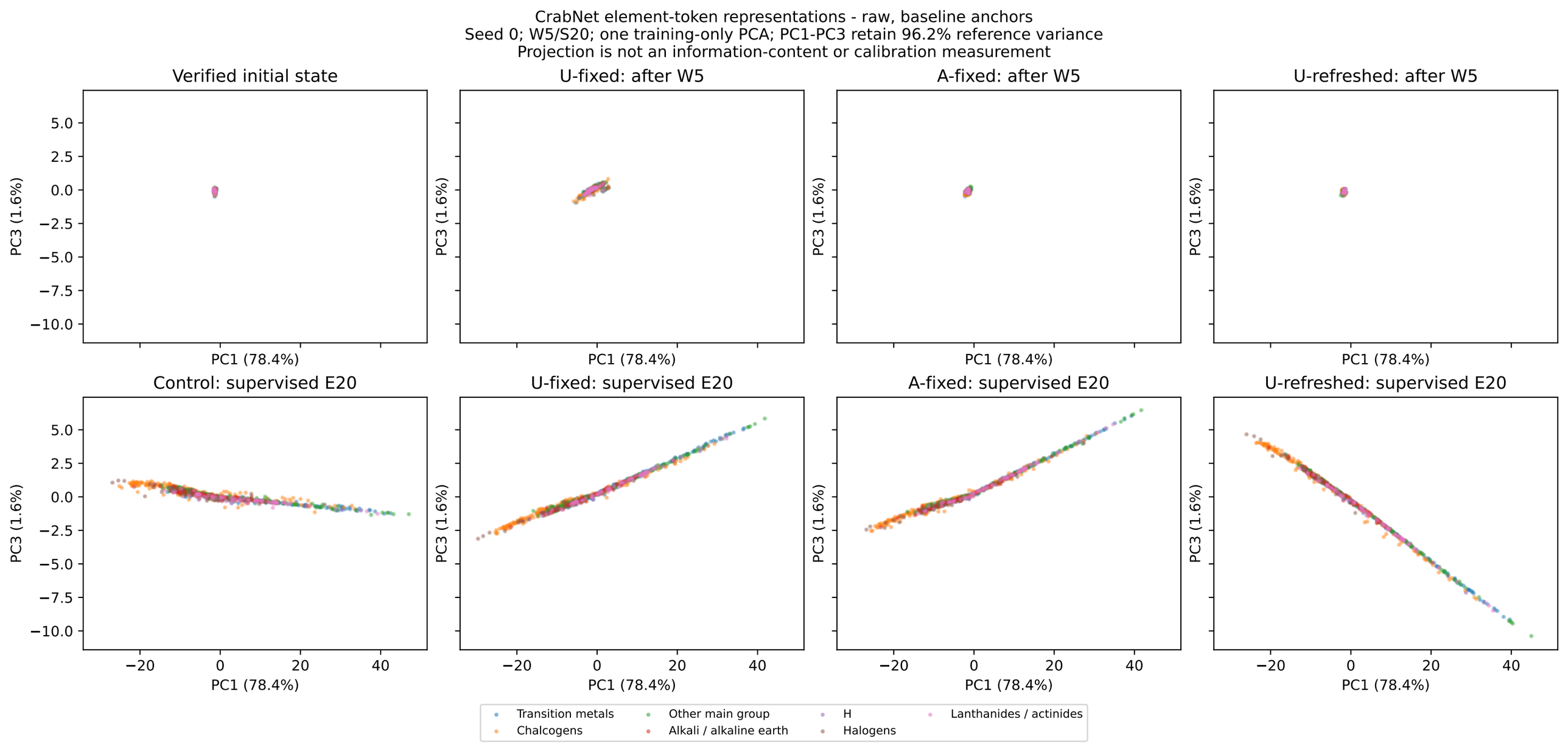
R: supervised E20

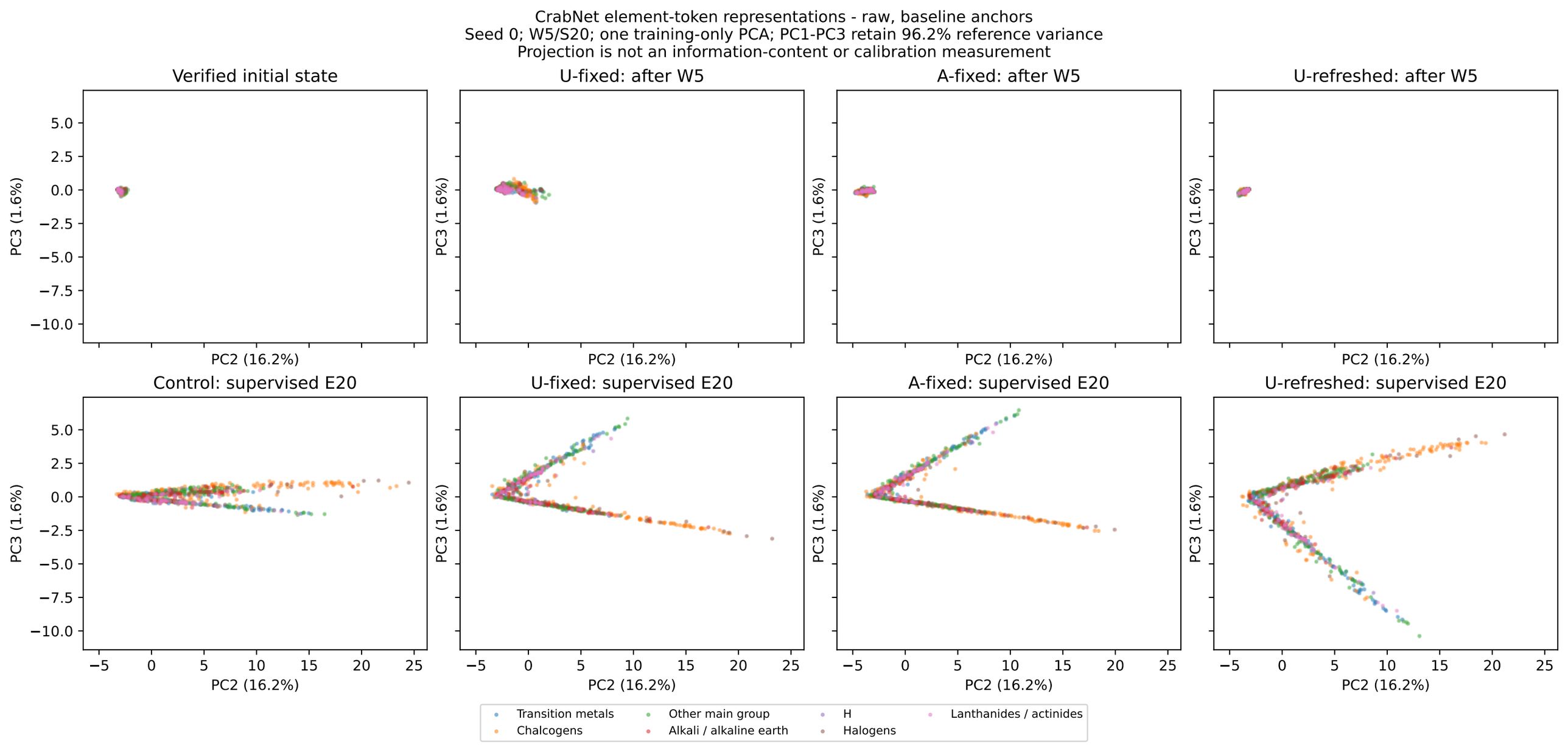


Non-metal Metal

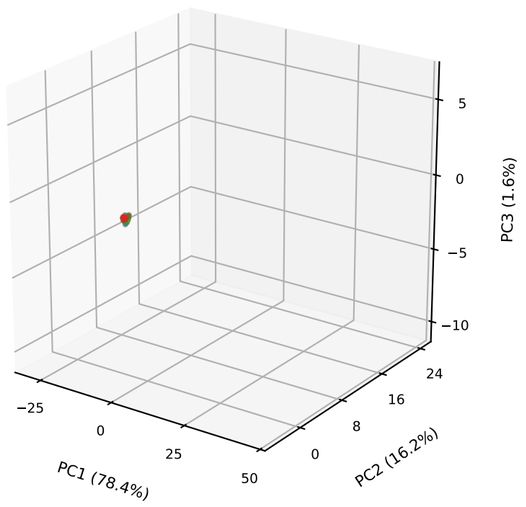
Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

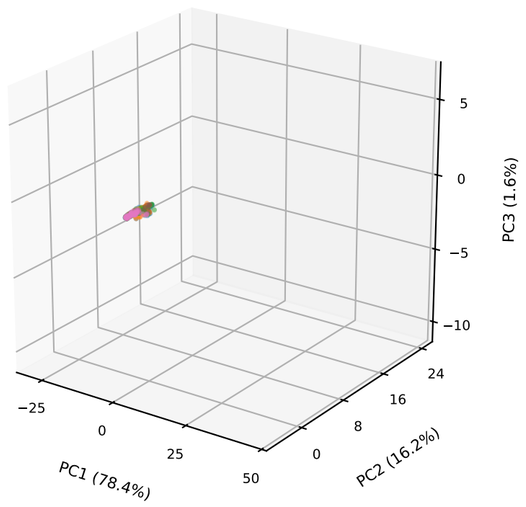

CrabNet element-token representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 96.2% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (78.4%)
PC2 (16.2%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

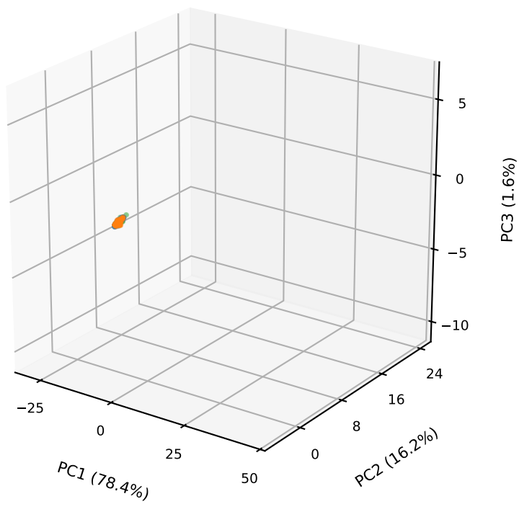

CrabNet element-token representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 96.2% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (78.4%)
PC3 (1.6%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

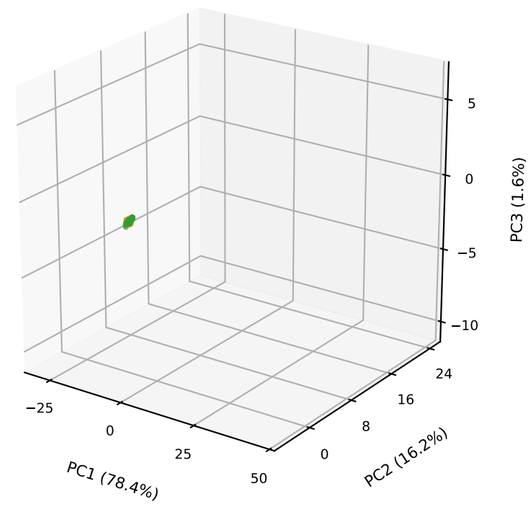

CrabNet element-token representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 96.2% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC3 (1.6%)
PC2 (16.2%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

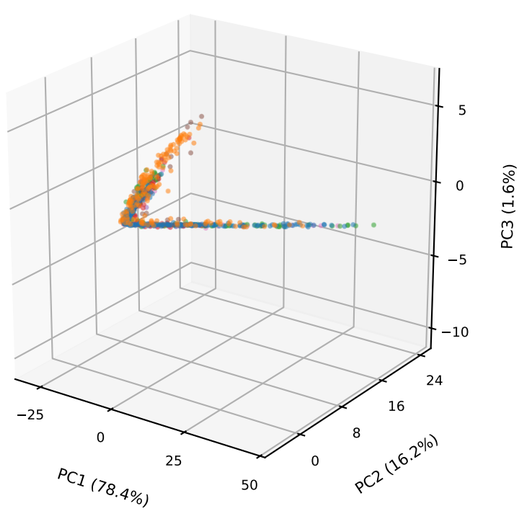
CrabNet element-token representations - raw: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 96.2% reference variance

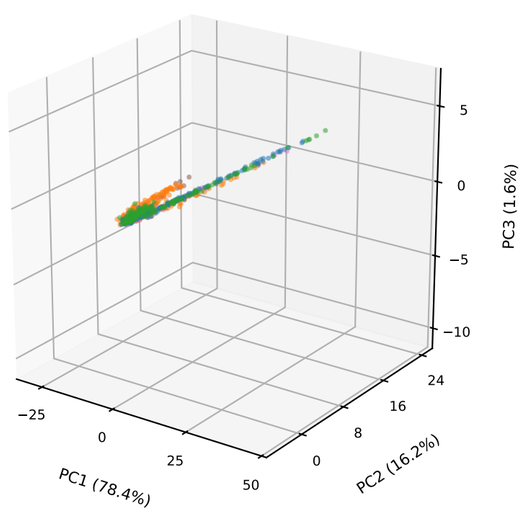


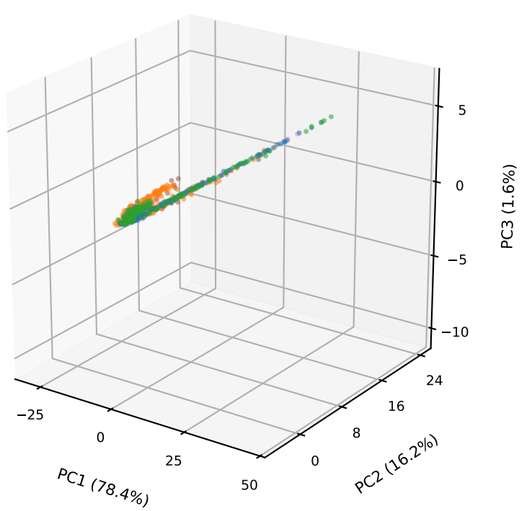


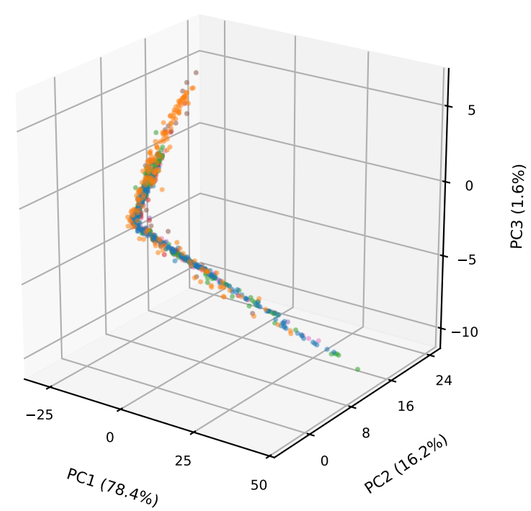






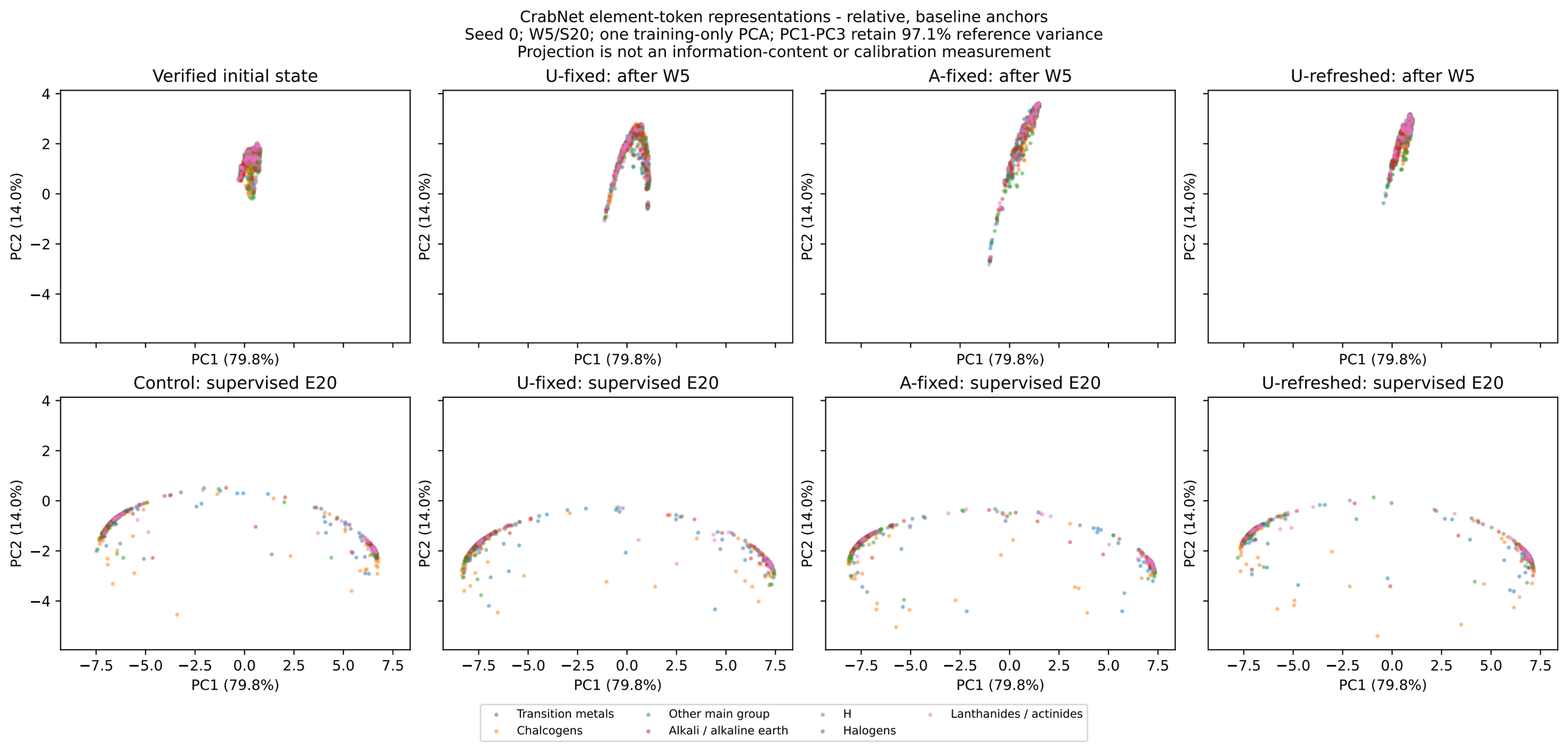


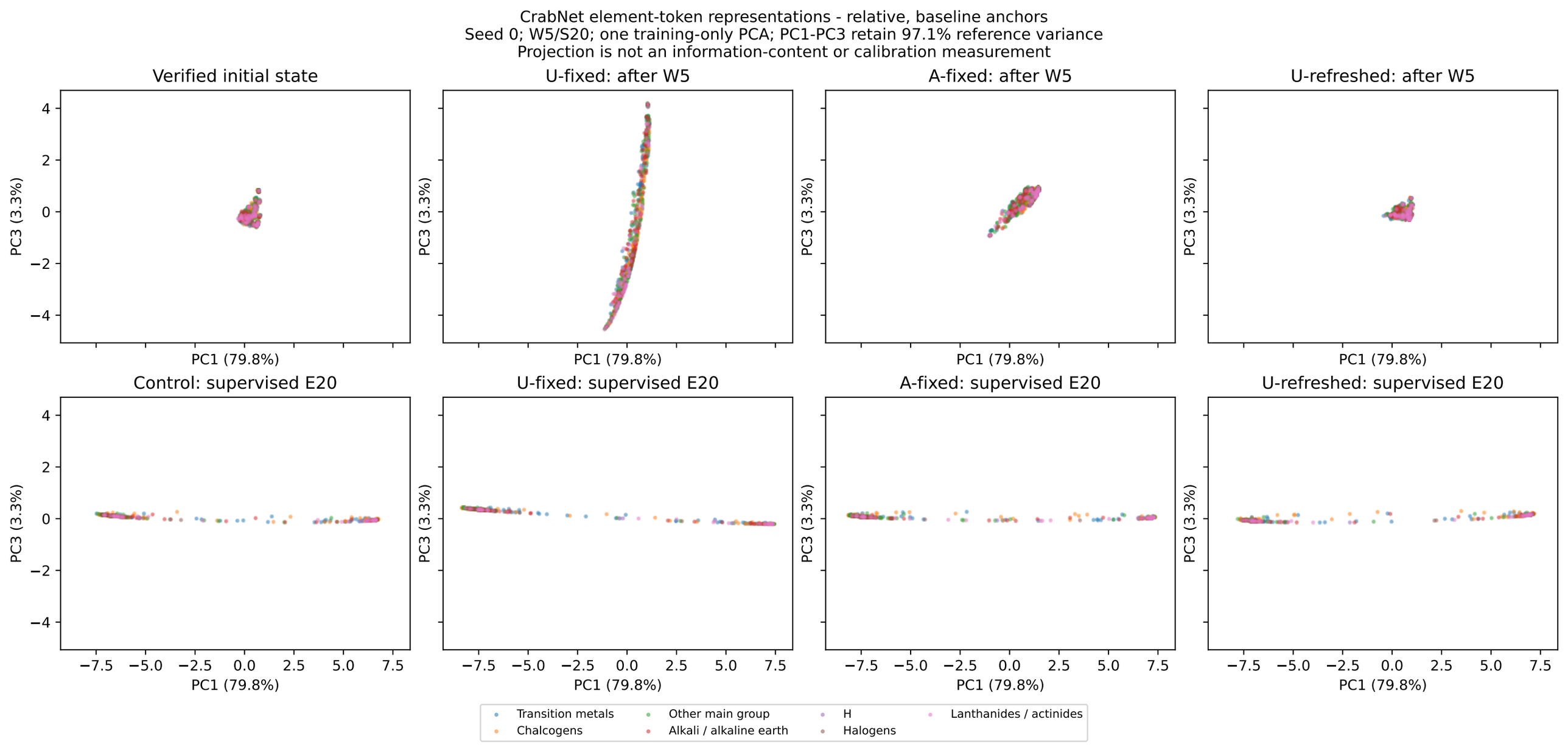


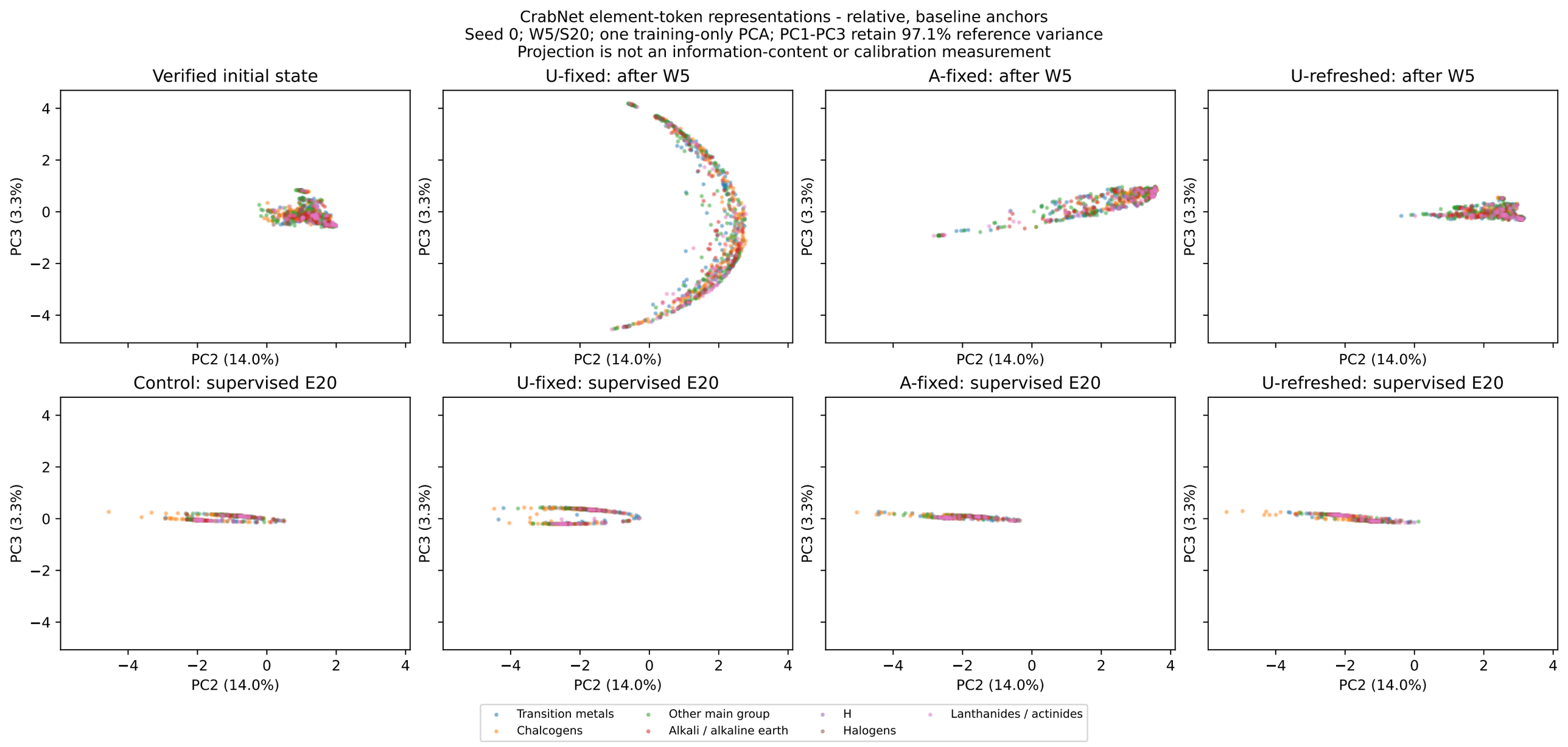


Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

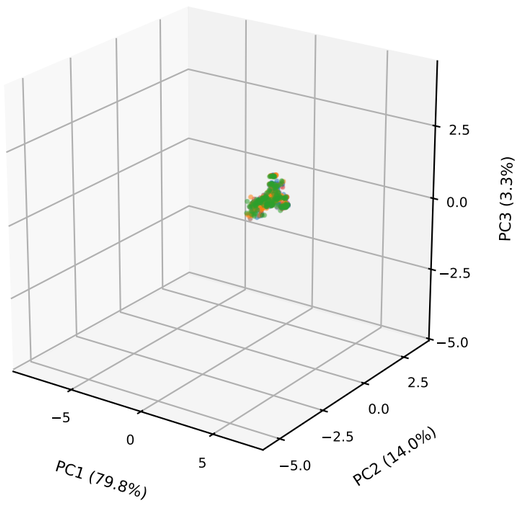

CrabNet element-token representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 97.1% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (79.8%)
PC2 (14.0%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides


CrabNet element-token representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 97.1% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (79.8%)
PC3 (3.3%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

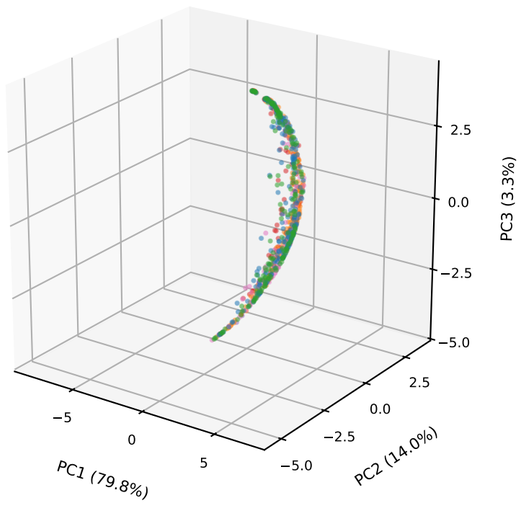

CrabNet element-token representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 97.1% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC2 (14.0%)
PC3 (3.3%)
Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

CrabNet element-token representations - relative: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 97.1% reference variance

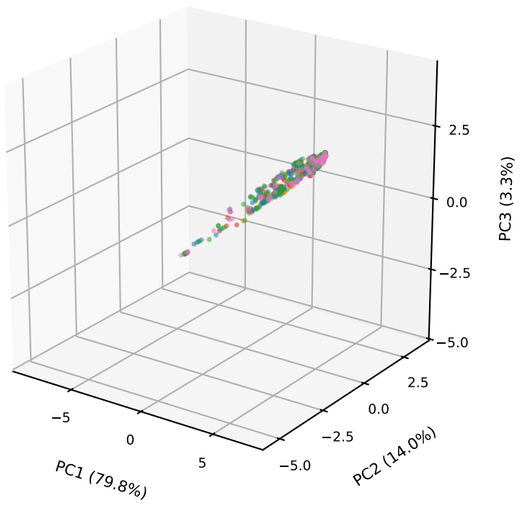
Verified initial state



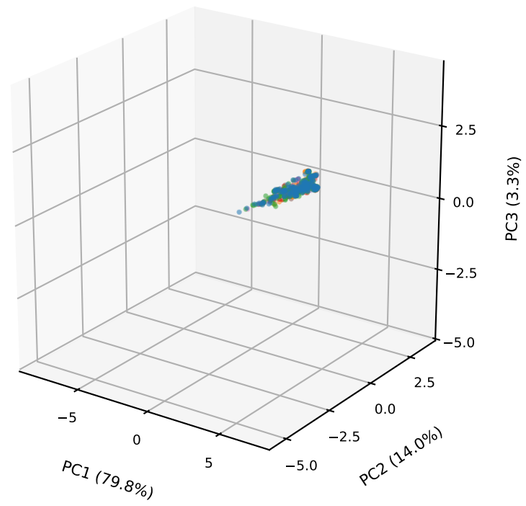
U-fixed: after W5



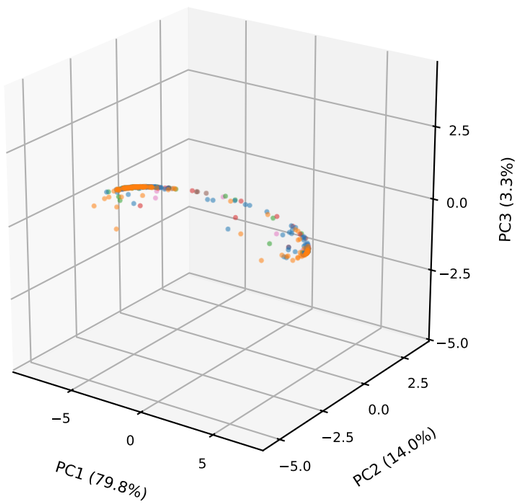
A-fixed: after W5



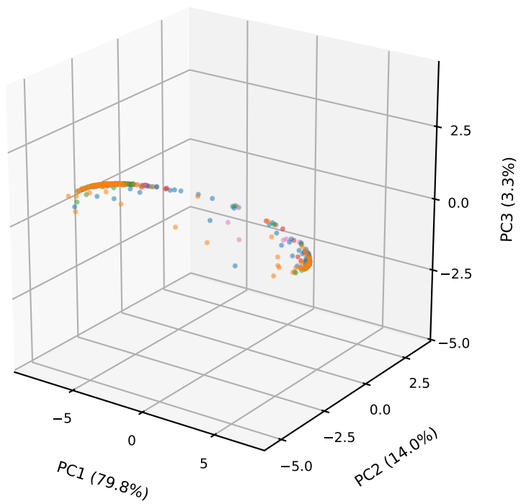
U-refreshed: after W5



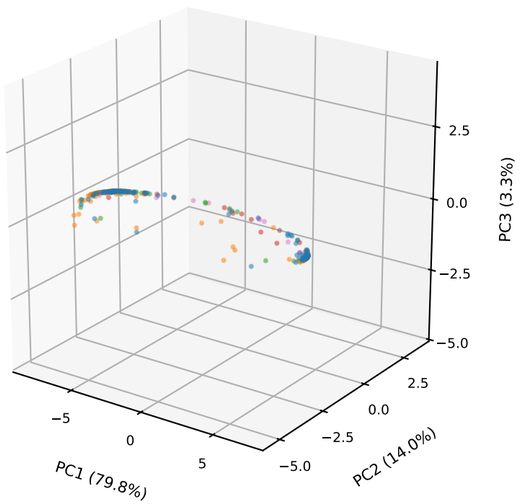
Control: supervised E20



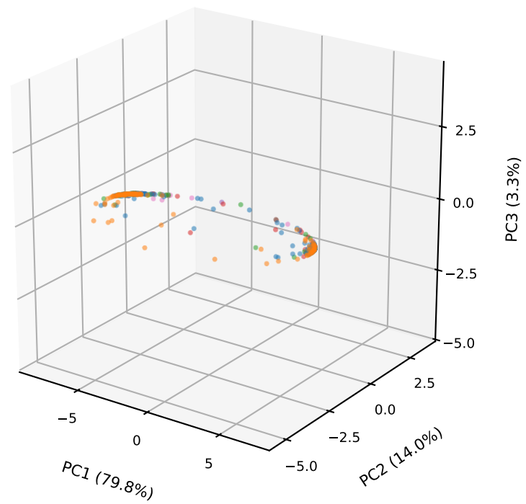
U-fixed: supervised E20



A-fixed: supervised E20

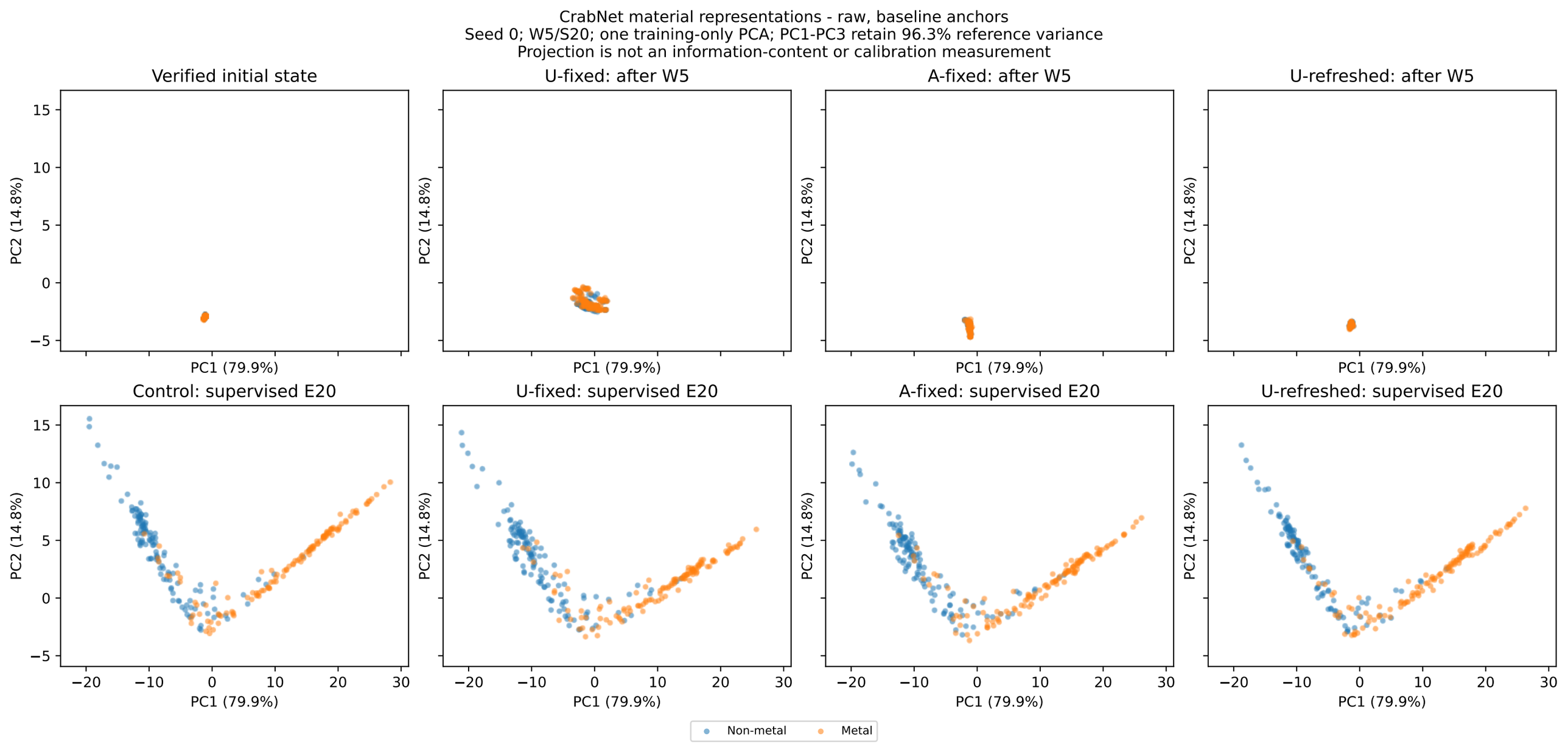


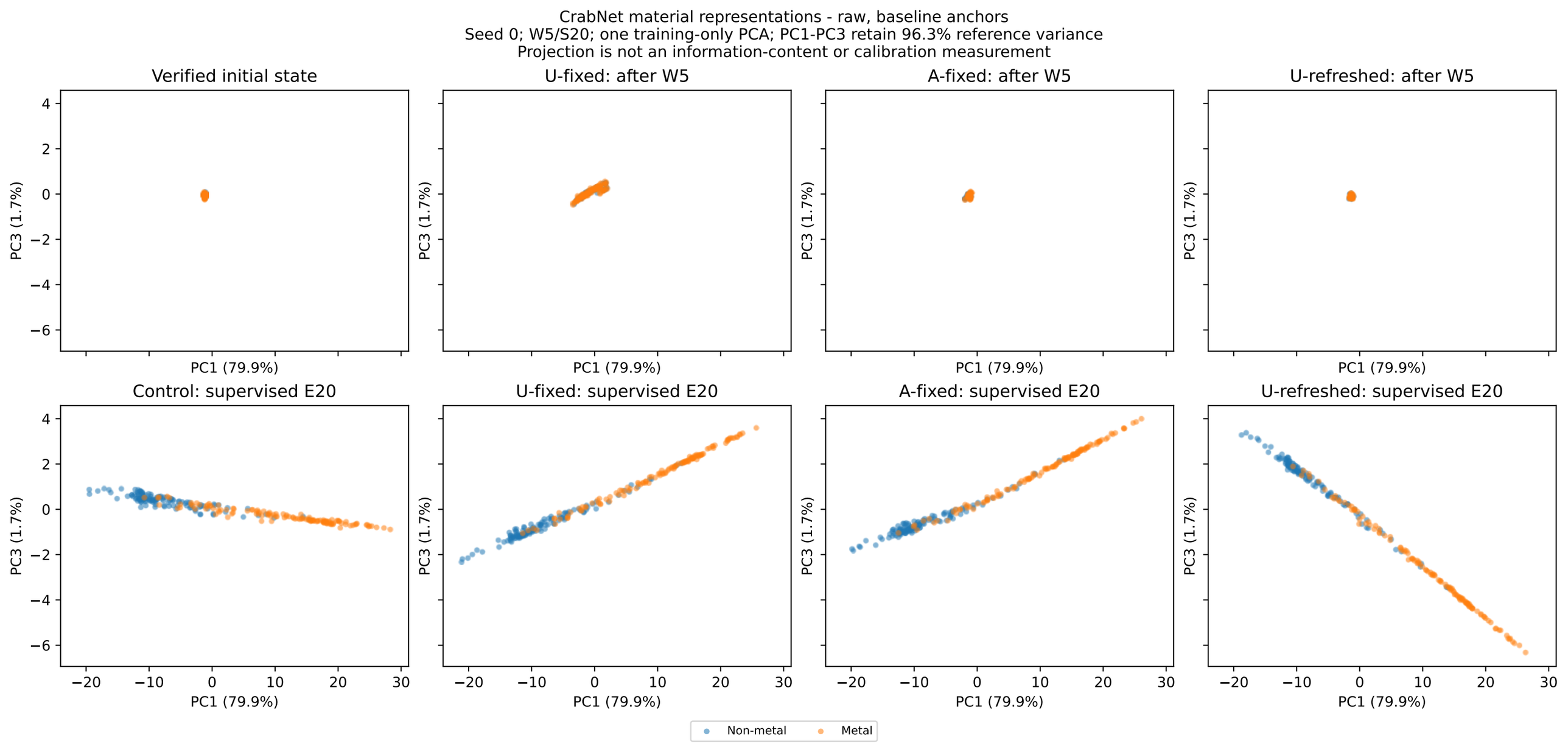
U-refreshed: supervised E20

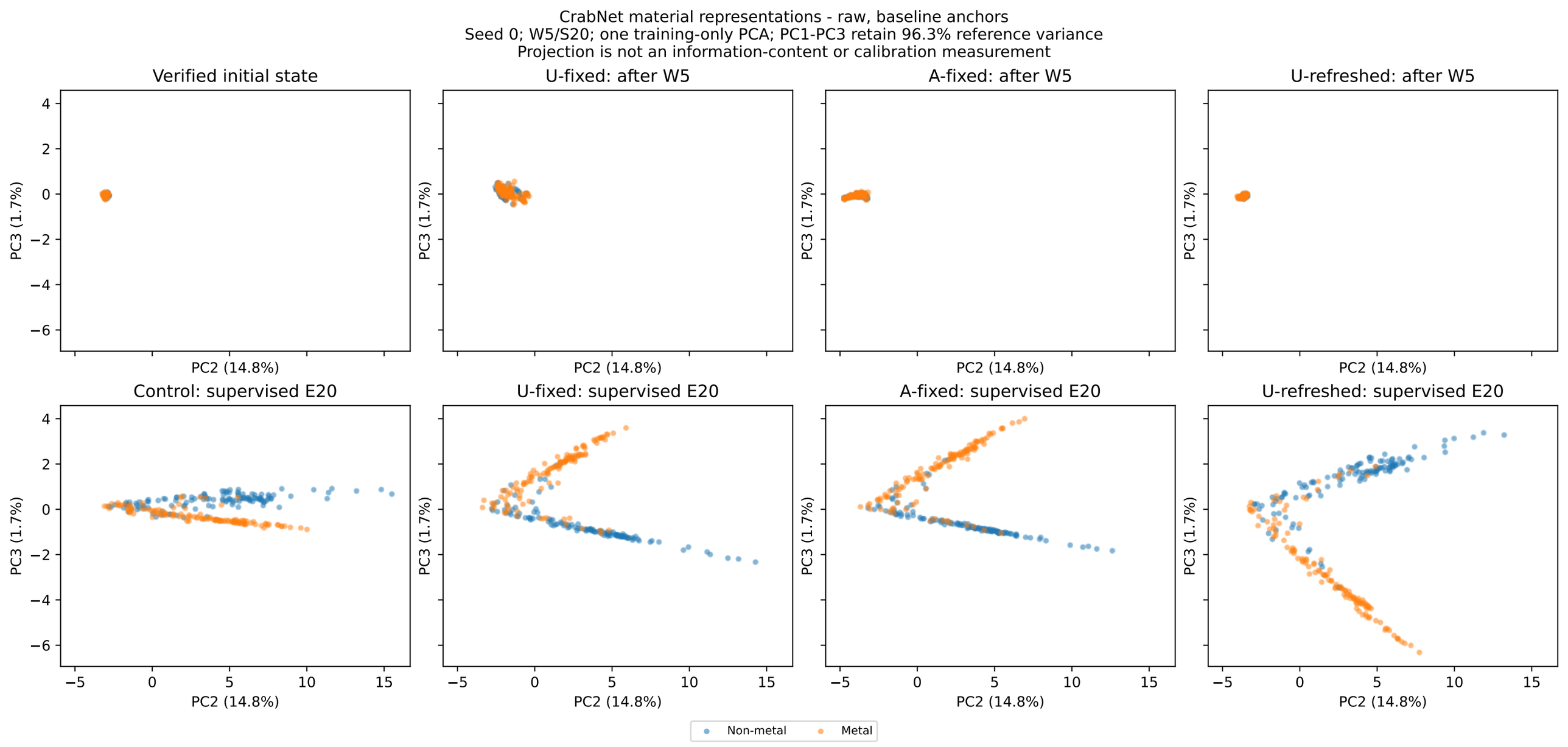


Transition metals
Chalcogens
Other main group
Alkali / alkaline earth
H
Halogens
Lanthanides / actinides

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

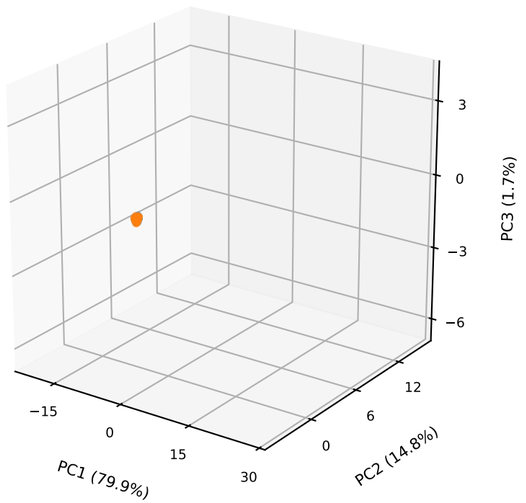

CrabNet material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 96.3% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (79.9%)
PC2 (14.8%)
Non-metal
Metal


CrabNet material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 96.3% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (79.9%)
PC3 (1.7%)
Non-metal
Metal

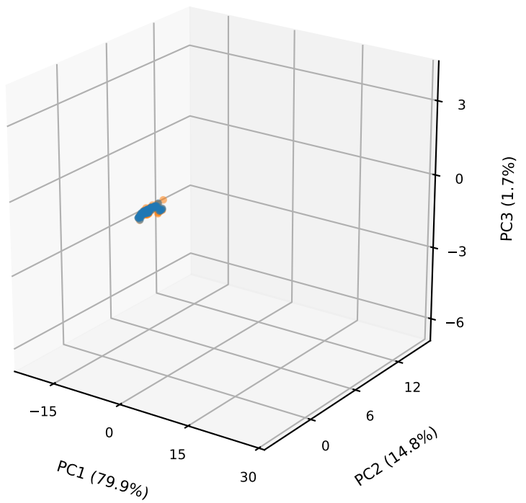

CrabNet material representations - raw, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 96.3% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC2 (14.8%)
PC3 (1.7%)
Non-metal
Metal

CrabNet material representations - raw: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 96.3% reference variance

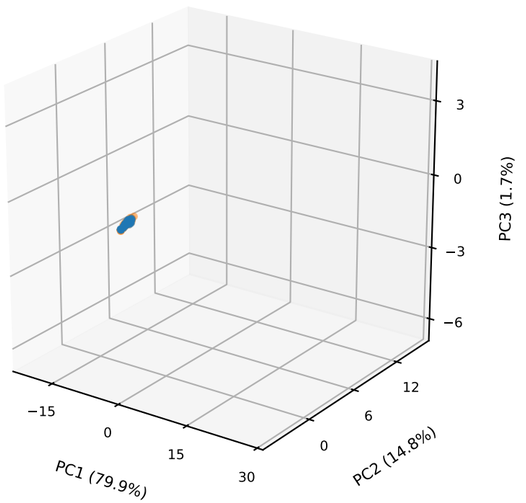
Verified initial state



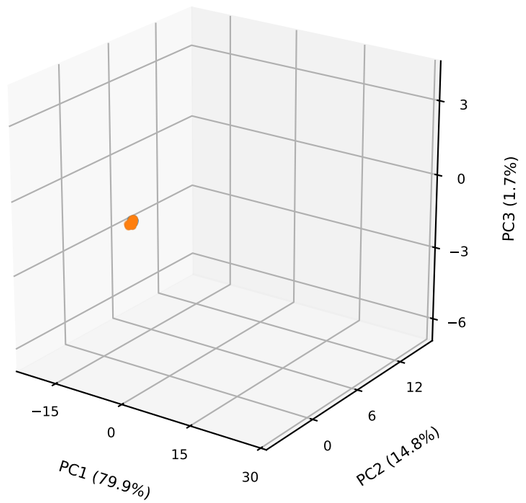
U-fixed: after W5



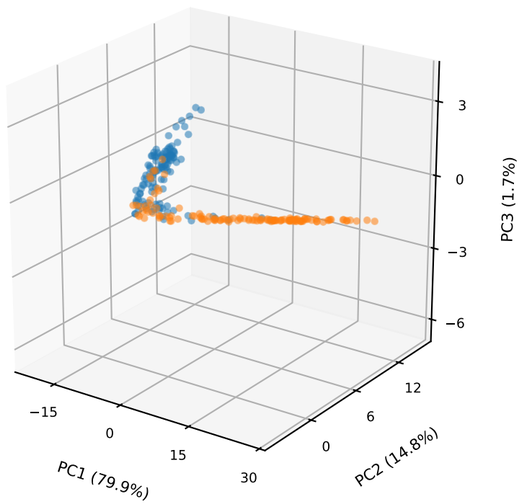
A-fixed: after W5



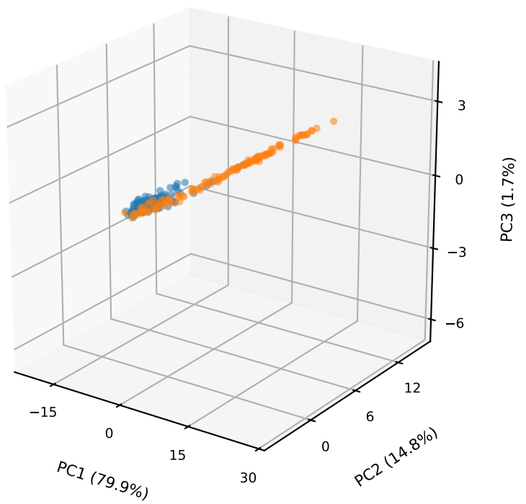
U-refreshed: after W5



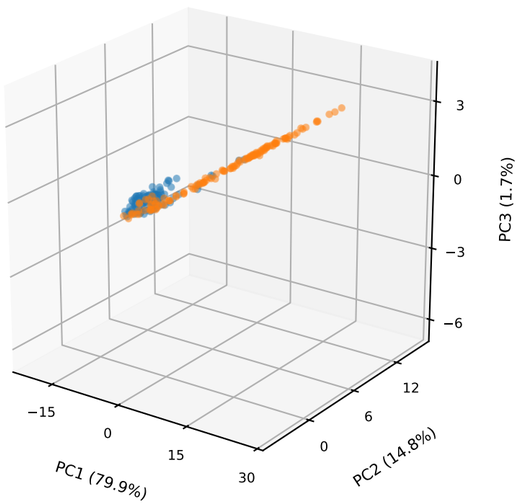
Control: supervised E20



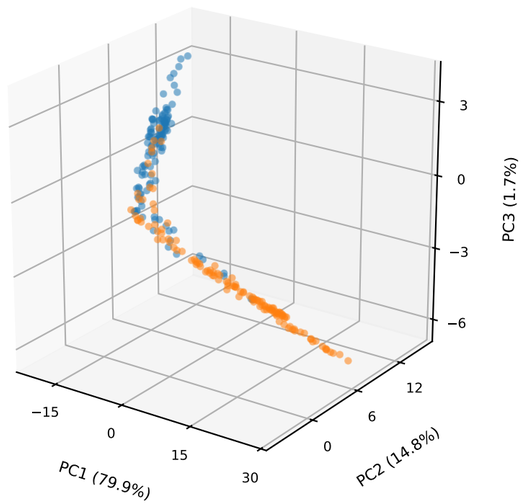
U-fixed: supervised E20



A-fixed: supervised E20

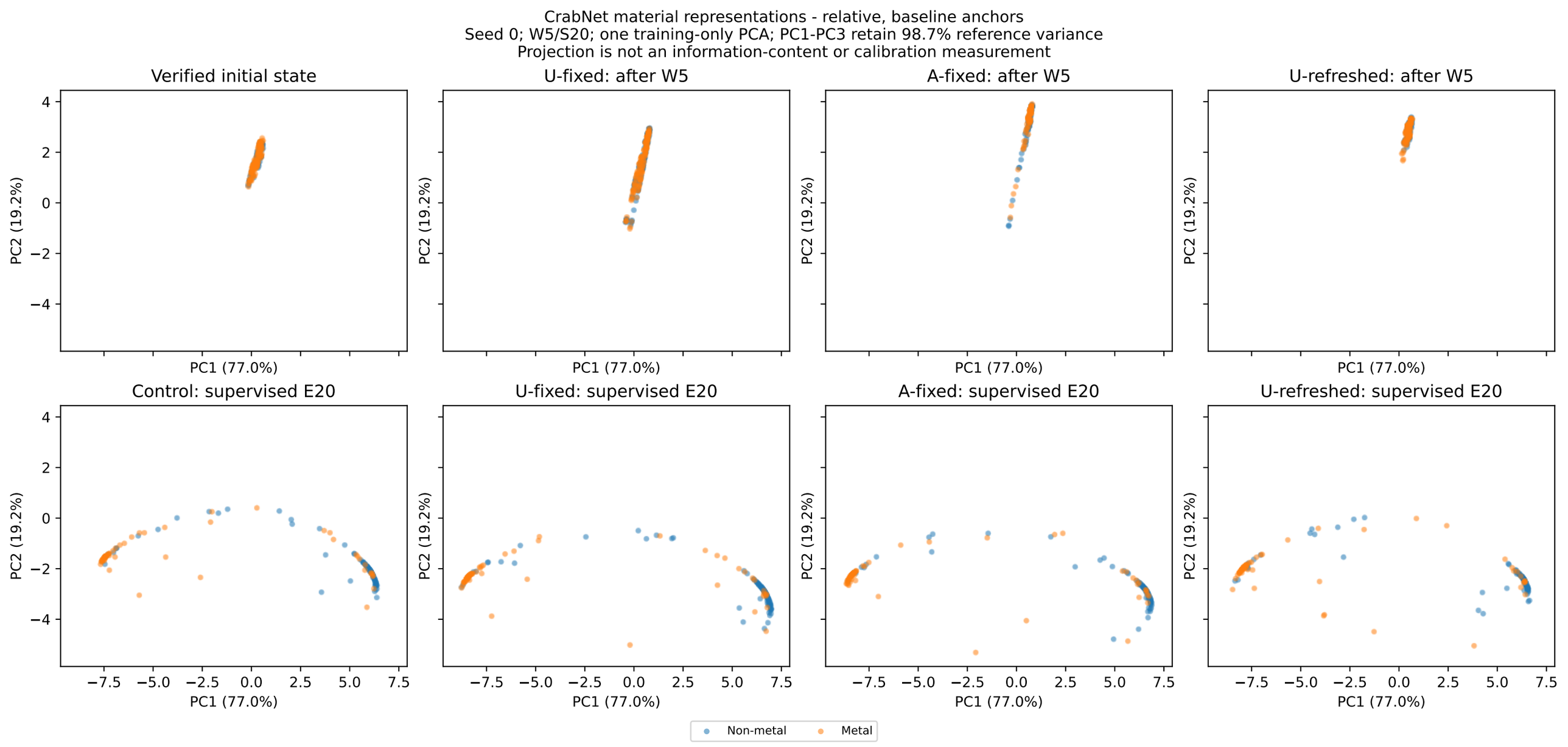


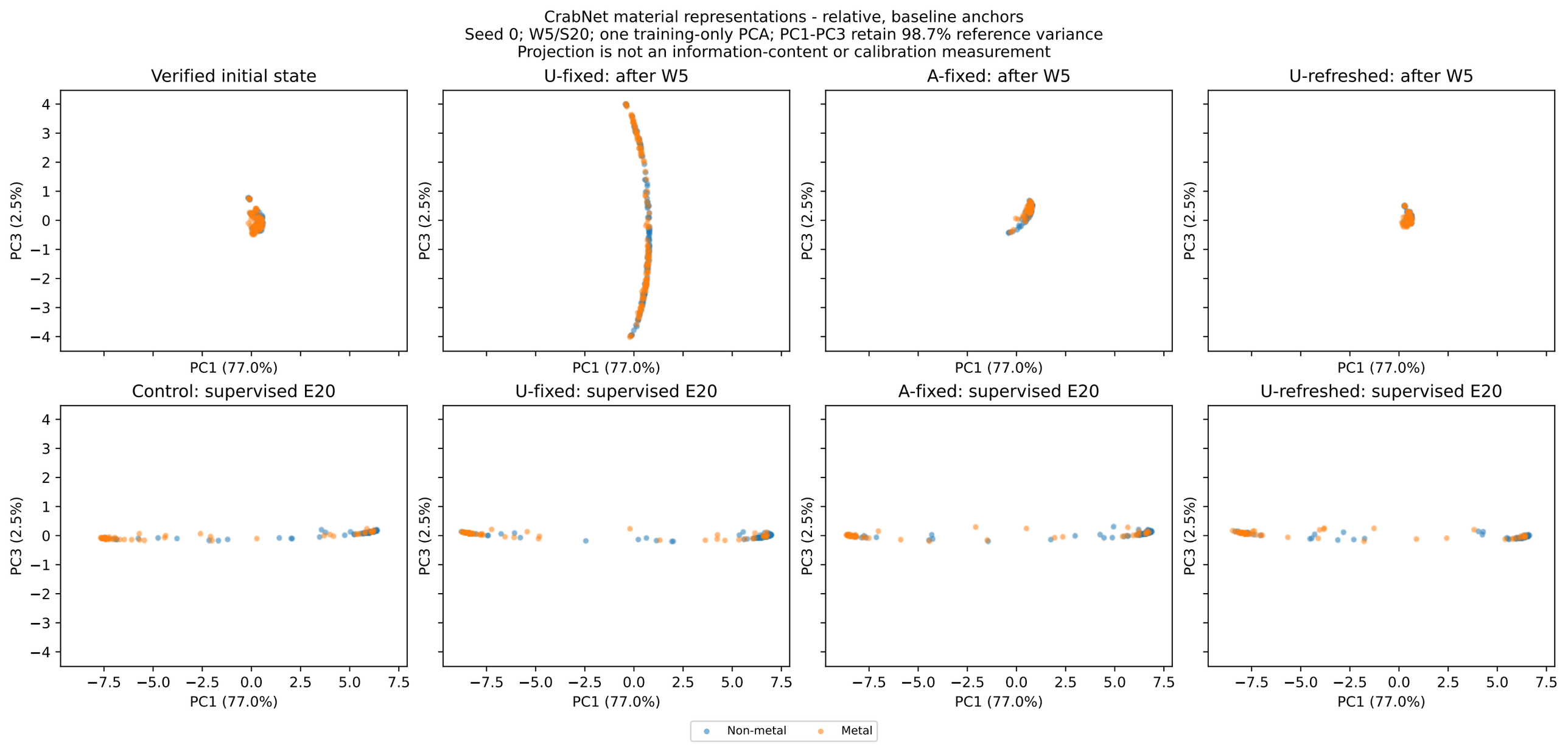
U-refreshed: supervised E20

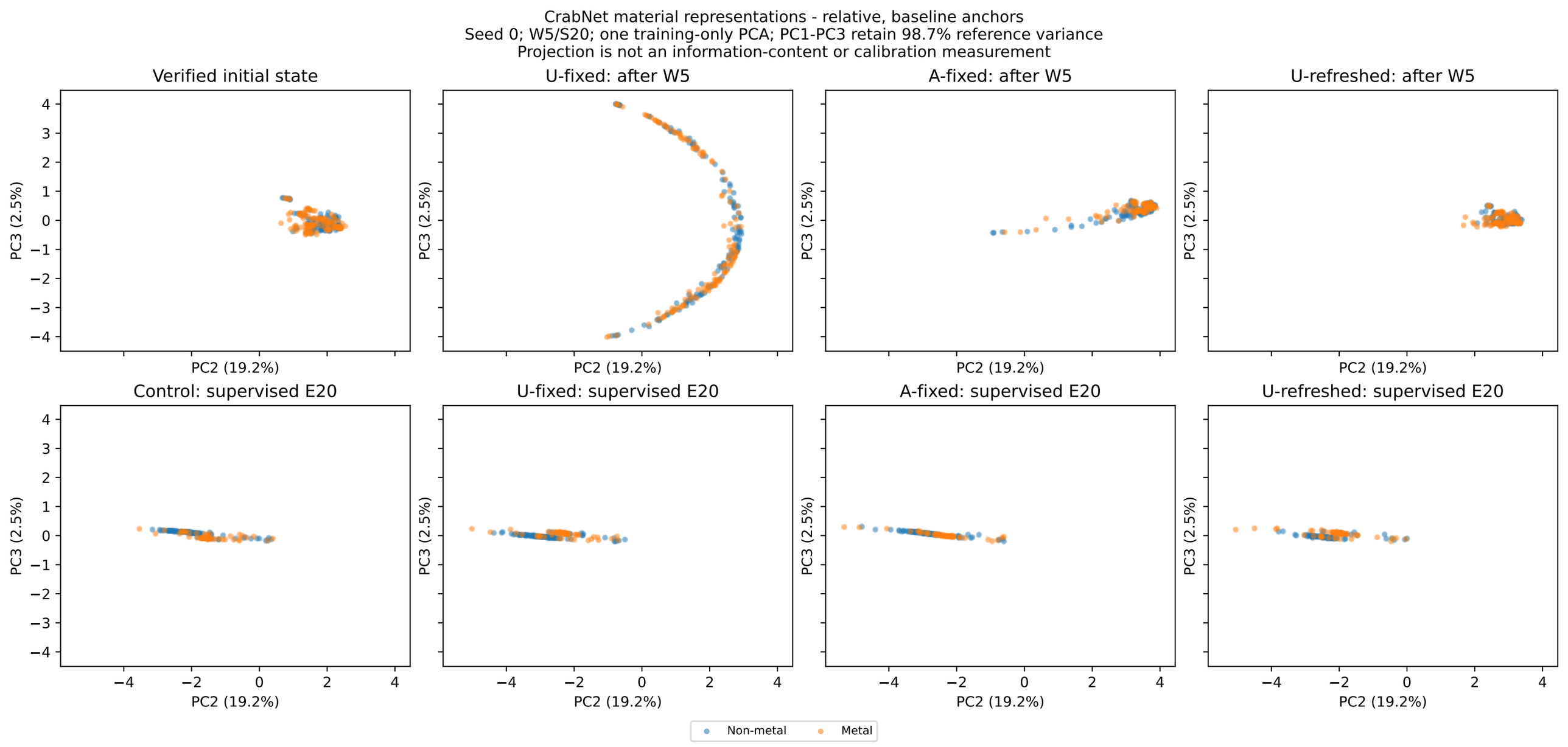


Non-metal Metal

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.

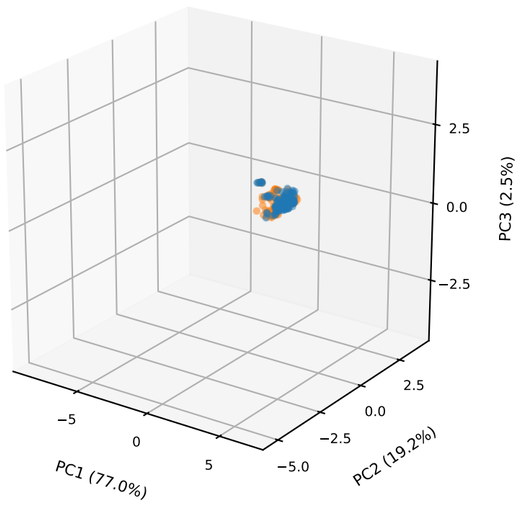

CrabNet material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 98.7% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (77.0%)
PC2 (19.2%)
Non-metal
Metal


CrabNet material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 98.7% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC1 (77.0%)
PC3 (2.5%)
Non-metal
Metal

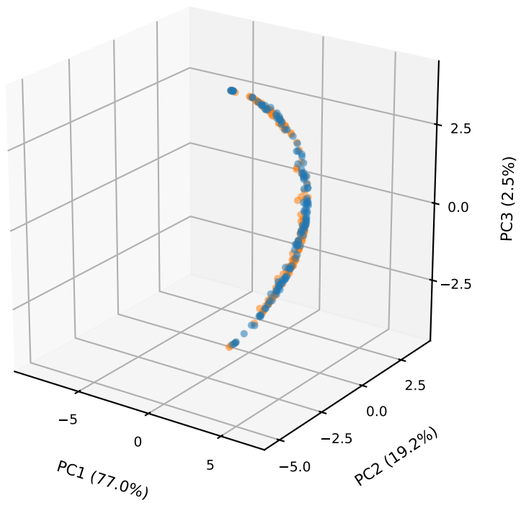

CrabNet material representations - relative, baseline anchors
Seed 0; W5/S20; one training-only PCA; PC1-PC3 retain 98.7% reference variance
Projection is not an information-content or calibration measurement
Verified initial state
U-fixed: after W5
A-fixed: after W5
U-refreshed: after W5
Control: supervised E20
U-fixed: supervised E20
A-fixed: supervised E20
U-refreshed: supervised E20
PC2 (19.2%)
PC3 (2.5%)
Non-metal
Metal

CrabNet material representations - relative: static 3D
Seed 0; W5/S20; same saved training-only PCA; PC1-PC3 retain 98.7% reference variance

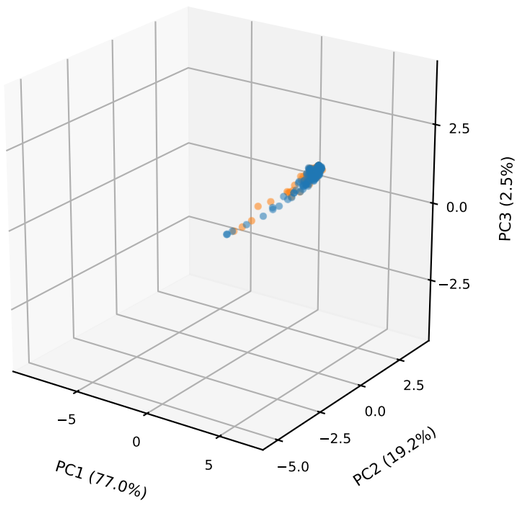
Verified initial state



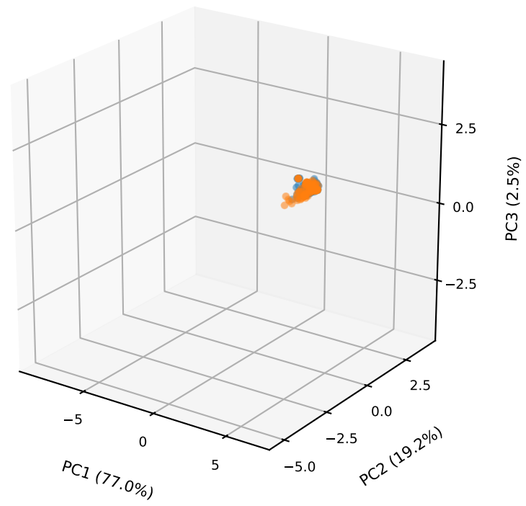
U-fixed: after W5



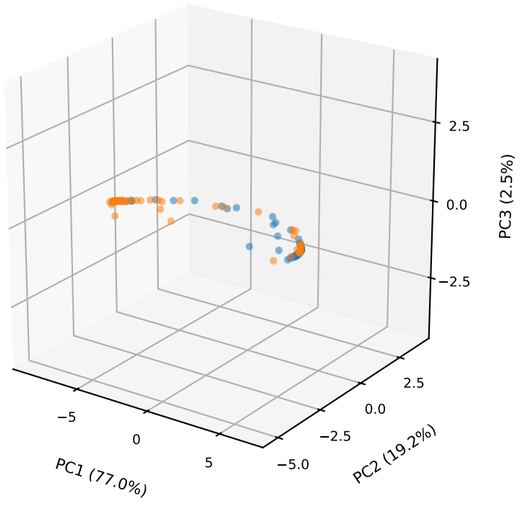
A-fixed: after W5



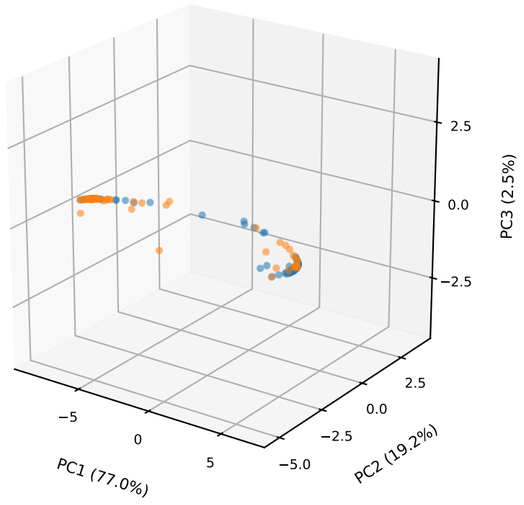
U-refreshed: after W5



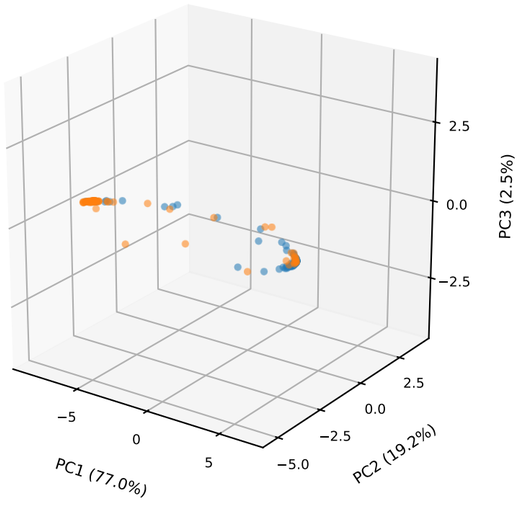
Control: supervised E20



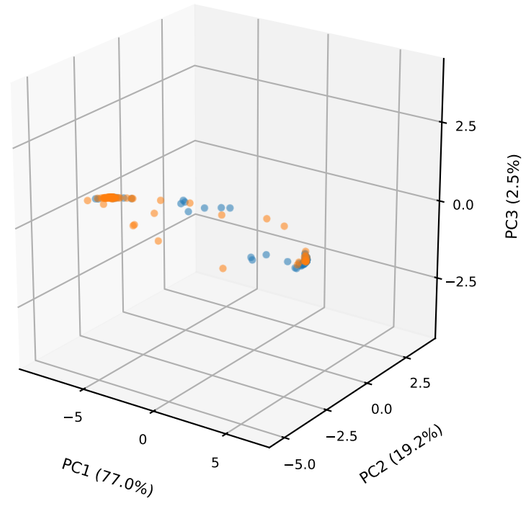
U-fixed: supervised E20



A-fixed: supervised E20



U-refreshed: supervised E20



Non-metal Metal

Fixed camera and common limits across checkpoints; percentages refer to the combined training reference.
Static projection only: overlapping points can hide structure; PCA variance is not prediction accuracy.